NEW SUMMIT COLLEGE

(Affiliated to Tribhuvan University)

Lab Report of

Computer Networking

CACS 303

Bachelors of Computer Application

Faculty of Humanities and Social Sciences

Submitted by:

Name: Sudha Pantha

Semester: V

Program: BCA

Submitted to:

Name: Apil Adhikari

Lab Program Number: …09… ***Date: 2080-10-08***

Title: To configure DHCP and Provide Dynamic Ip to the nodes in network using cisco packet tracer.

Theory:

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to automate the process of configuring devices on IP networks, thus allowing them to use network services such as DNS, NTP, and any communication protocol based on UDP or TCP.

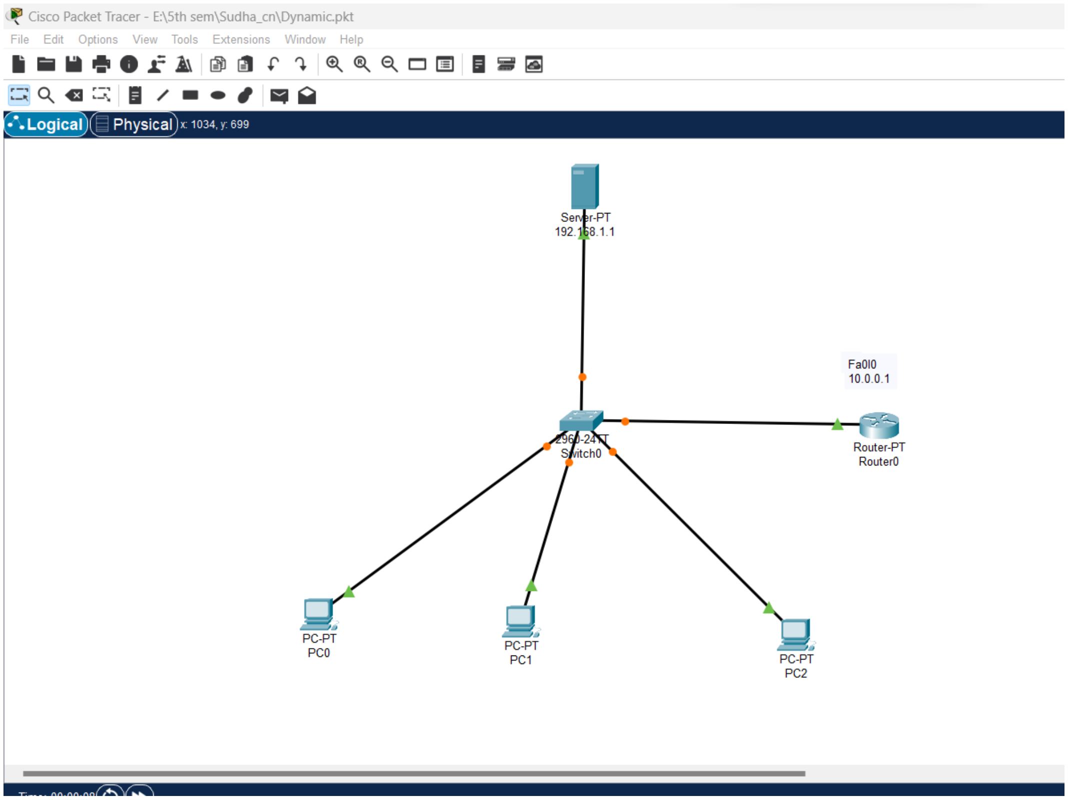
Procedure:

Step 1: Open Cisco Packet Tracer desktop and select all the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	03
02	Switch	PT-Switch	01
03	Server	Server-PT	01
04	Router	Router-PT	01

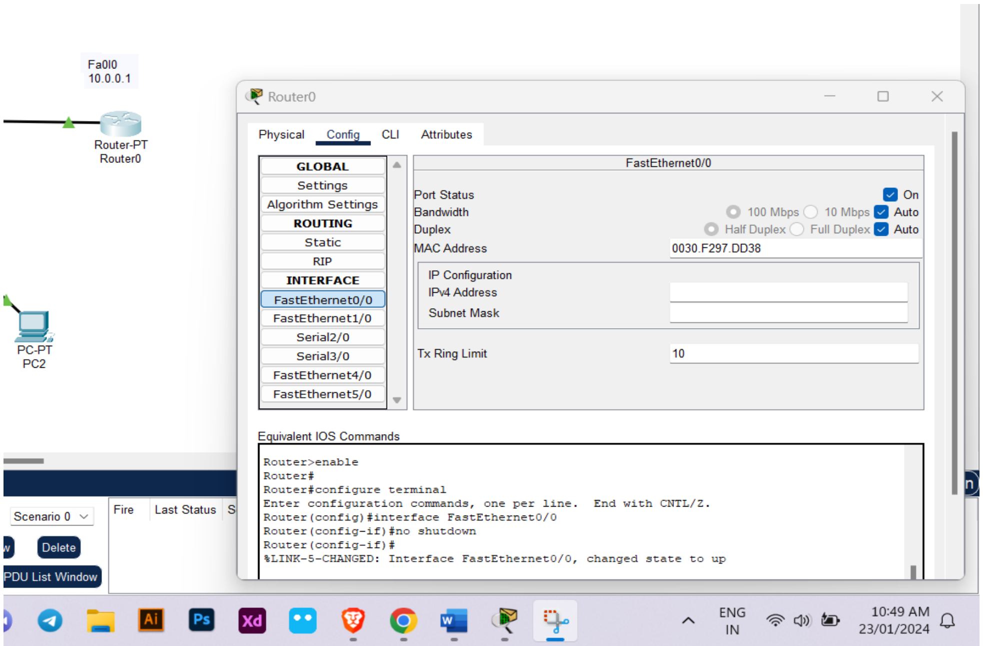
- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch”, “Server” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

Name: Sudha Pantha

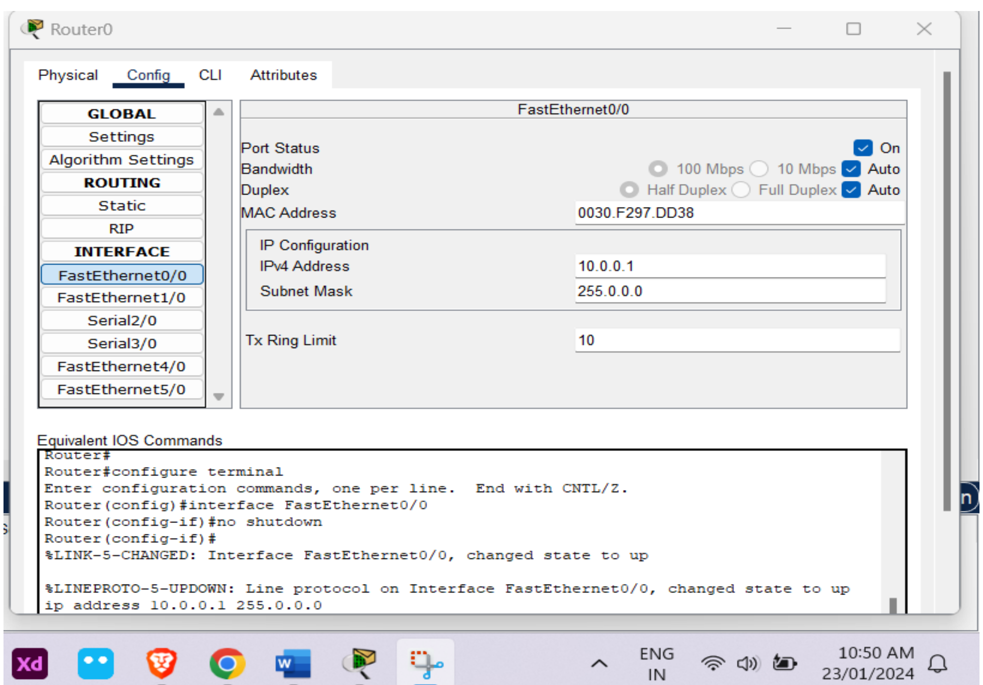


Step 2: Assigning Ip to the Router

- Click on the router in your topology.
- Go to the config and select FastEthernet0/0.
- Click on Port status on.



- Then, Assign Ipv4 address to the Router.



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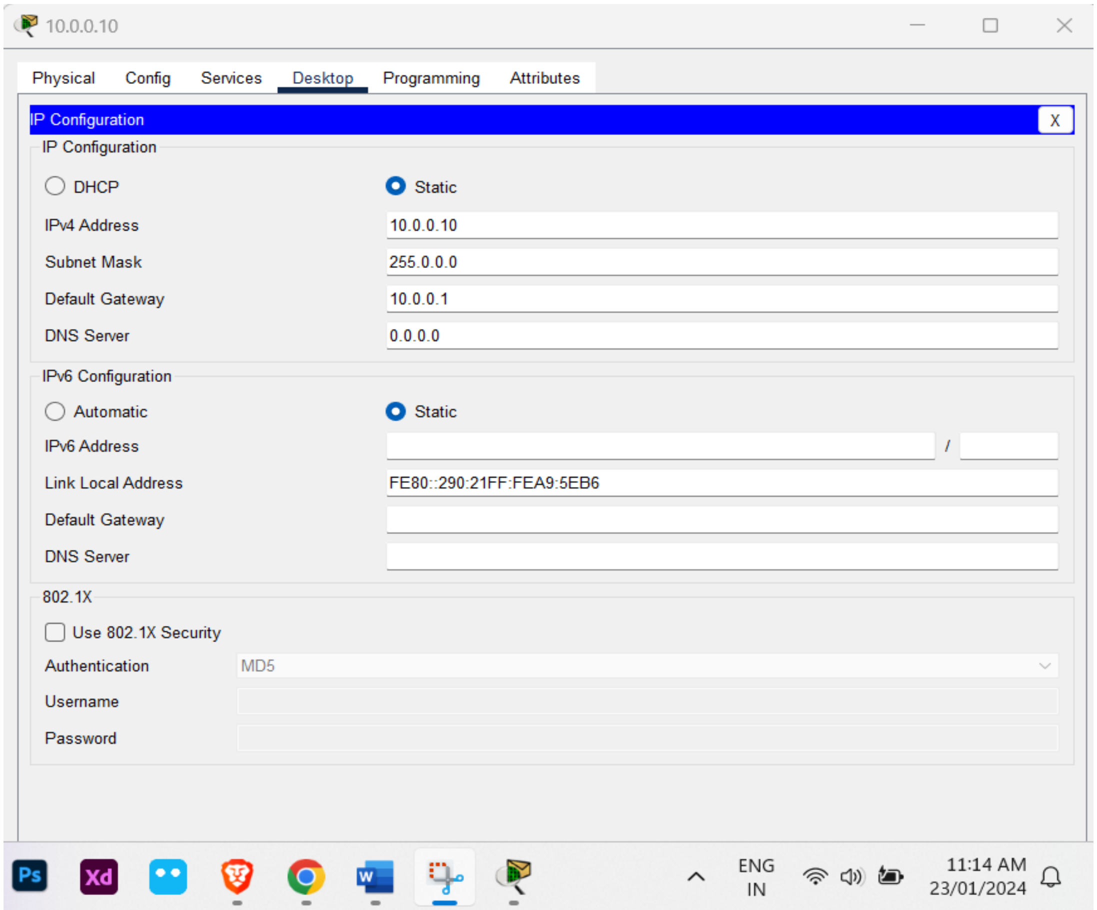
Step 3: Assigning Ip address to the Server

- Click on the server of your topology
- Go to the Ip Config and assign static Ip to the server.



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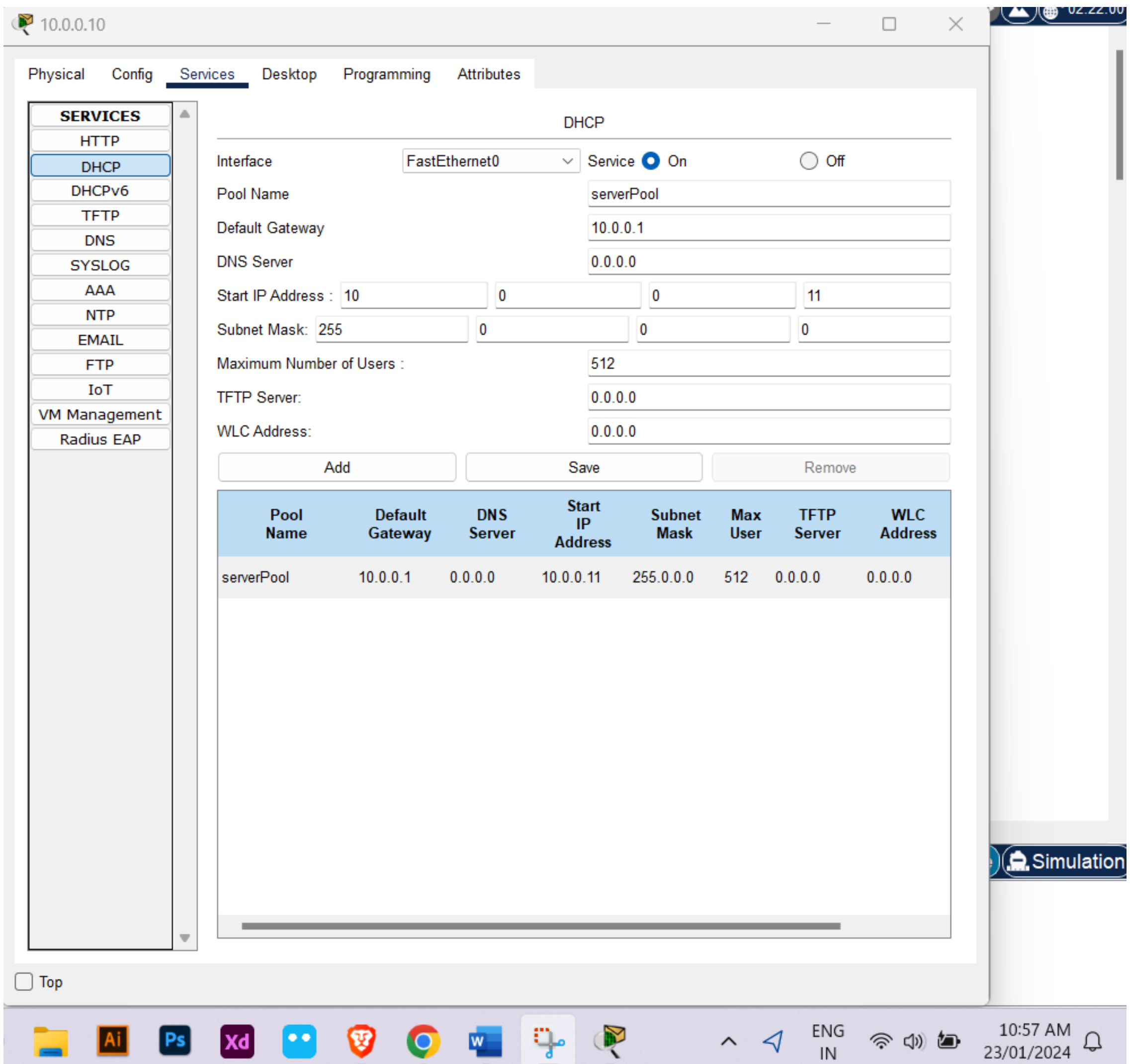
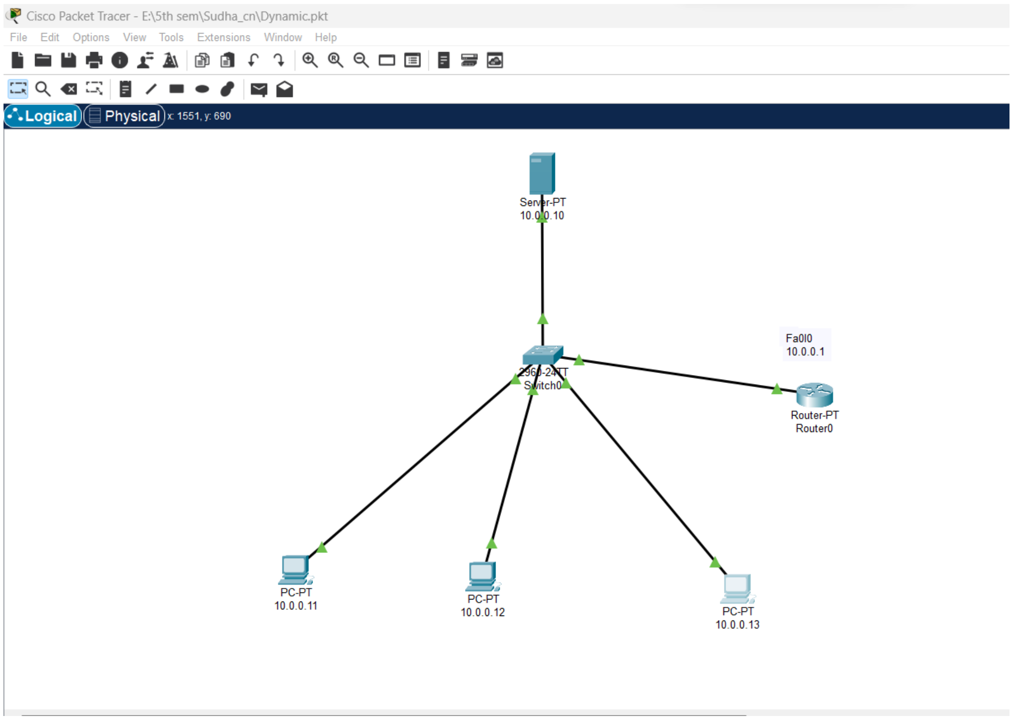
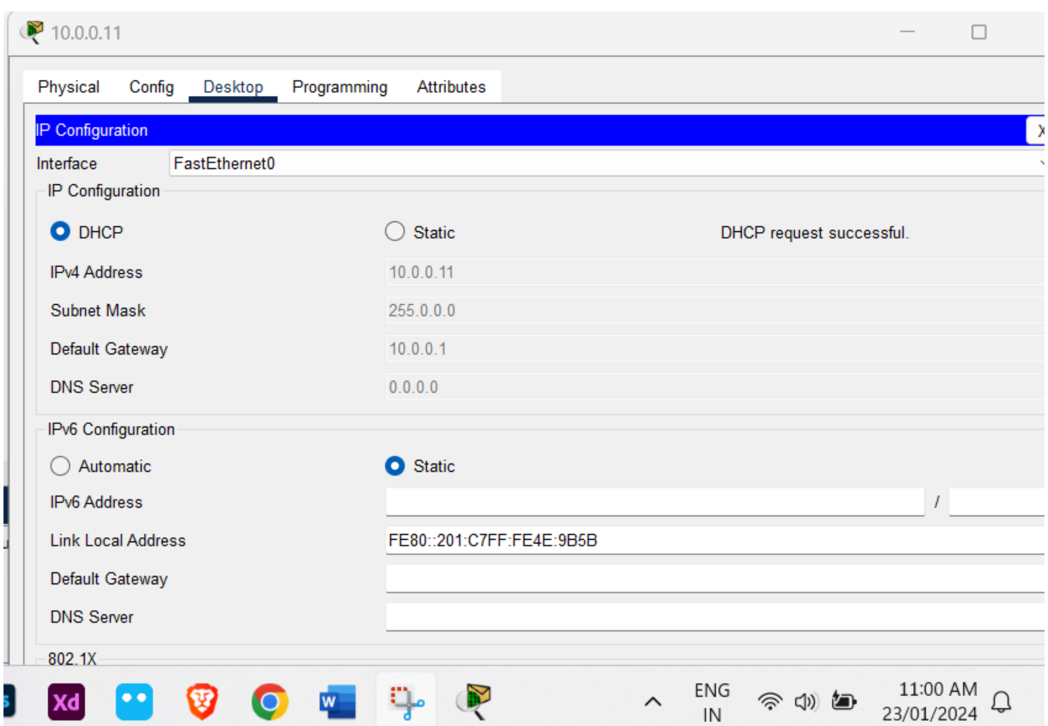
- Then go to the services, click on the DHCP and turn on the service.



Step 4: After assigning Ip to the router and server, we setup IP in PC's as:





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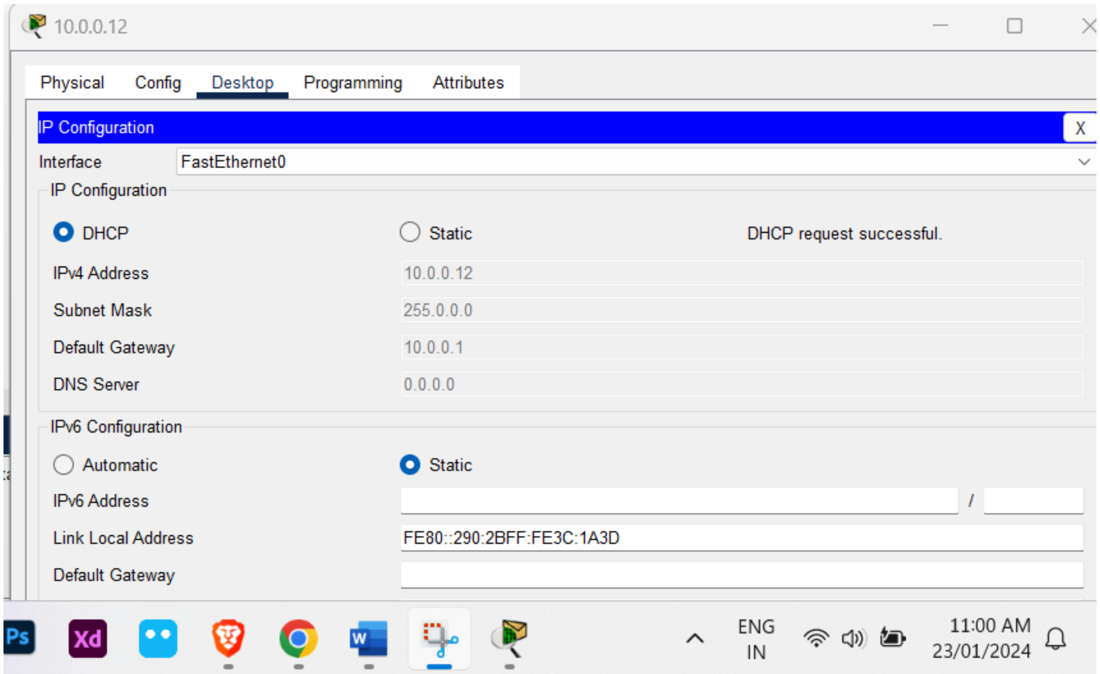

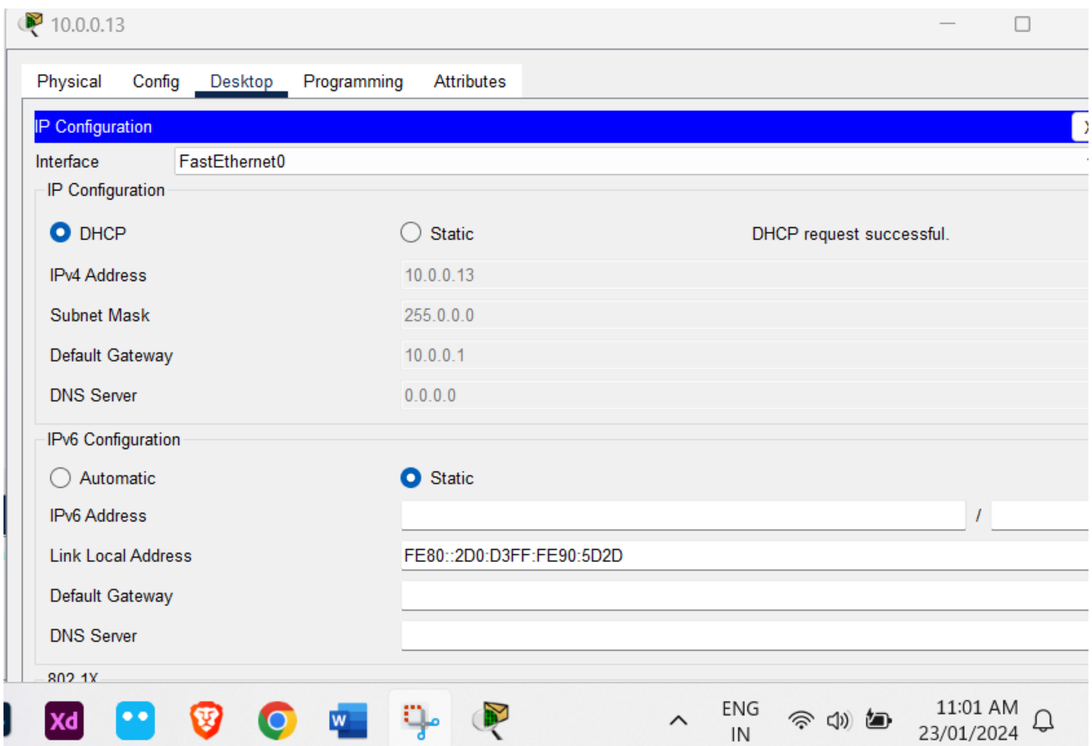

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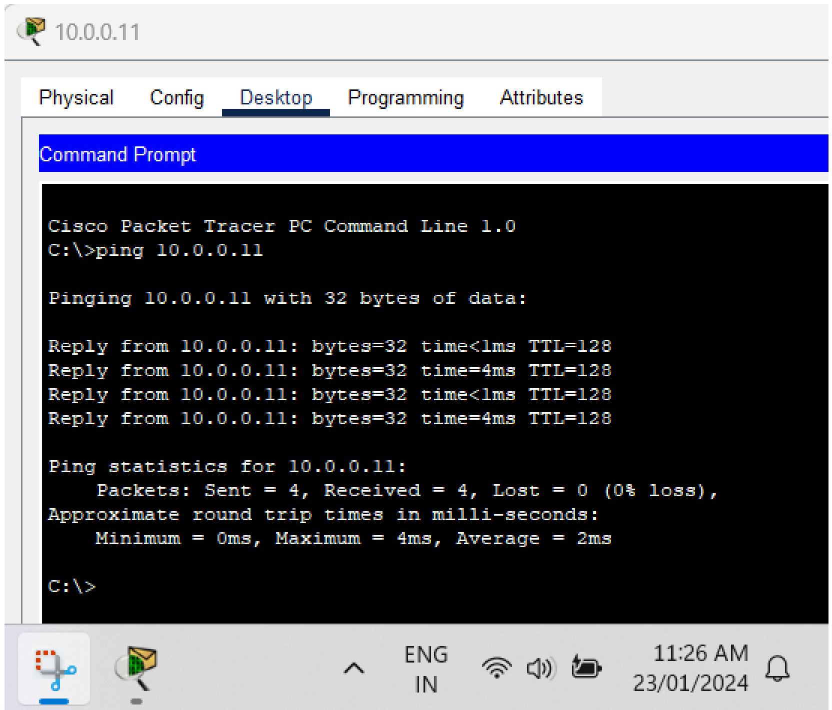
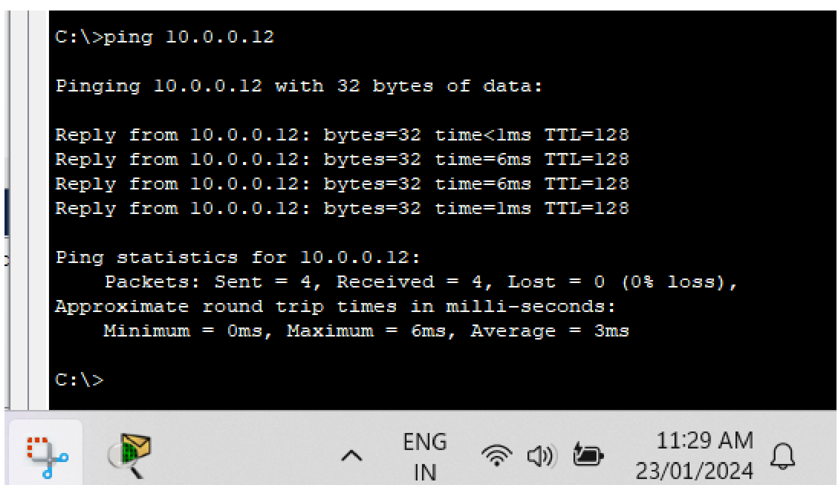
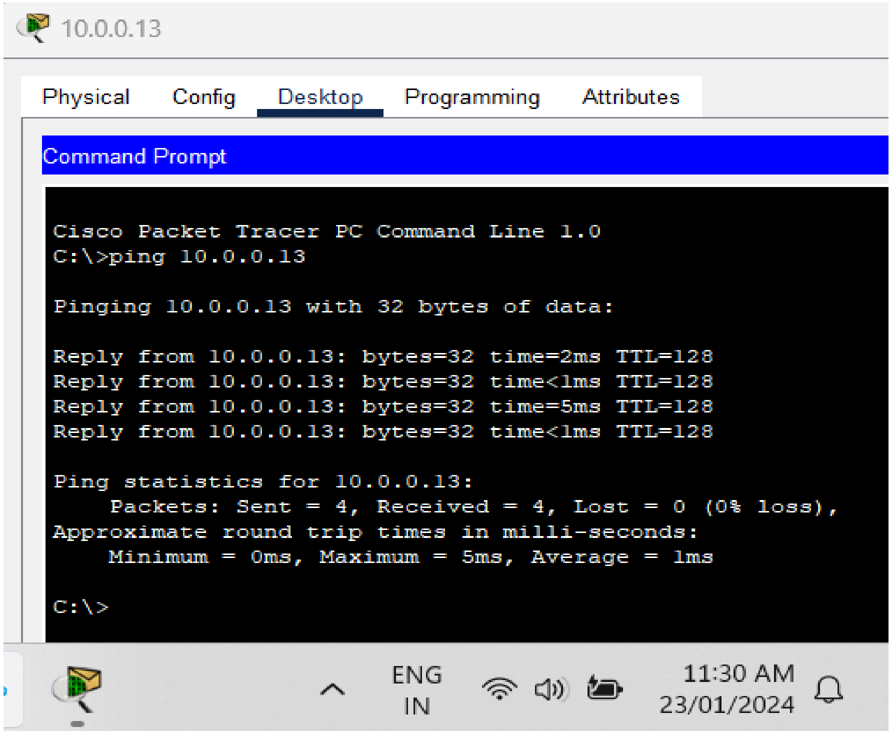
Step 5: At last, we ping each nodes to verify the network.







Conclusion:

Hence, we configured DHCP in router and distributed dynamic IP address.

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Lab Program Number: …10… ***Date: 2080-10-08***

Title: To configure and understand about Routing Information Protocol Using Cisco Packet Tracer.

Theory:

Routing Information Protocol (RIP) is a protocol that routers can use to exchange network topology information. It is characterized as an interior gateway protocol, and is typically used in small to medium-sized networks.

Procedure:

Step 1: First, open the Cisco Packet Tracer desktop and select the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	06
02	Switch	PT-Switch	03
03	Router	Router-PT	03

Ip Addressing Table:

S.No	Device	IPv4 Address	Subnet mask	Default Gateway
1	PC0	192.168.10.2	255.255.255.0	192.168.10.1
2	PC1	192.168.10.3	255.255.255.0	192.168.10.1
3	PC2	192.168.20.2	255.255.255.0	192.168.20.1
4	PC3	192.168.20.3	255.255.255.0	192.168.20.1
5	PC4	192.168.30.2	255.255.255.0	192.168.30.1
6	PC5	192.168.30.3	255.255.255.0	192.168.30.1

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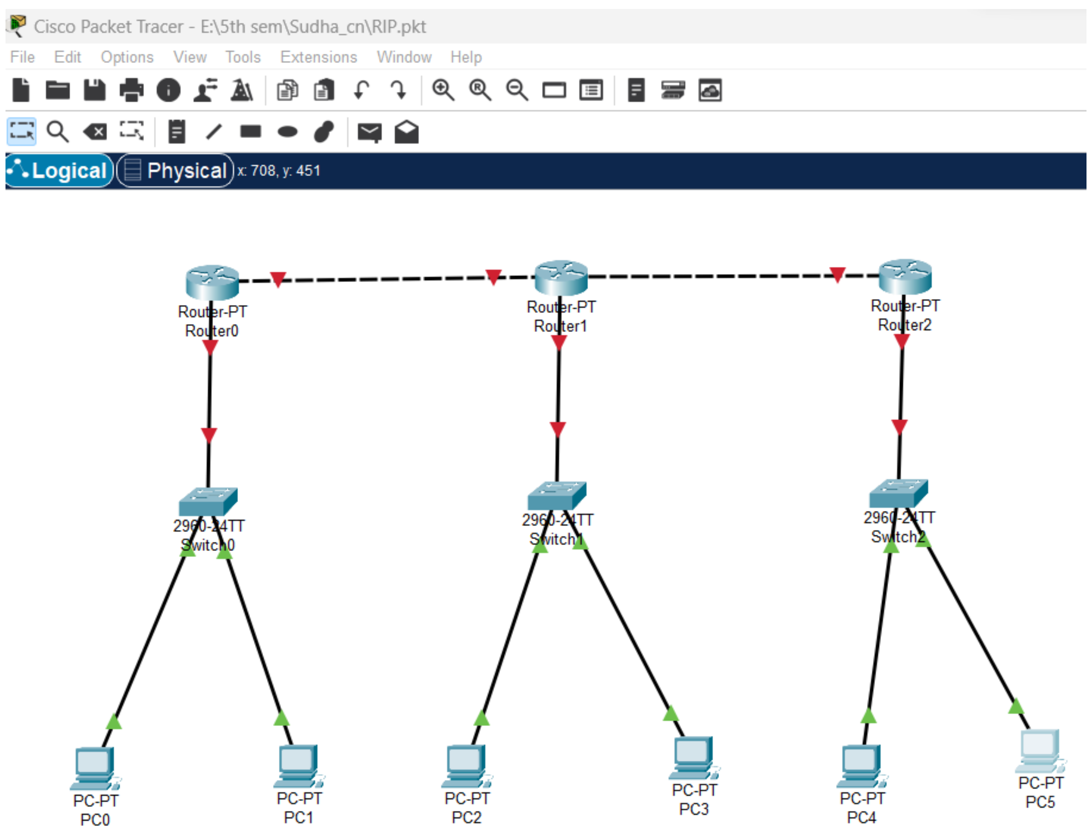
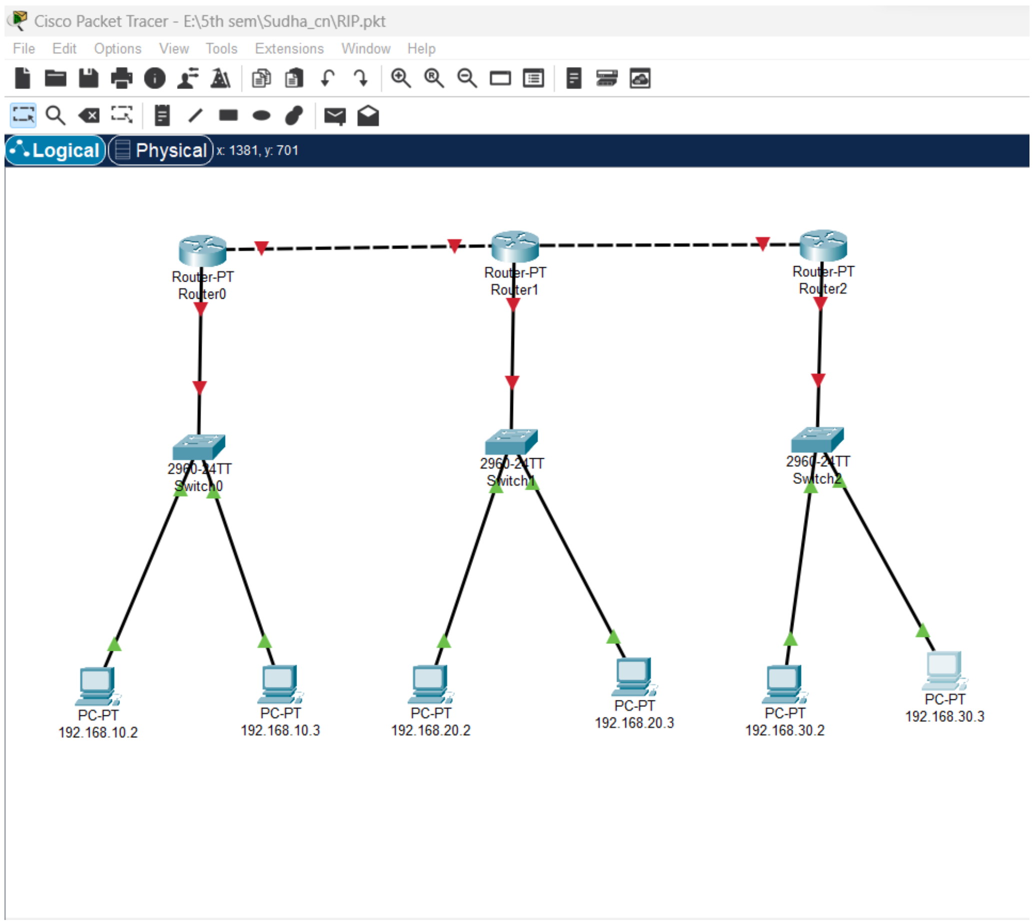
- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables for different devices and cross over for same devices.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

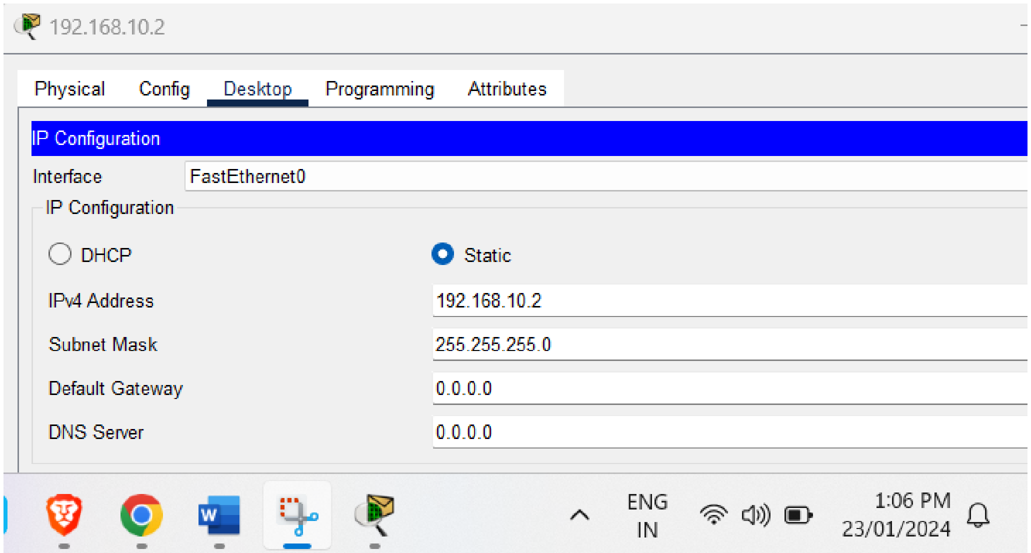


Step 2**:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

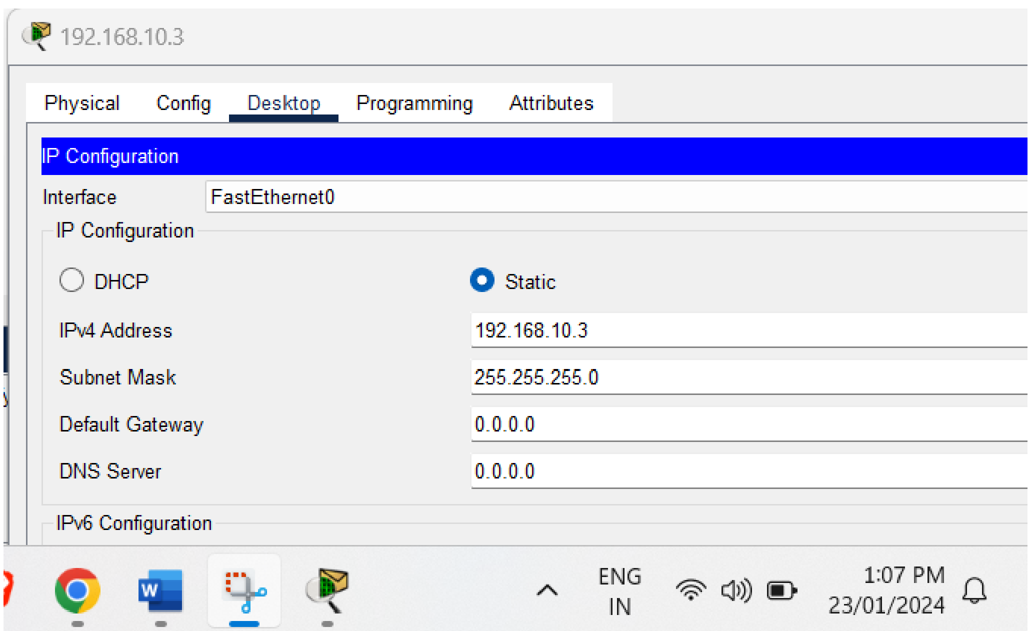
- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.



- PC0:



- PC1:



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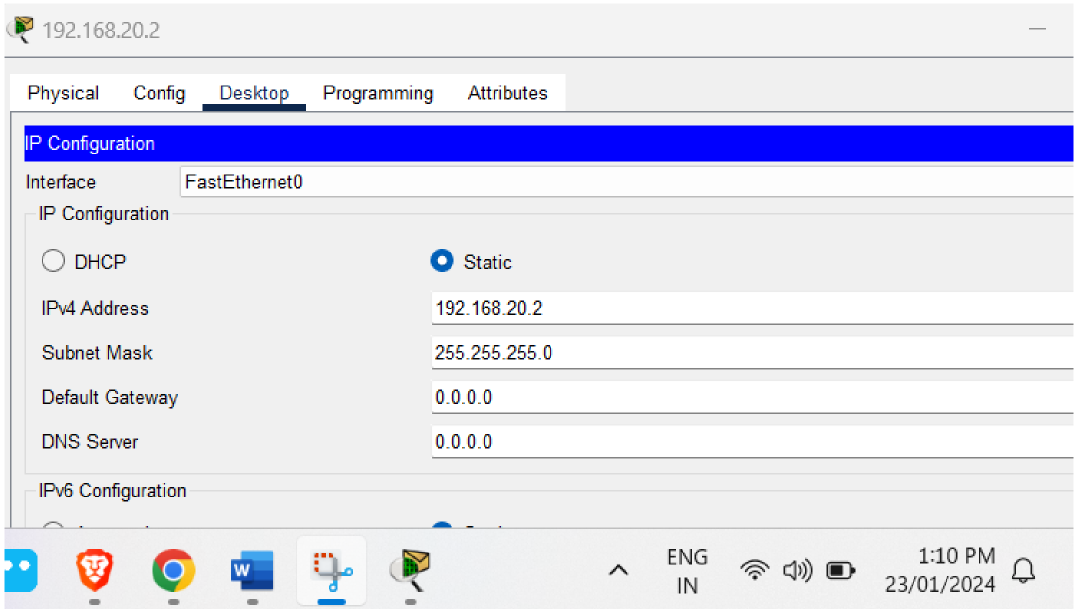
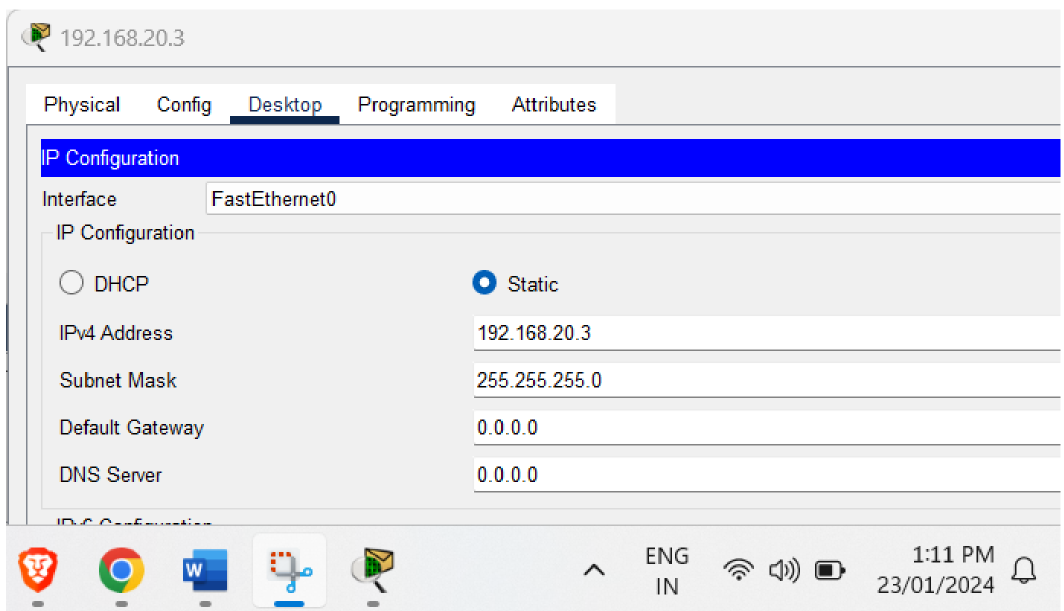
- PC 2:



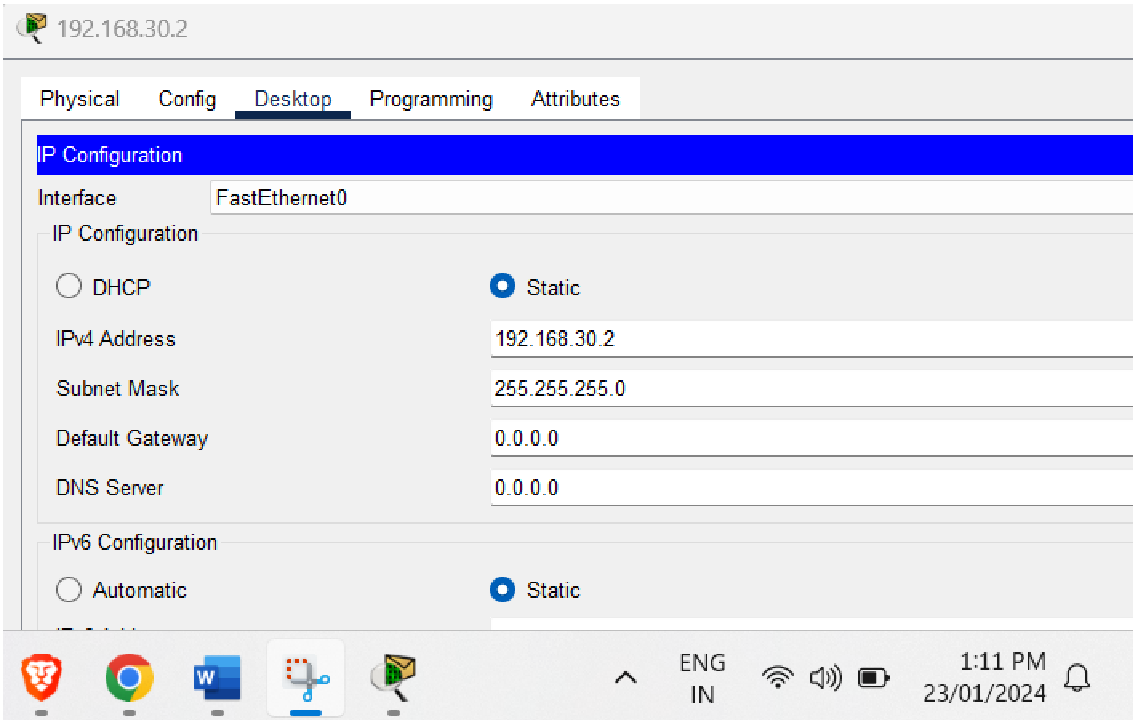
192.168.20.2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
ENG
IN
1:10 PM
23/01/2024

- PC3:

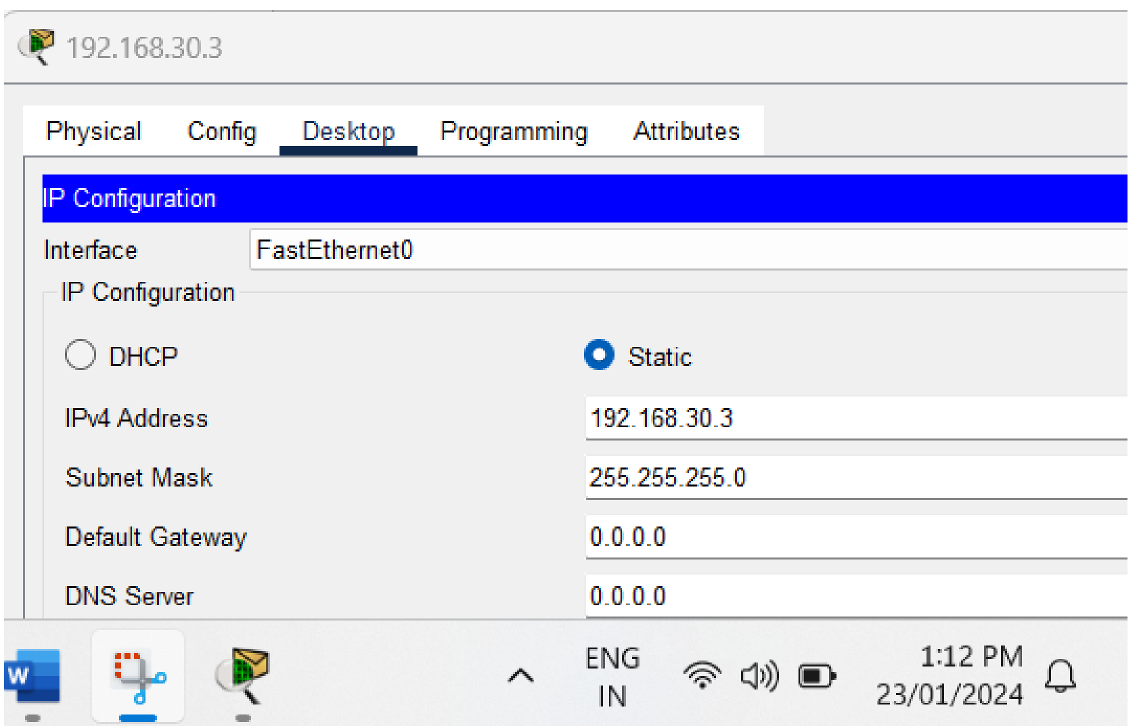


192.168.20.3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
ENG
IN
1:11 PM
23/01/2024

- PC4:



- PC5:



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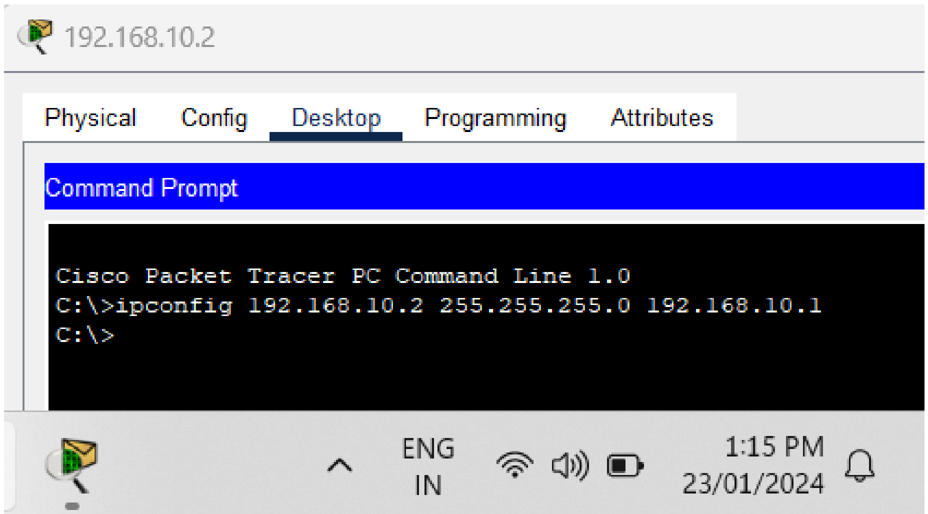
- Assigning an IP address using the ipconfig command, or we can also assign an IP address with the help of a command.
- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address><subnet mask><default gateway> (if needed)



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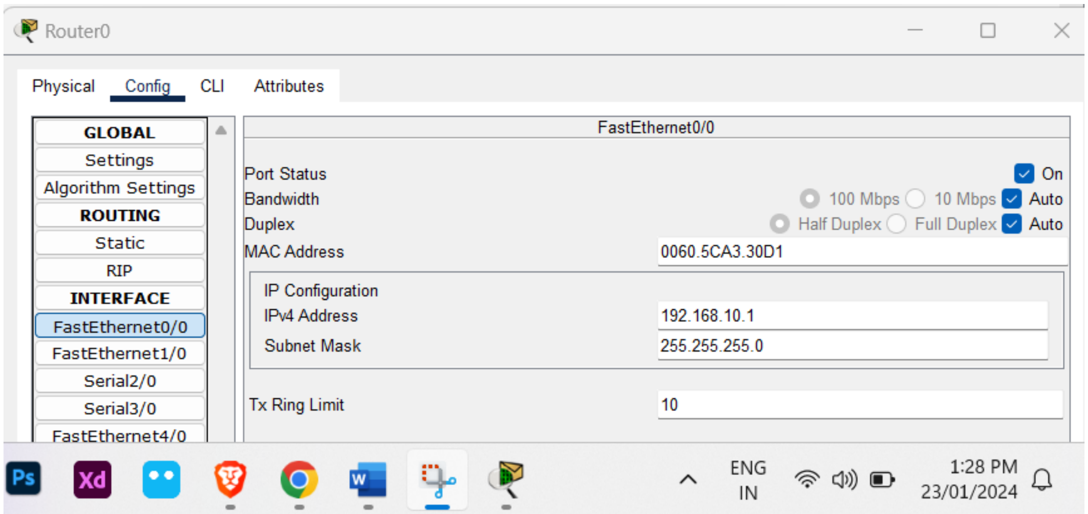
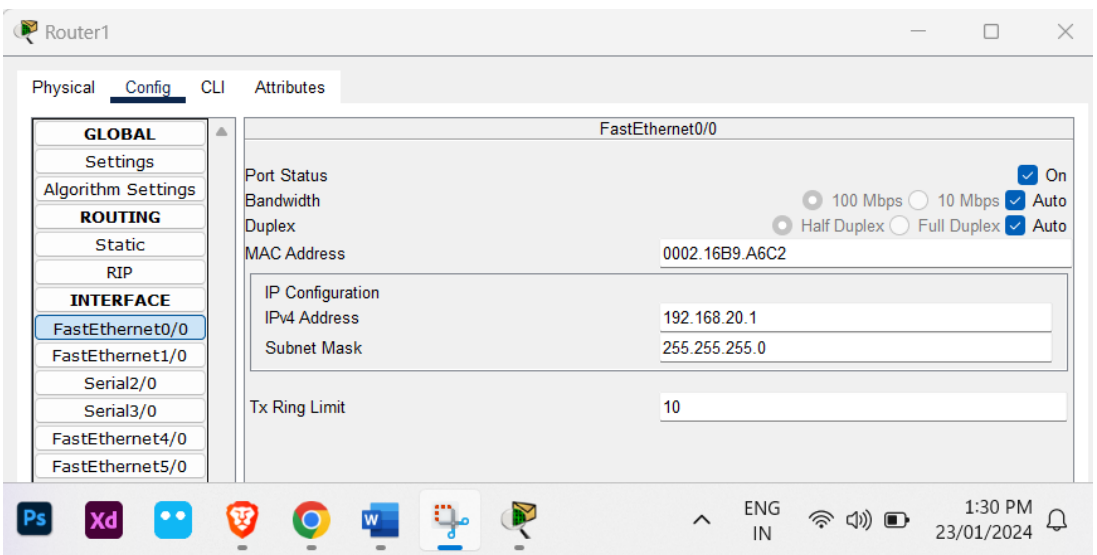
- Repeat the same procedure with other routers to configure them thoroughly.

Step 3: Configure router with IP address and Subnet mask.

- To assign an IP address in router, click on router.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in FastEthernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask for all router.





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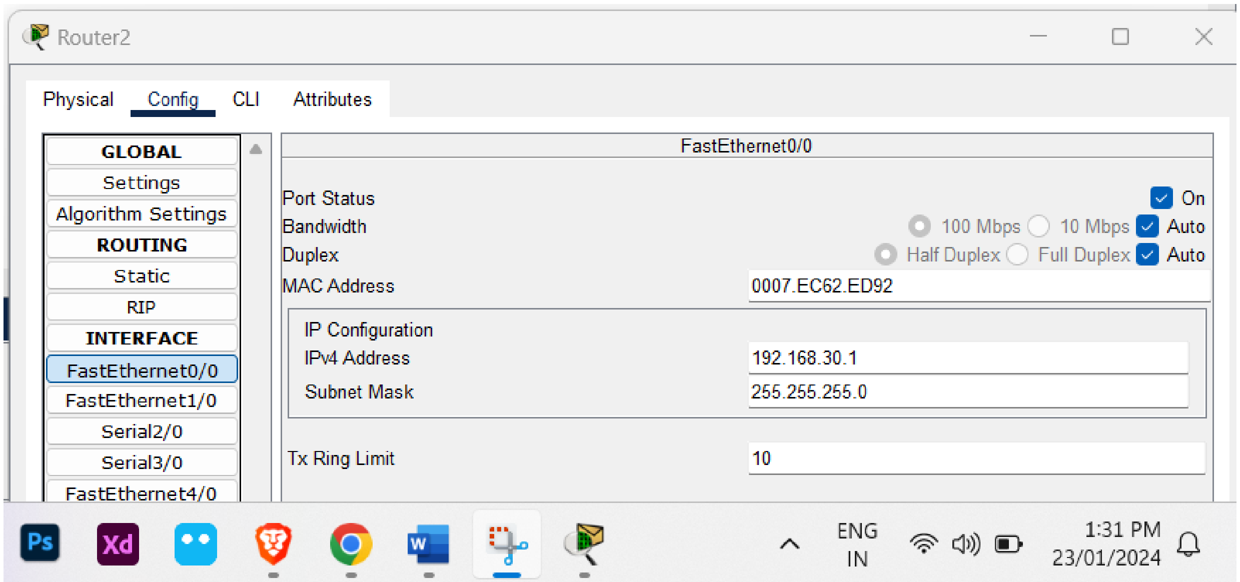

Step 4: After configuring all of the devices we need to assign the routes to the router.

To assign RIP routes to the particular router:

- First, click on router then Go to CLI.
- Then type the commands and IP information given below.
  - CLI command: router rip
  - CLI command: network <network id>

RIP Routes for Router0 are given below:

```
Router>en
Router#conf
Configuring from terminal, memory, or network [terminal]? y
?Must be "terminal", "memory" or "network"
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.10.1
Router(config-router)#network 192.168.30.1
Router(config-router)#do write
Building configuration...
[OK]
Router(config-router)#
```

ENG
IN
1:49 PM
23/01/2024

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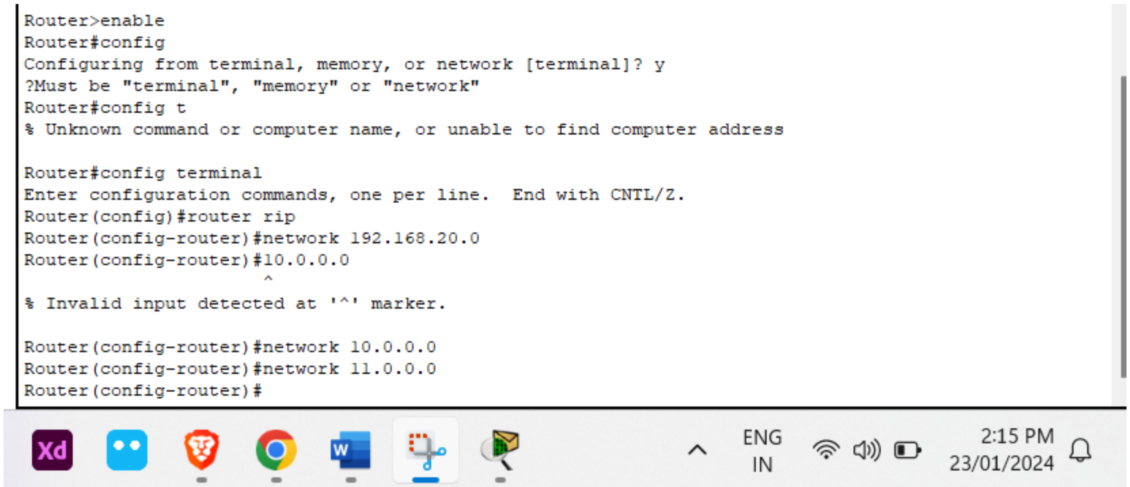
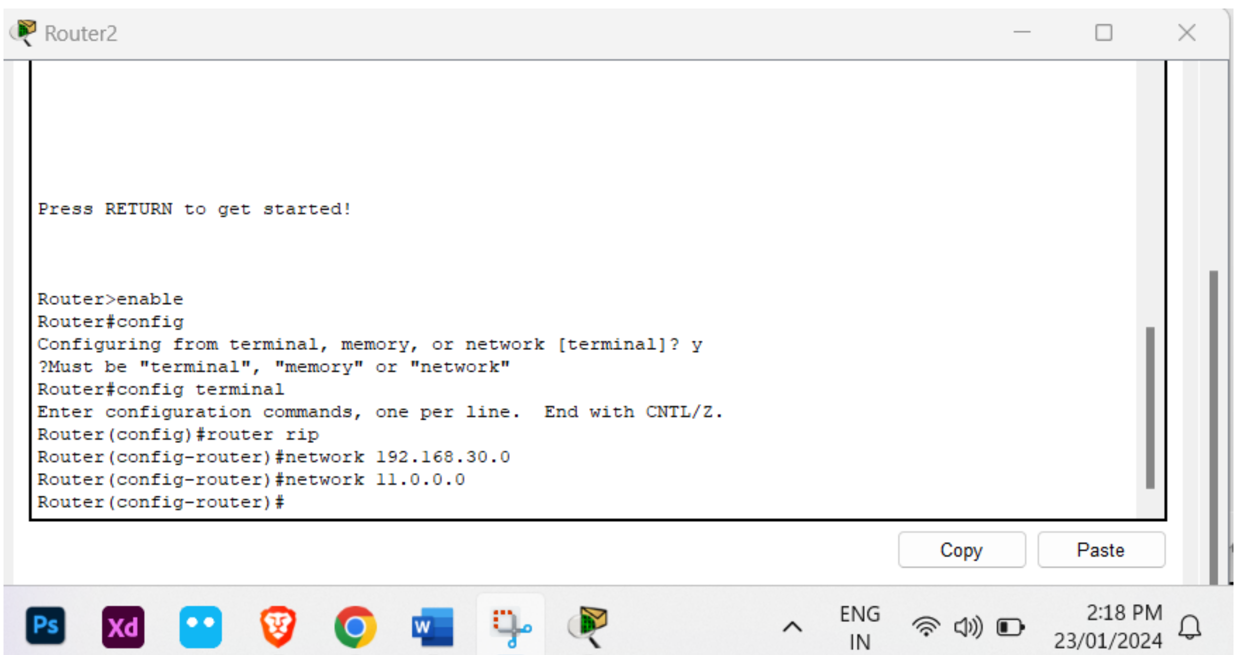
RIP Routes for Router1 are given below:




```
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]? y
?Must be "terminal", "memory" or "network"
Router#config t
% Unknown command or computer name, or unable to find computer address

Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.20.0
Router(config-router)#10.0.0.0
                       ^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.0.0.0
Router(config-router)#network 11.0.0.0
Router(config-router)#
```

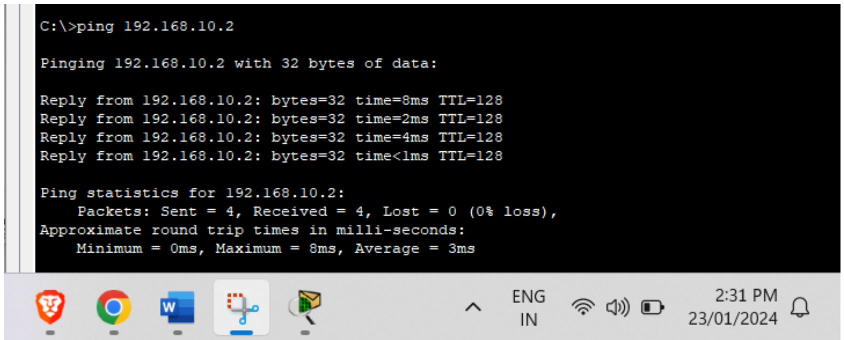
RIP Routes for Router2 are given below:



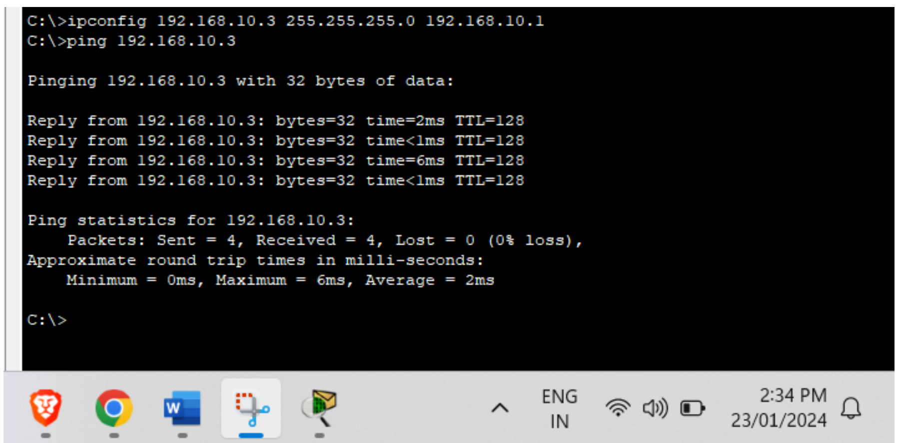

```
Press RETURN to get started!


Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]? y
?Must be "terminal", "memory" or "network"
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.30.0
Router(config-router)#network 11.0.0.0
Router(config-router)#
```

Step 5: Verifying the network by pinging the IP address of any PC.

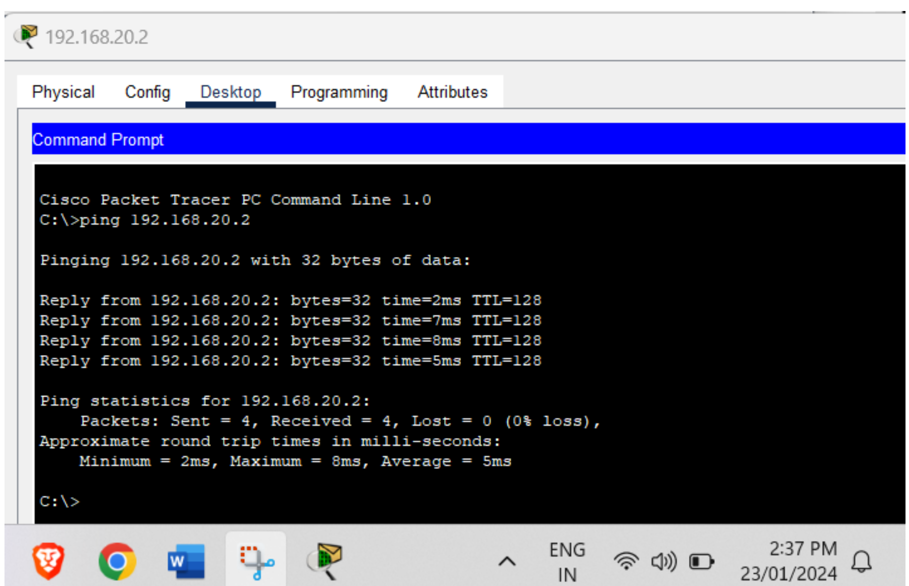
- PC0:



- PC1:



- PC2:



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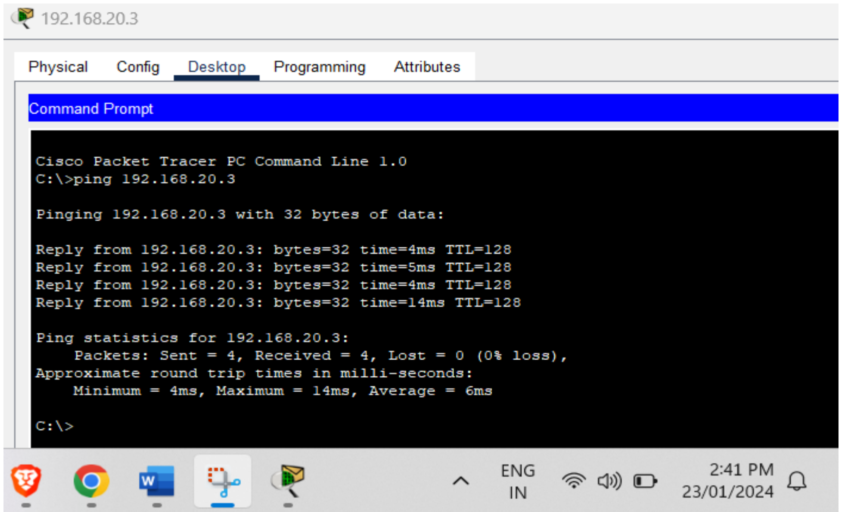
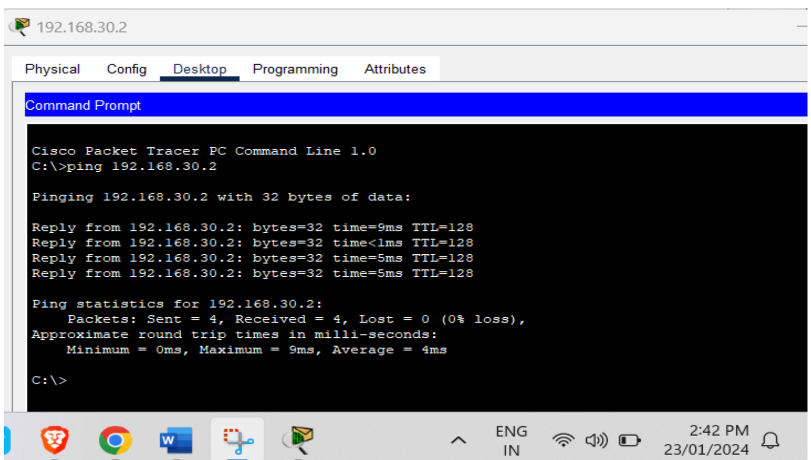
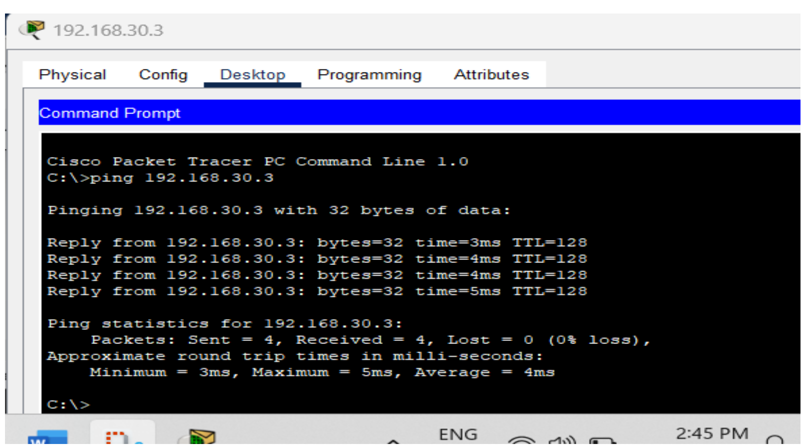
- PC3:



- PC4:



- PC5:



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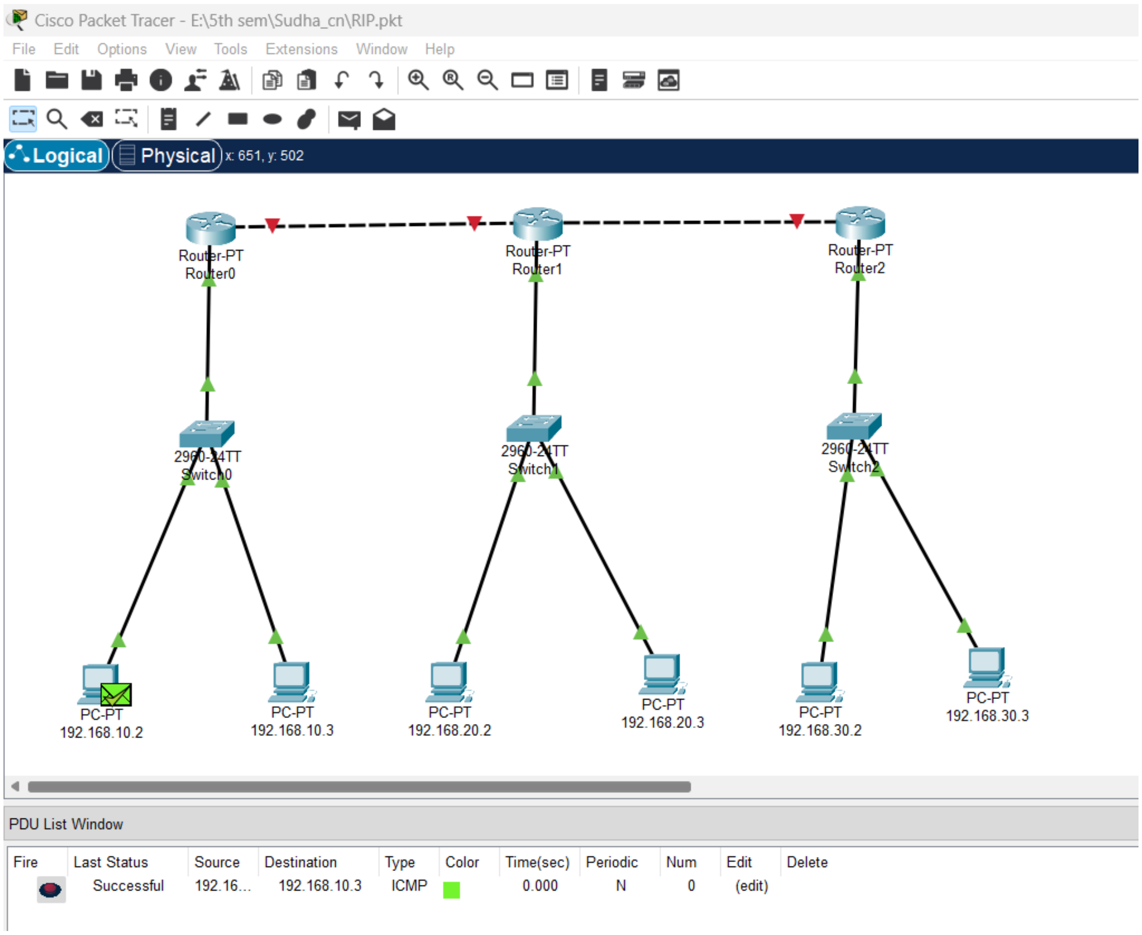
Step 6: Simulation

- A simulation of the experiment is given below we are sending PDU from PC0 to PC1:



- Verifying the network by pinging the IP address of any PC.

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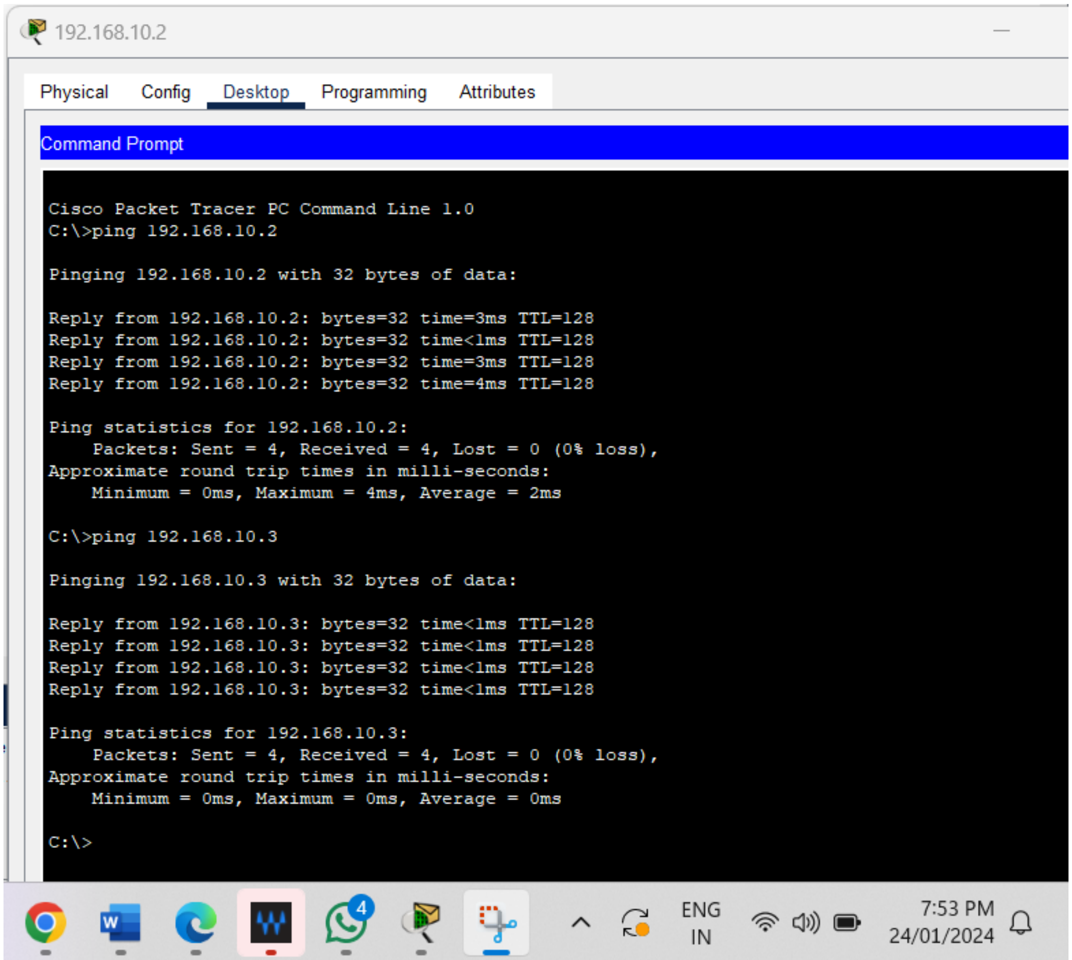

Conclusion:

Hence, we configured RIP protocol successfully verified the connection.

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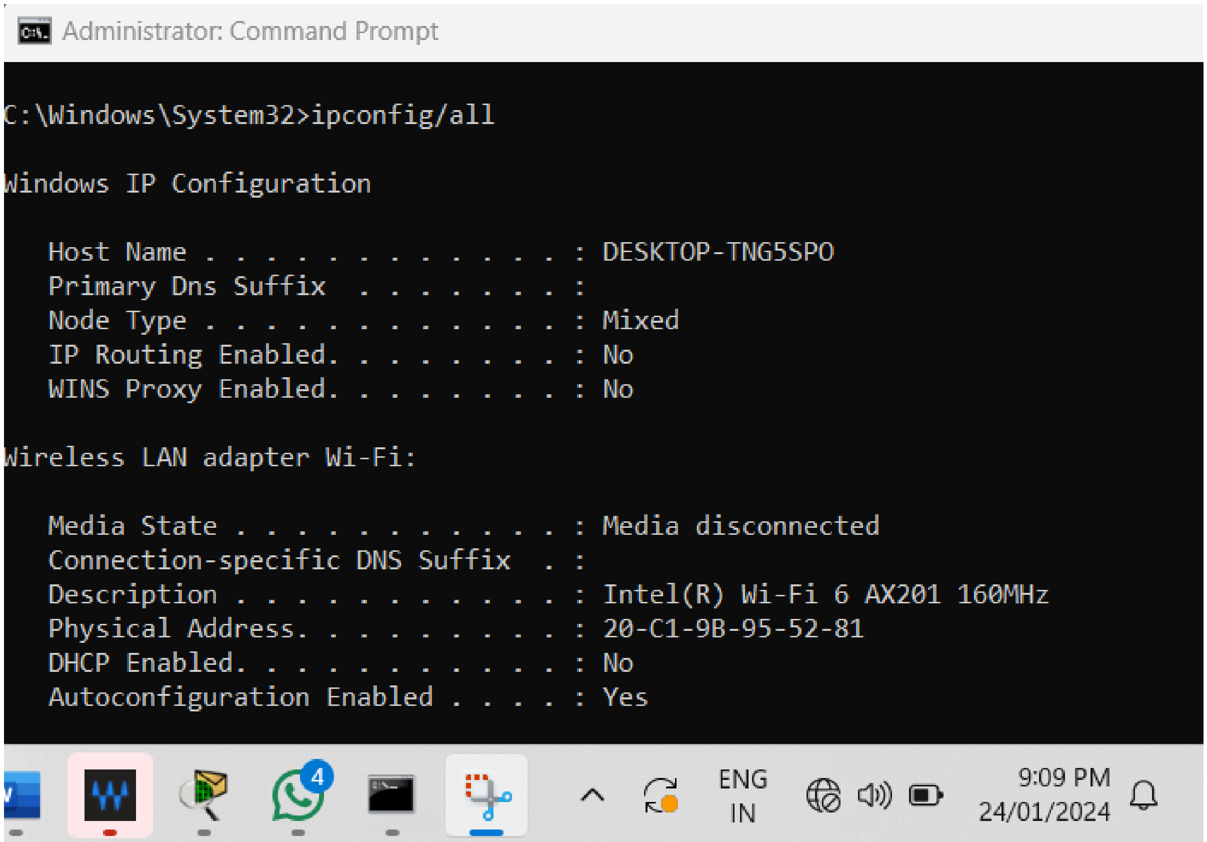


Name: Sudha Pantha

Lab Program Number: …11… ***Date: 2080-10-08***

Title: To configure and understand about VLAN using cisco packet tracer.

Theory:

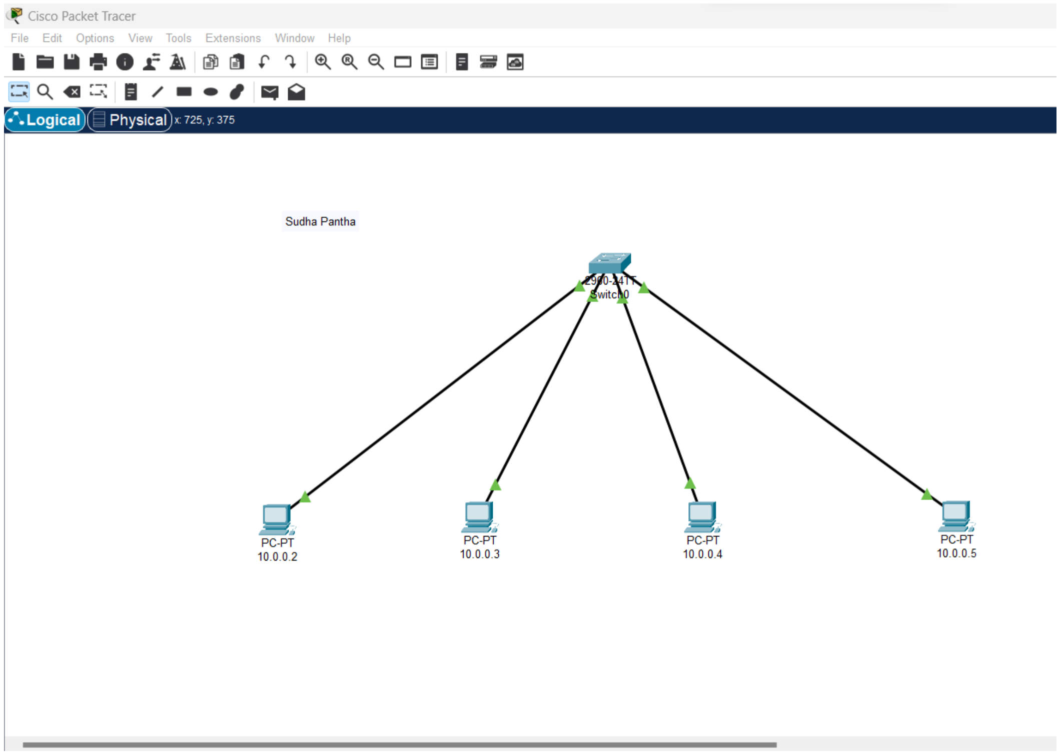
VLANs allow network administrators to automatically limit access to a specified group of users by dividing workstations into different isolated LAN segments. When users move their workstations, administrators don't need to reconfigure the network or change VLAN groups.

Procedure:

Step 1: Open Cisco Packet Tracer desktop and select all the devices given below:

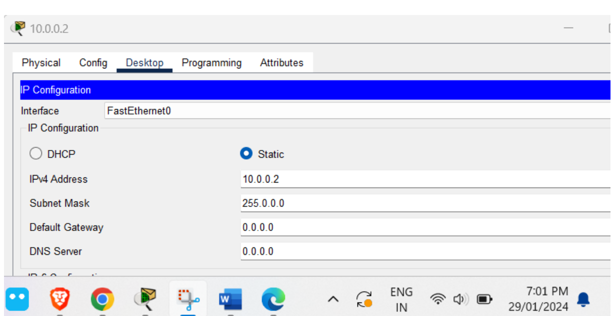
S.No	Device	Model Name	Qty.
01	PC	PC	04
02	Switch	PT-Switch	01

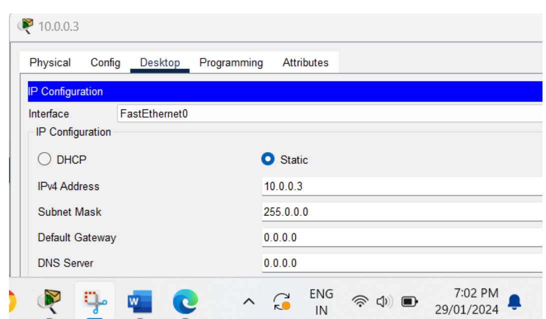
- In the "Devices" tab on the left, select "End Devices" and drag a "Generic" Pc or laptop to the workspace.
- Similarly, drag a "Switch" to the workspace.
- Click on the "Connections" icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device's interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.



Step 2**:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.





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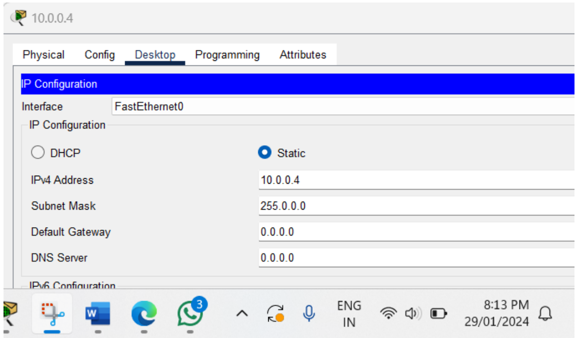
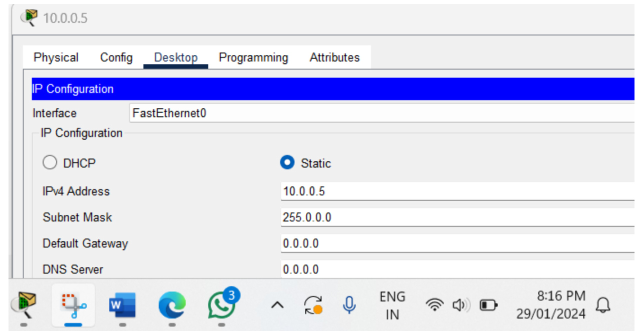
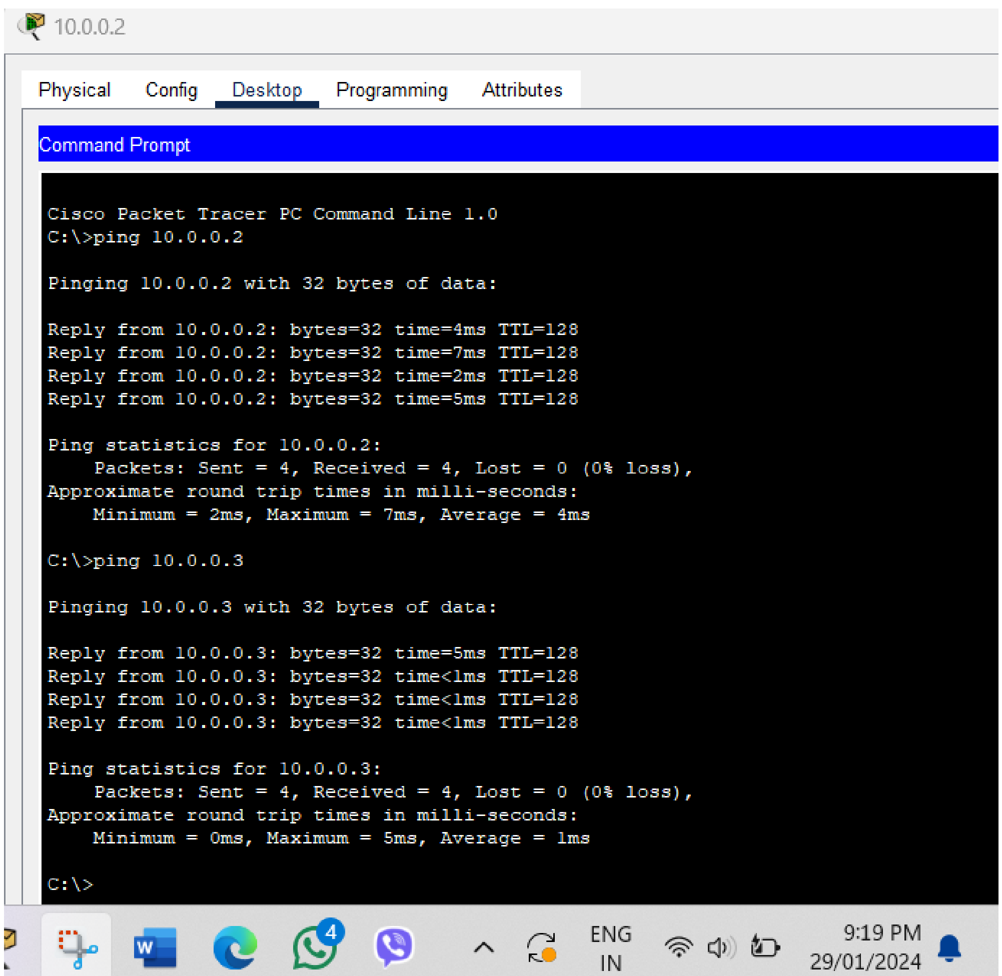




Step 3: Verifying the network by pinging the IP address of any PC.

In PC0 pinging the Ip of PC1 which is reliable.



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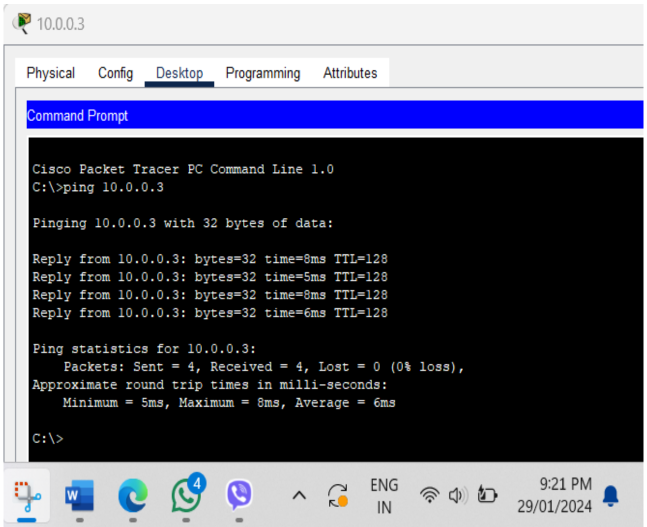

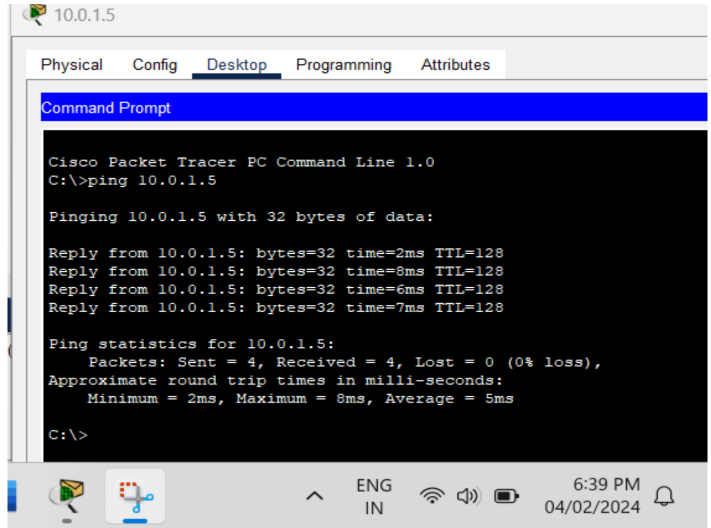

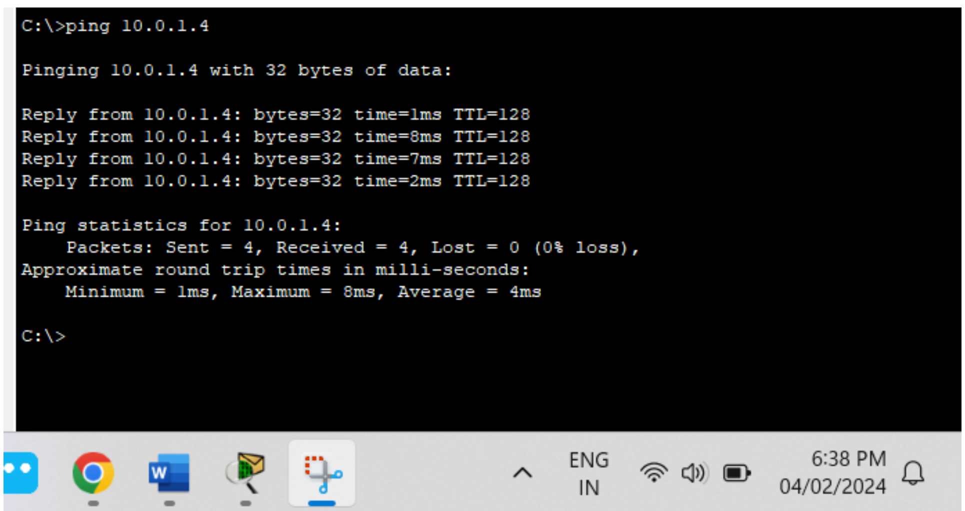

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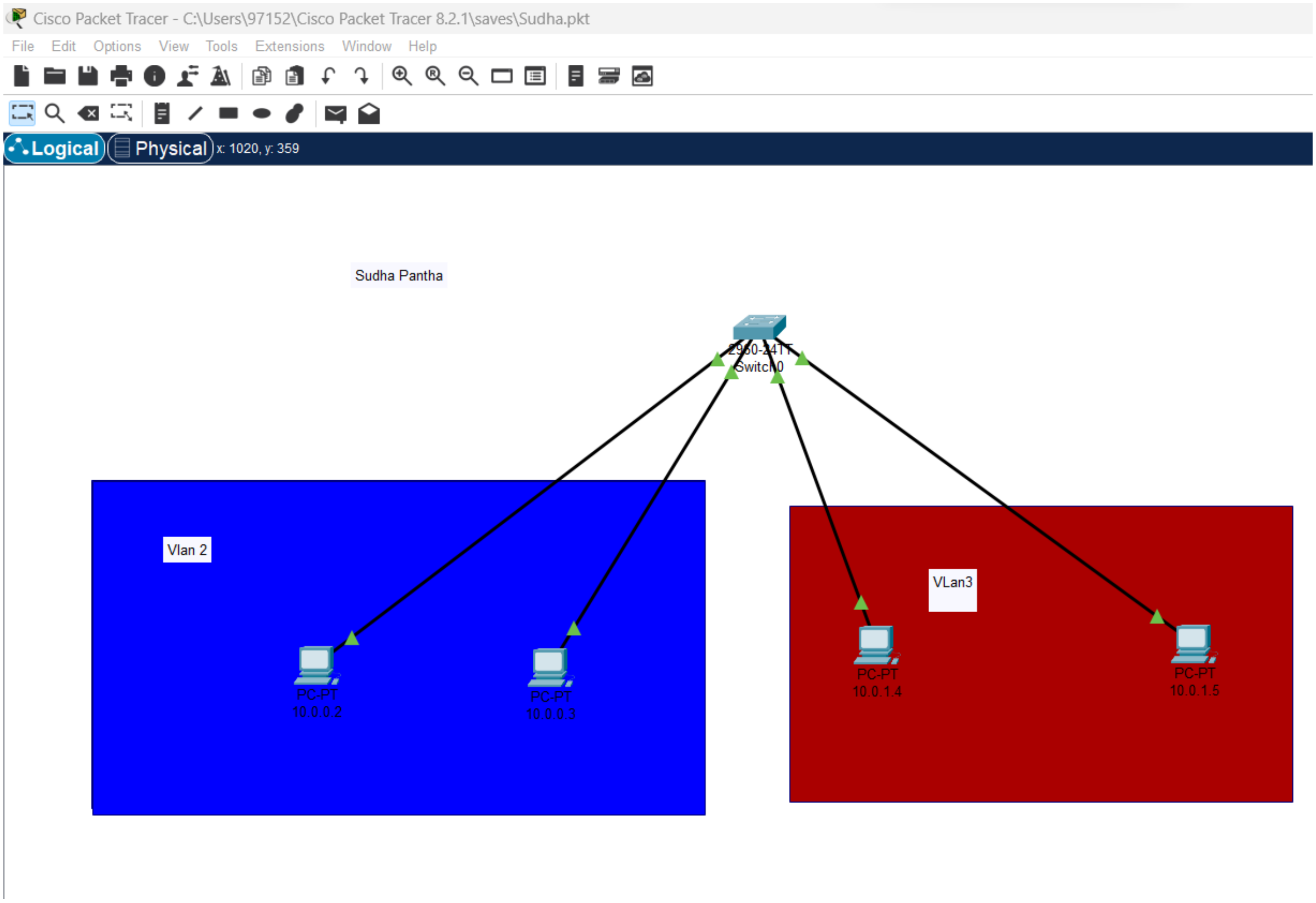
Step 4: Creating VLan 2 and Vlan 3



Step 5: Configuring the setting as:

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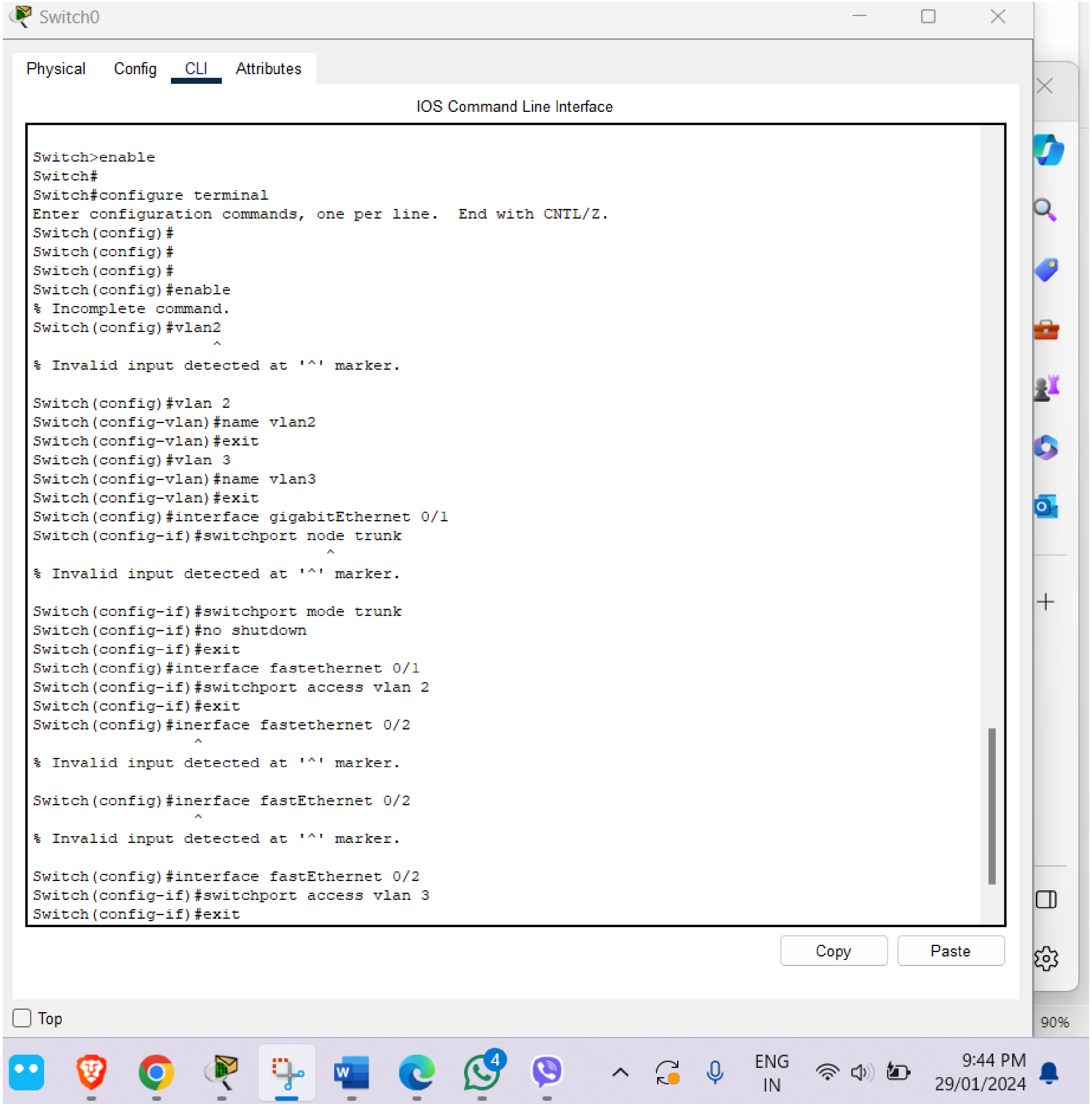
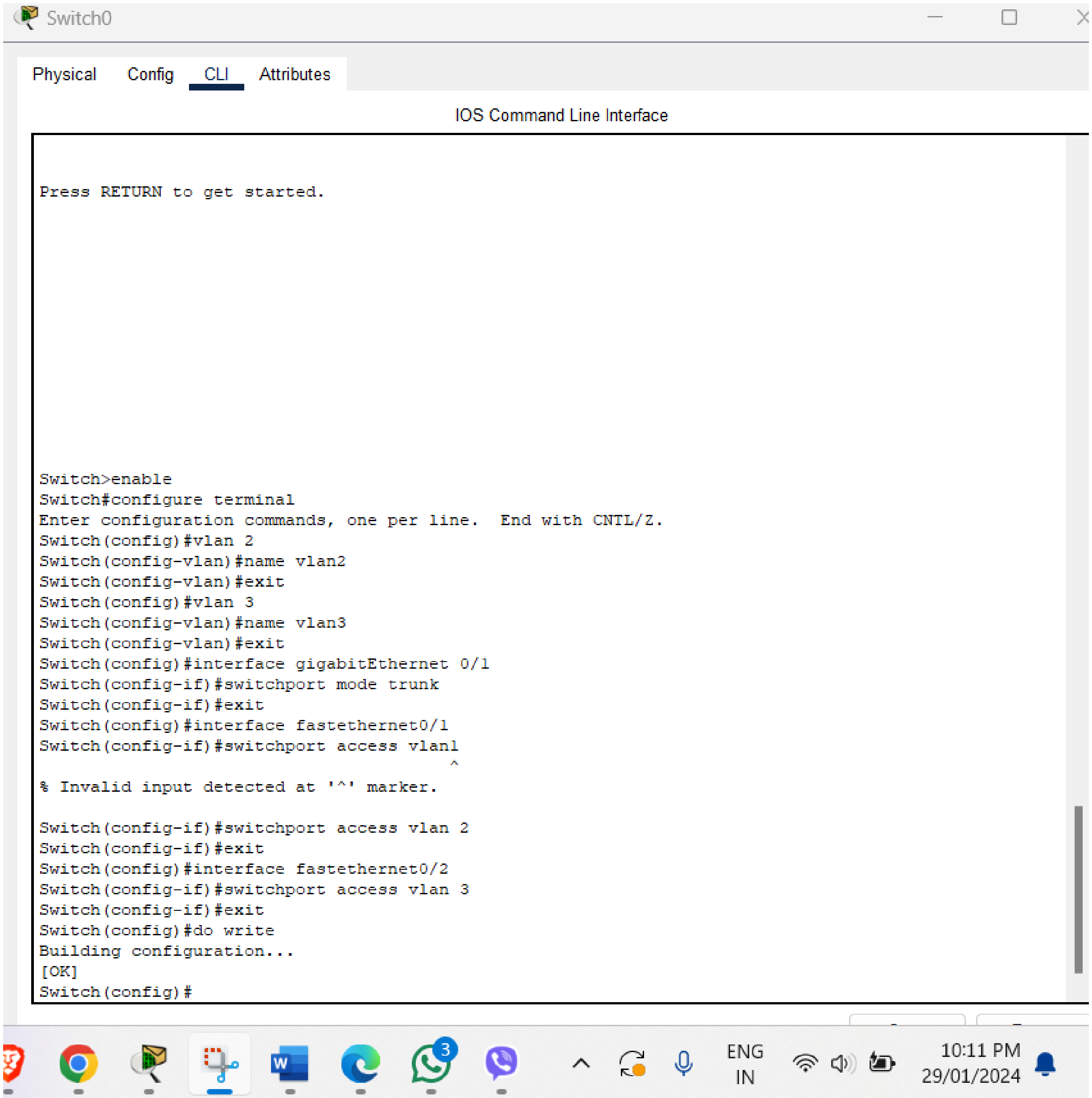


Name: Sudha Pantha



Step 6: Finally, we ping computers and verity network in vlan

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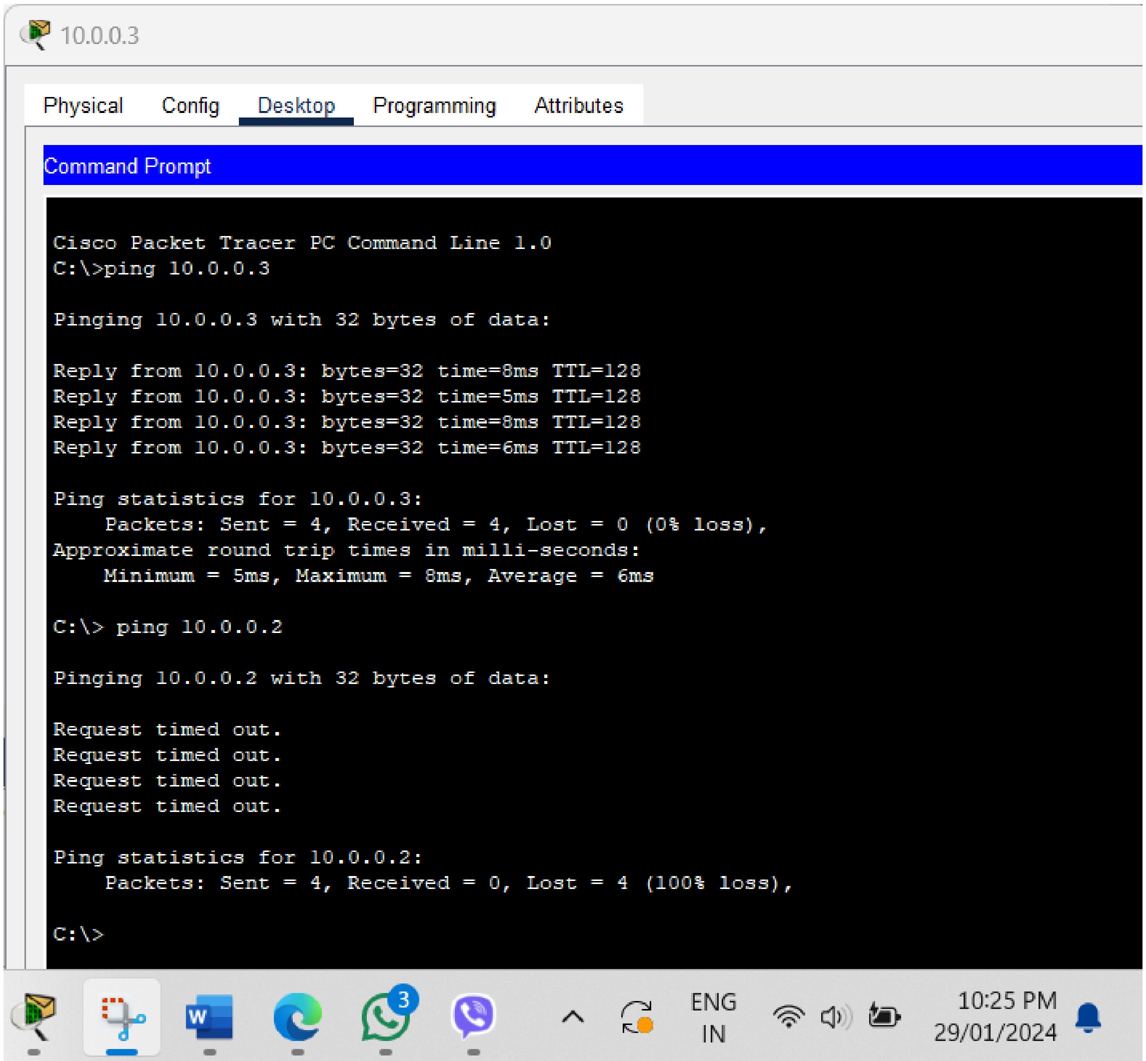
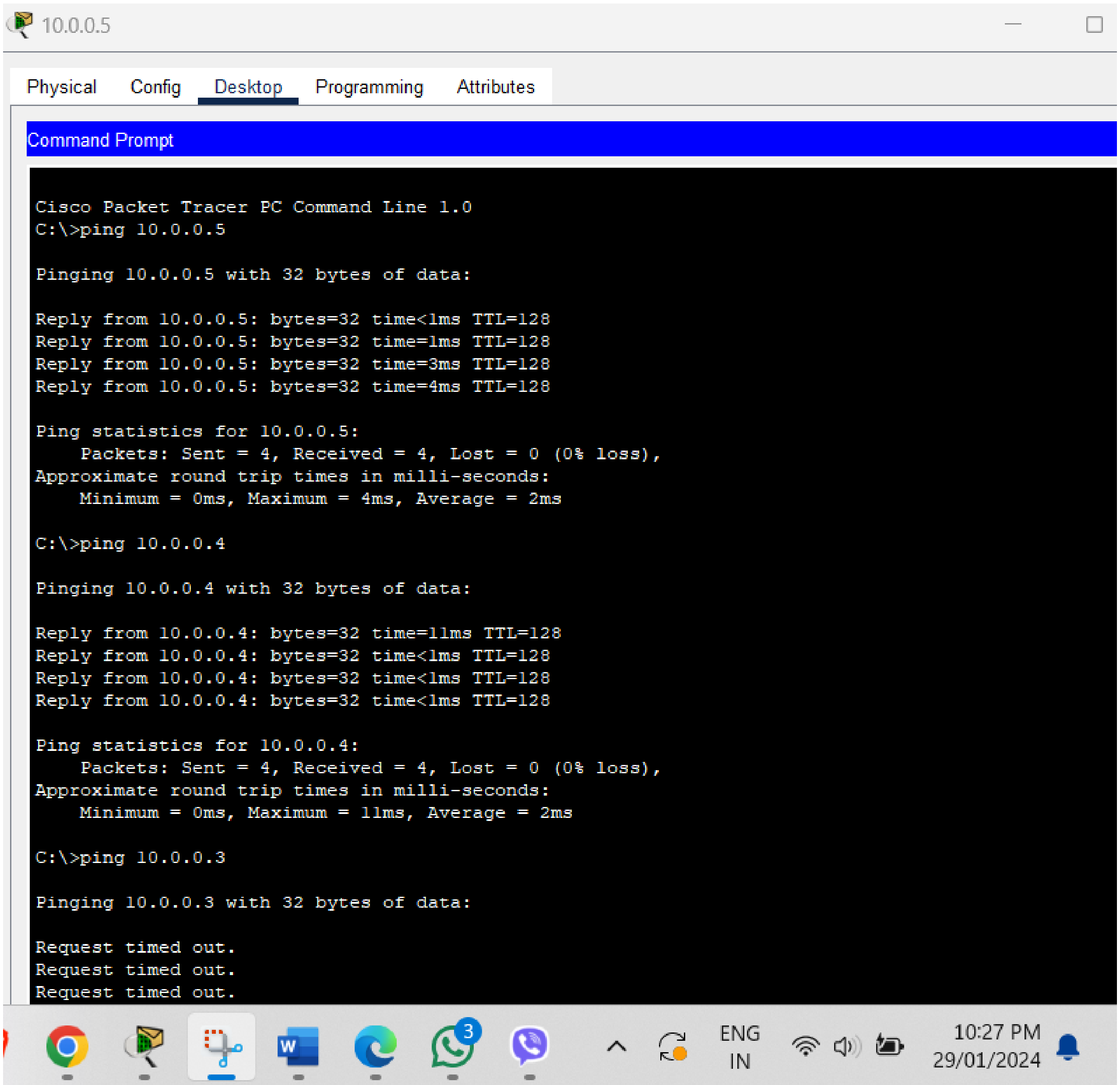


Name: Sudha Pantha



Conclusion:

Hence, we configured vlan and confirmed the connection.

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Lab Program Number: …12… ***Date: 2080-10-08***

Title: To configure and understand about router on a stick using cisco packet tracer.

Theory:

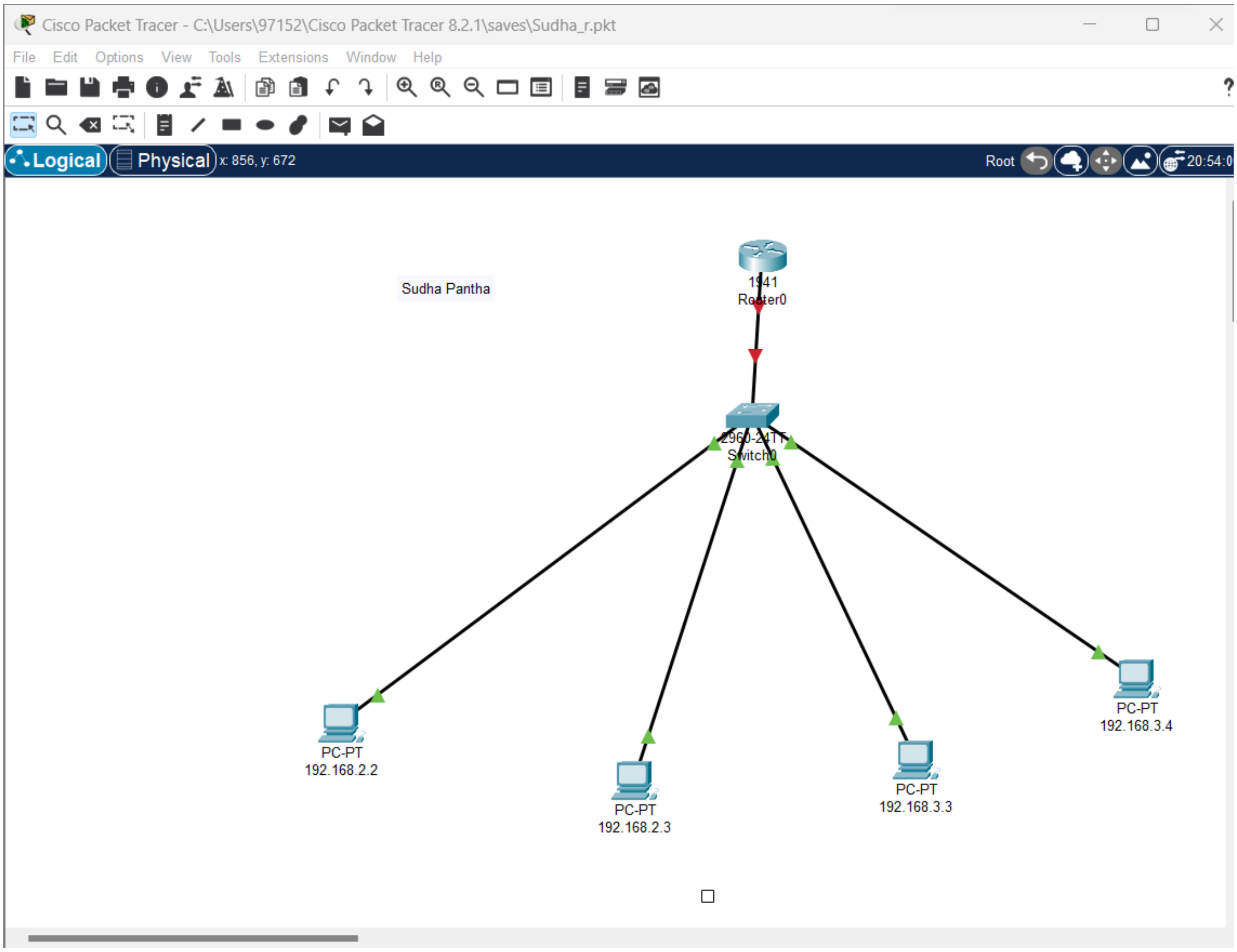
ROUTER-ON-A-STICK, also known as a "one-armed router" is a method for running multiple VLANs over a single connection in order to provide inter-VLAN routing. Essentially the router connects to a core switch with a single interface and acts as the relay point between networks.

Procedure:

Step 1: Open Cisco Packet Tracer desktop and select all the devices given below:

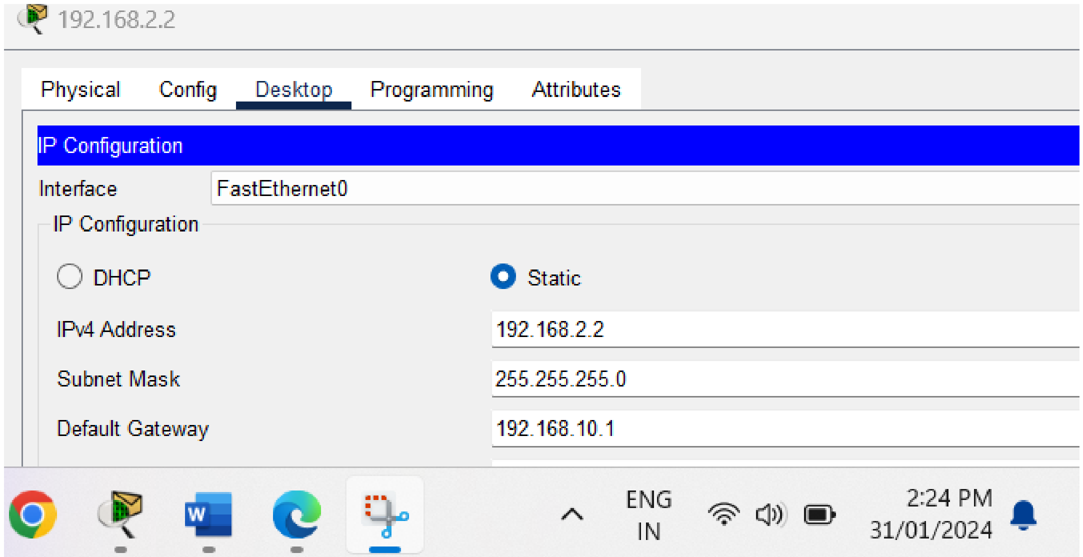
S.No	Device	Model Name	Qty.
01	PC	PC	04
02	Switch	PT-Switch	01
03	Router	Router-PT	01

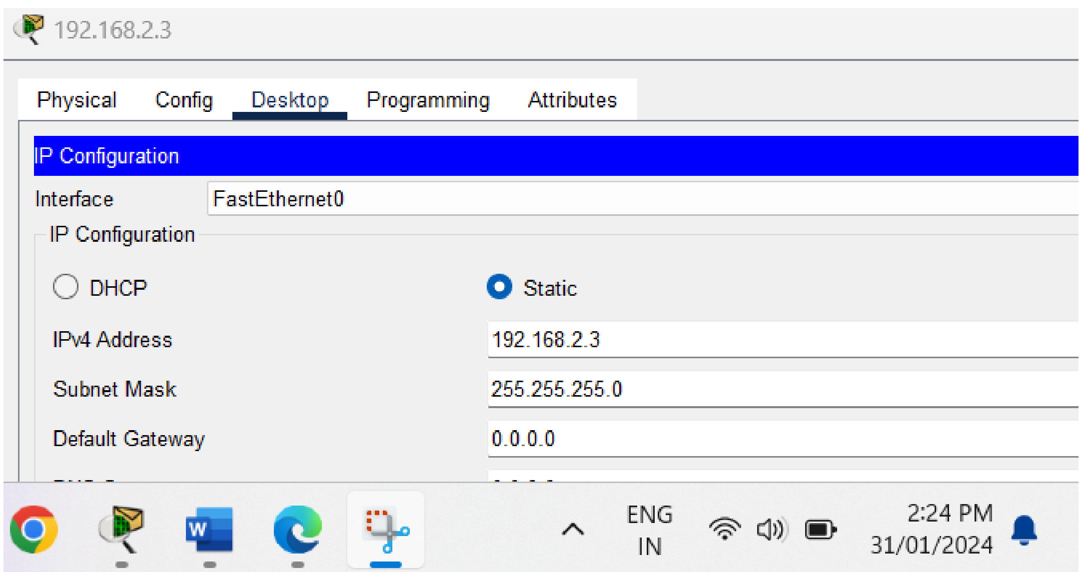
- In the "Devices" tab on the left, select "End Devices" and drag a "Generic" Pc or laptop to the workspace.
- Similarly, drag a "Switch" and "Router" to the workspace.
- Click on the "Connections" icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device's interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.



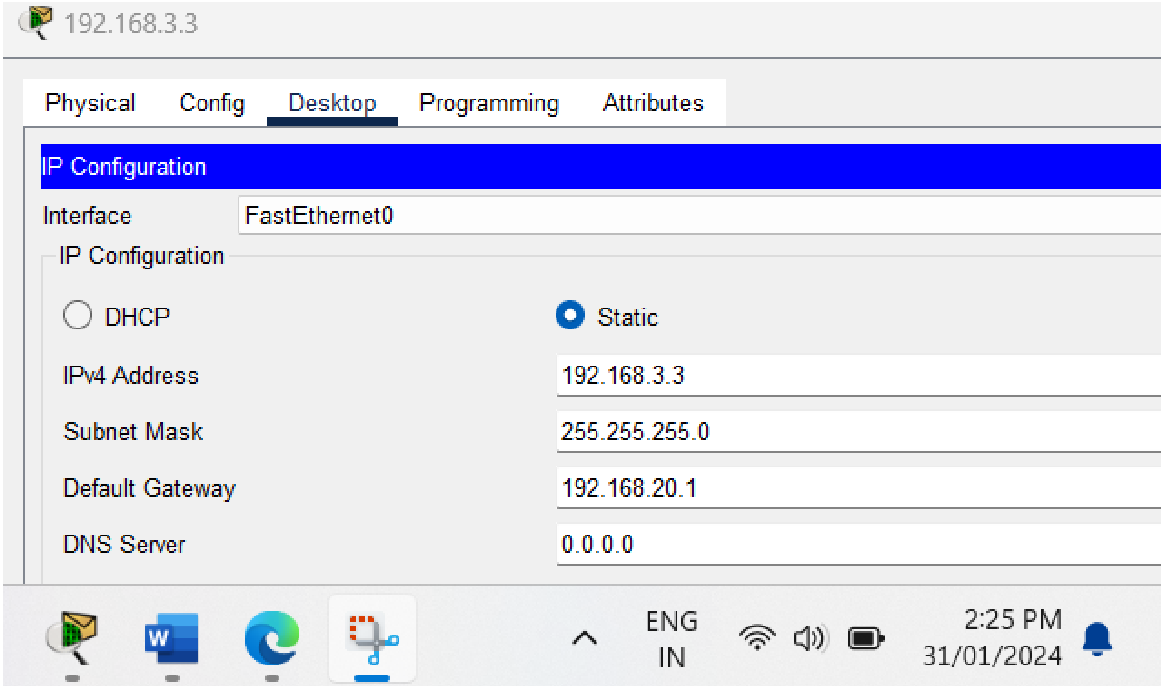
Step 2**:** Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

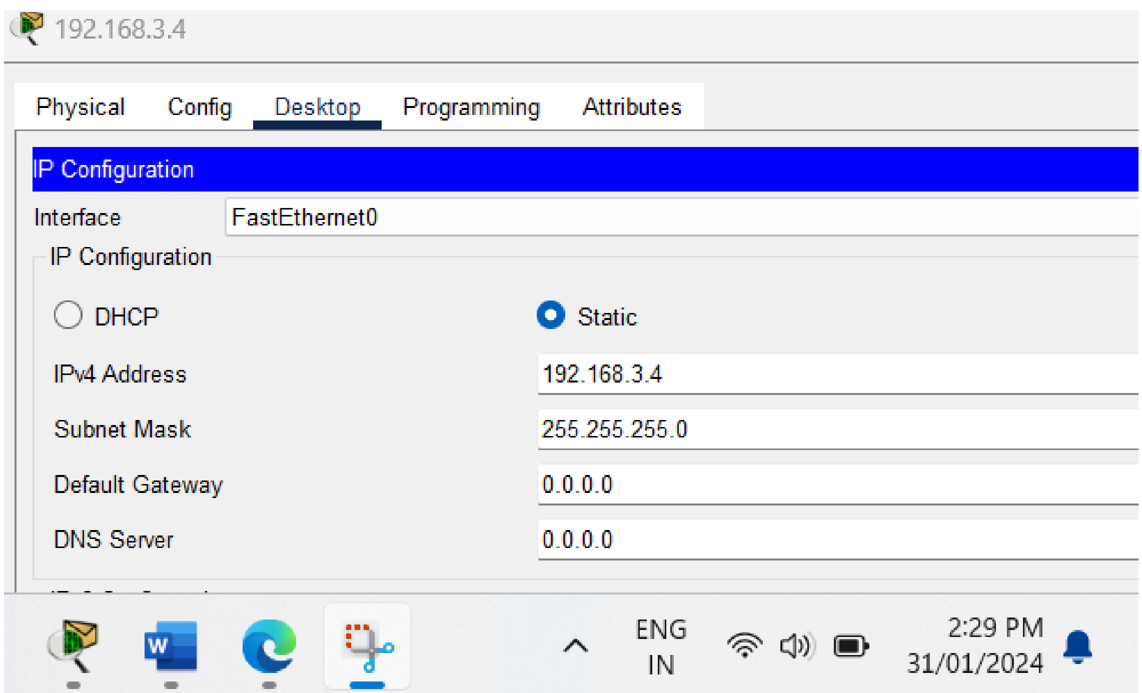
- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.





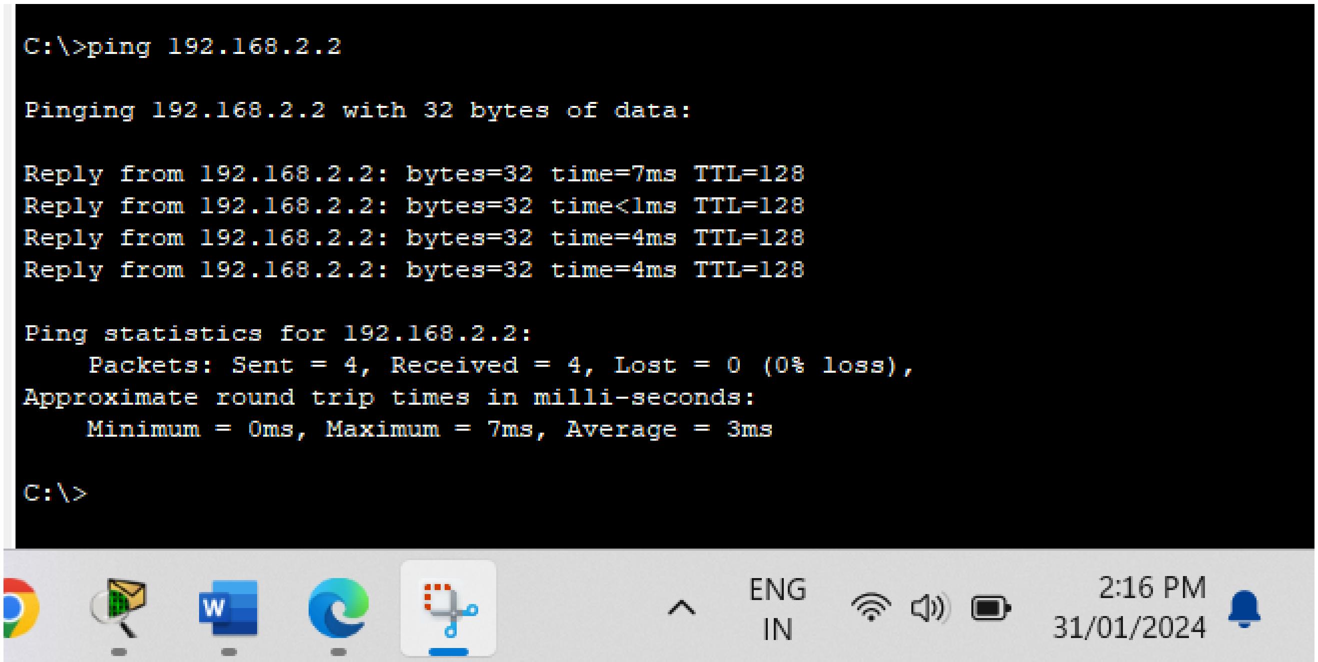
Name: Sudha Pantha



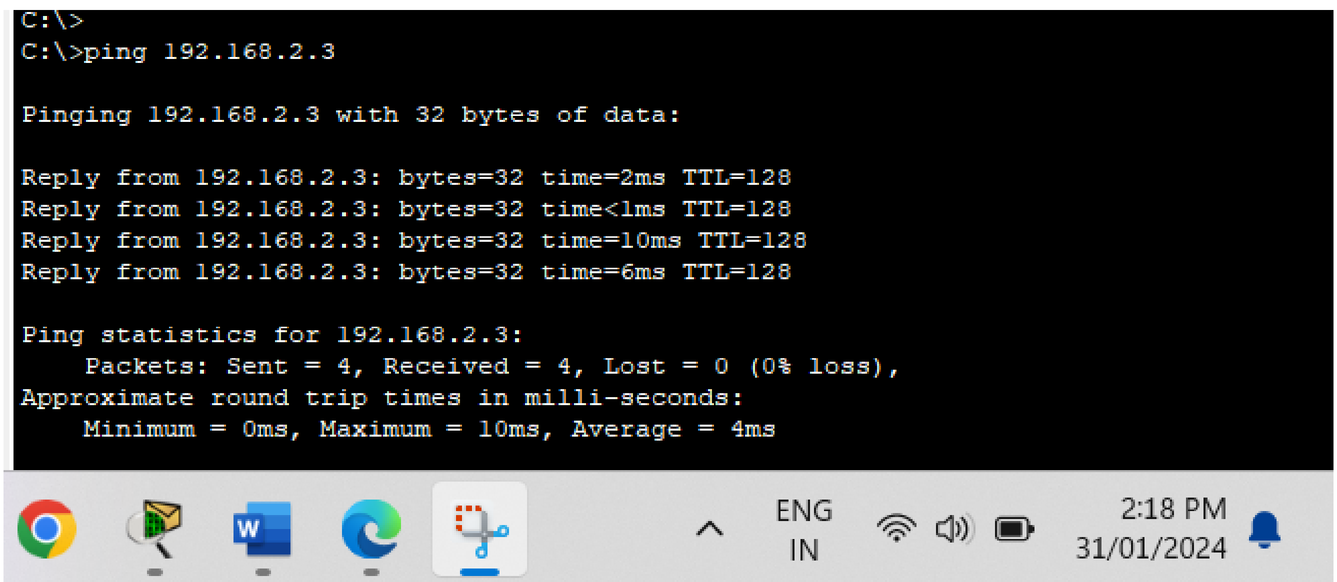


Name: Sudha Pantha

Step 3: Verifying the network by pinging the IP address of any PC.

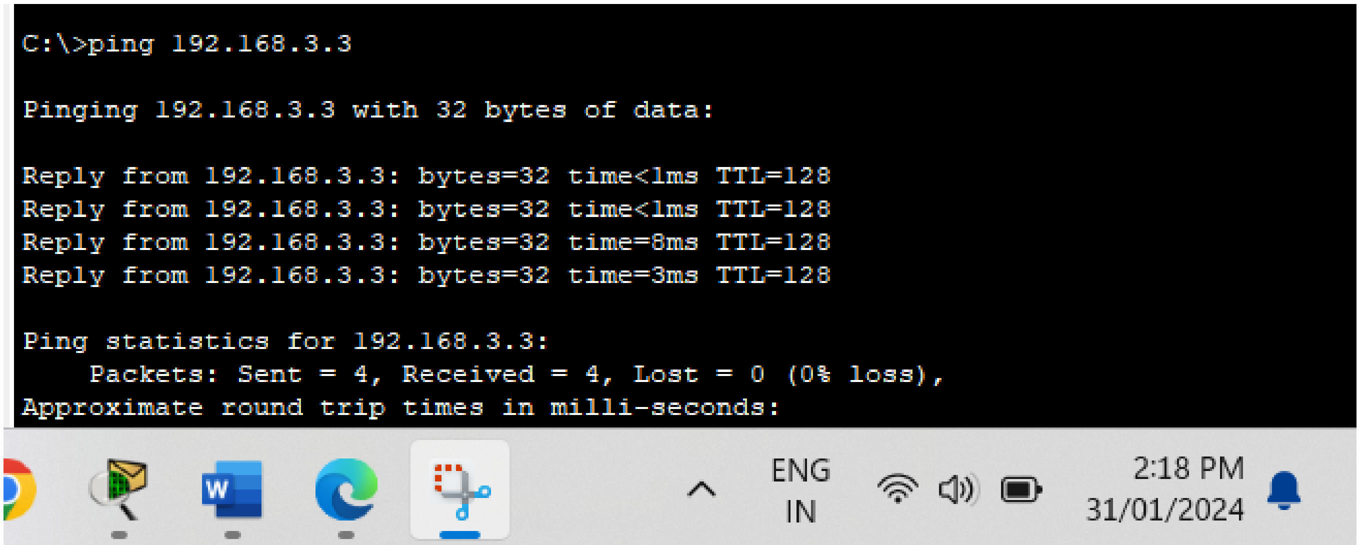


```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=7ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```

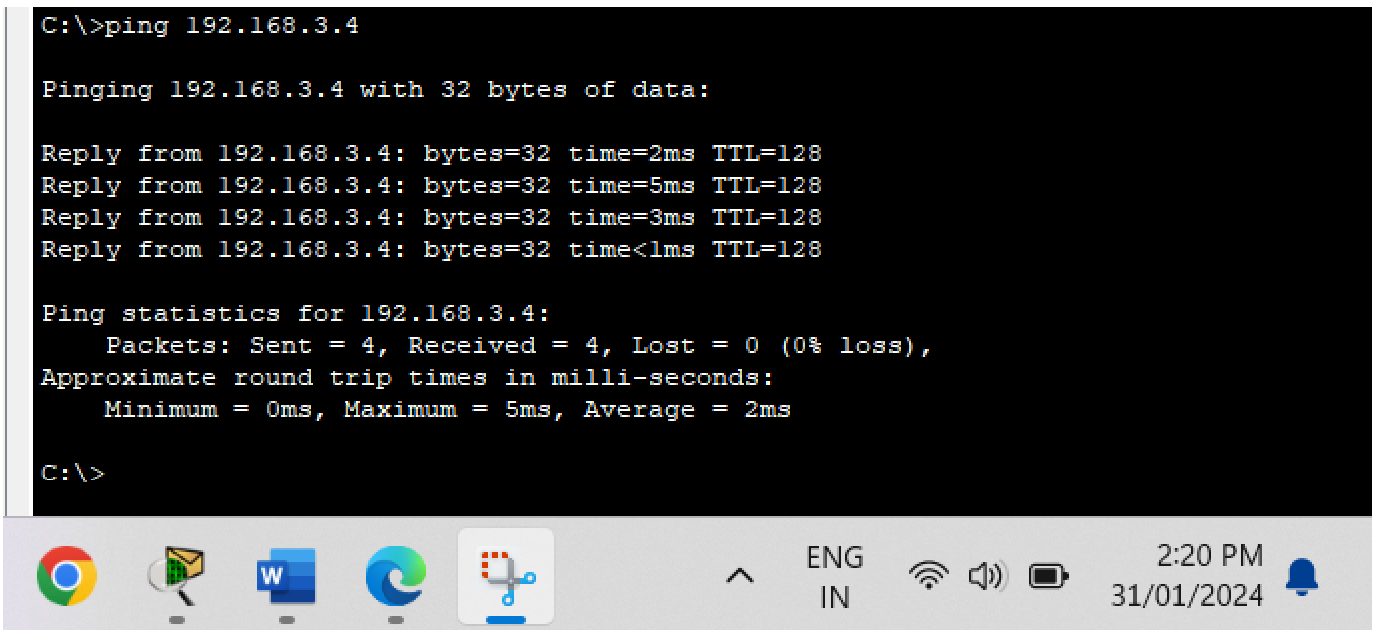



```
C:\>
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=2ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=10ms TTL=128
Reply from 192.168.2.3: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 4ms
```


Name: Sudha Pantha



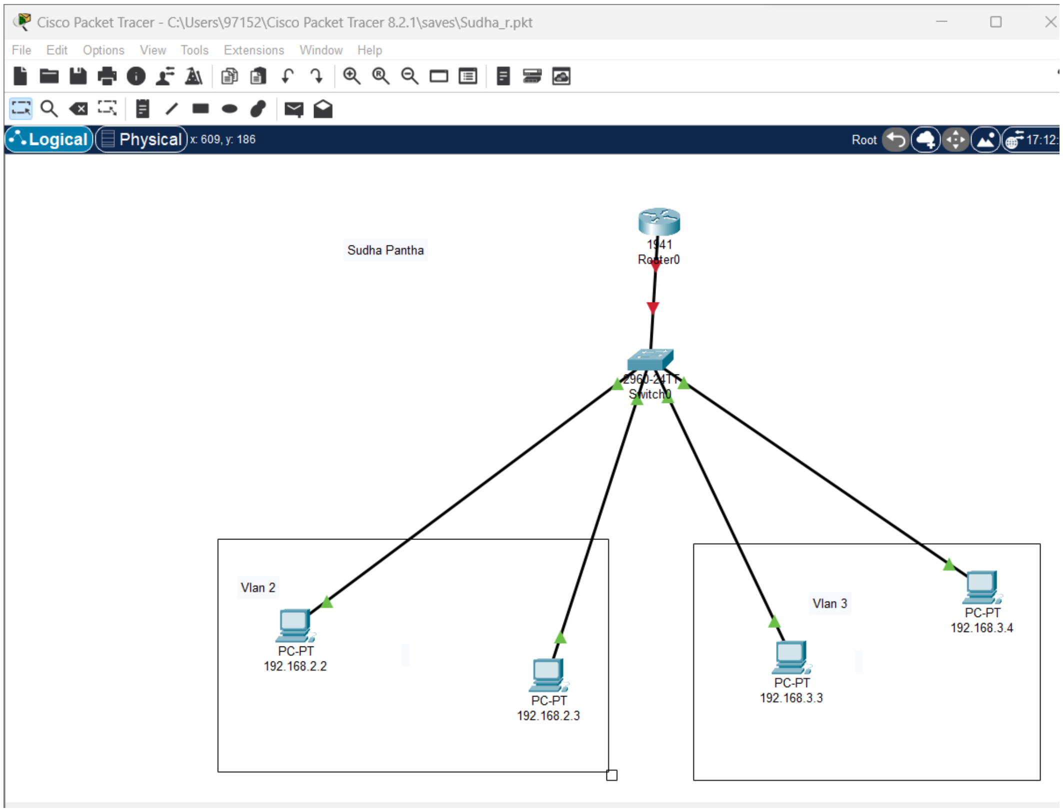
```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time=8ms TTL=128
Reply from 192.168.3.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

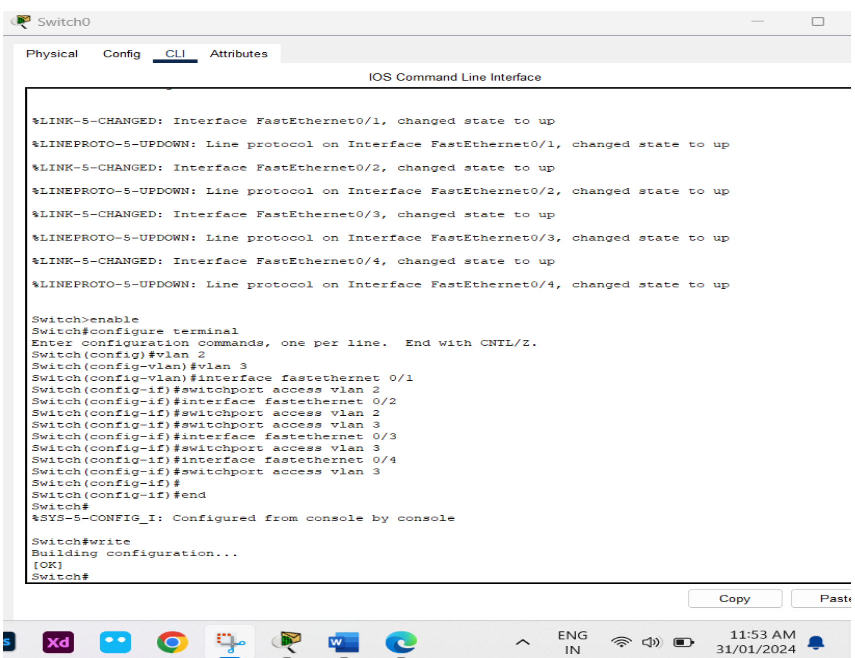
```
C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Reply from 192.168.3.4: bytes=32 time=2ms TTL=128
Reply from 192.168.3.4: bytes=32 time=5ms TTL=128
Reply from 192.168.3.4: bytes=32 time=3ms TTL=128
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 2ms

C:\>
```


Step 4: Then, we configure a topology for our Router-on- stick, as shown below:

Name: Sudha Pantha



- Then, we setup a Vlan network



- After setting up vlan, we setup a trunk for our vlan traffic as:

Name: Sudha Pantha

```
Switch#write
Building configuration...
[OK]
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write
Building configuration...
[OK]
Switch#
```

Copy Paste

Top

- After the trunk, we set up the router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
                                         ^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 10
                                         ^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
% 192.168.10.0 overlaps with GigabitEthernet0/0.10
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown

Router(config-if)#
```

Conclusion:

Hence, we configured Router on a stick and verified.

Lab Program Number: …13… ***Date: 2080-10-08***

Title: To set up web DNS server in cisco packet Tracer.

Theory:

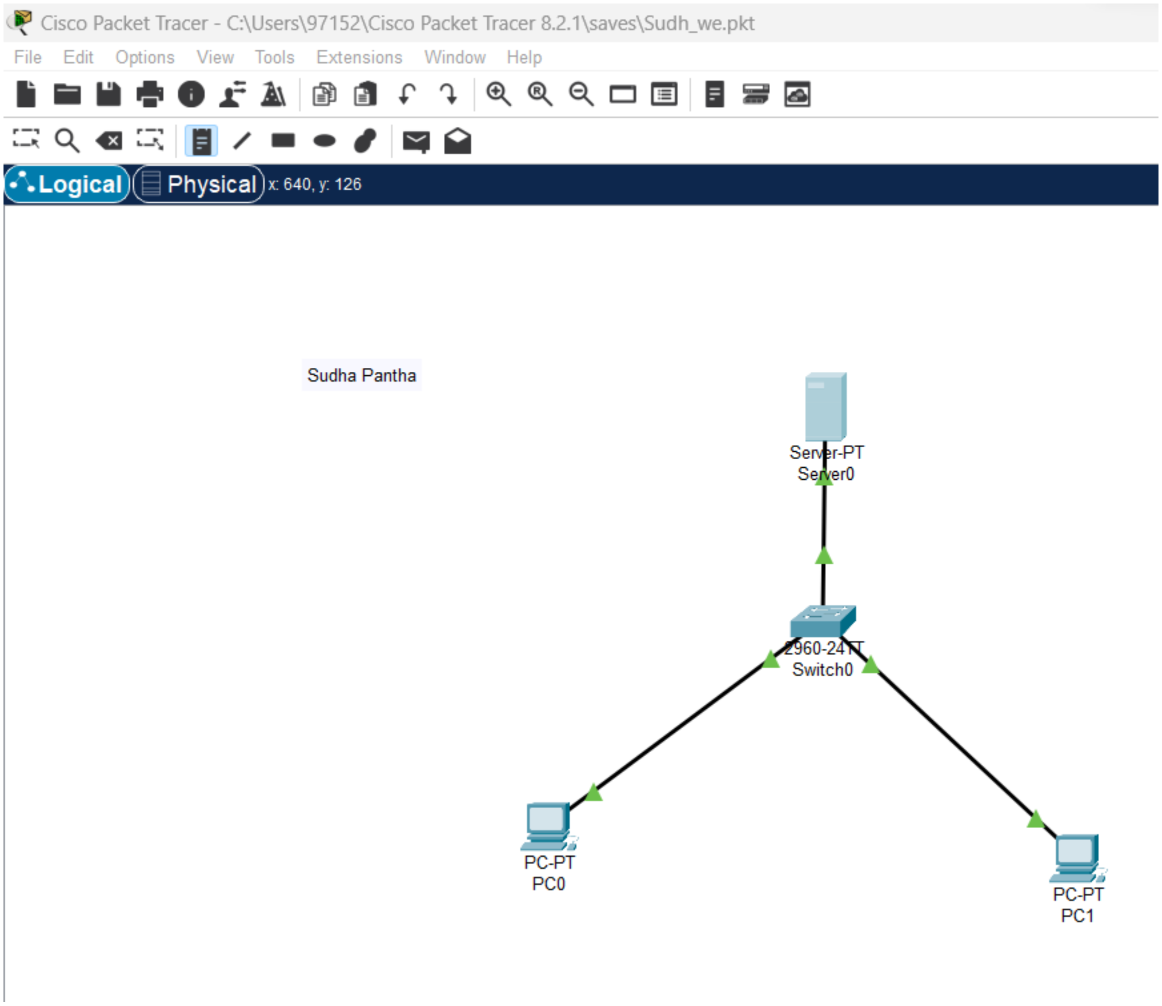
The Domain Name System (DNS) Server is a server that is specifically used for matching website hostnames (like example.com)to their corresponding Internet Protocol or IP addresses. The DNS server contains a database of public IP addresses and their corresponding domain names. Every device connected to the internet has a unique IP address that helps to identify it, according to the IPv4 or IPV6 protocols. The same goes for web servers that host websites.

Procedure:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

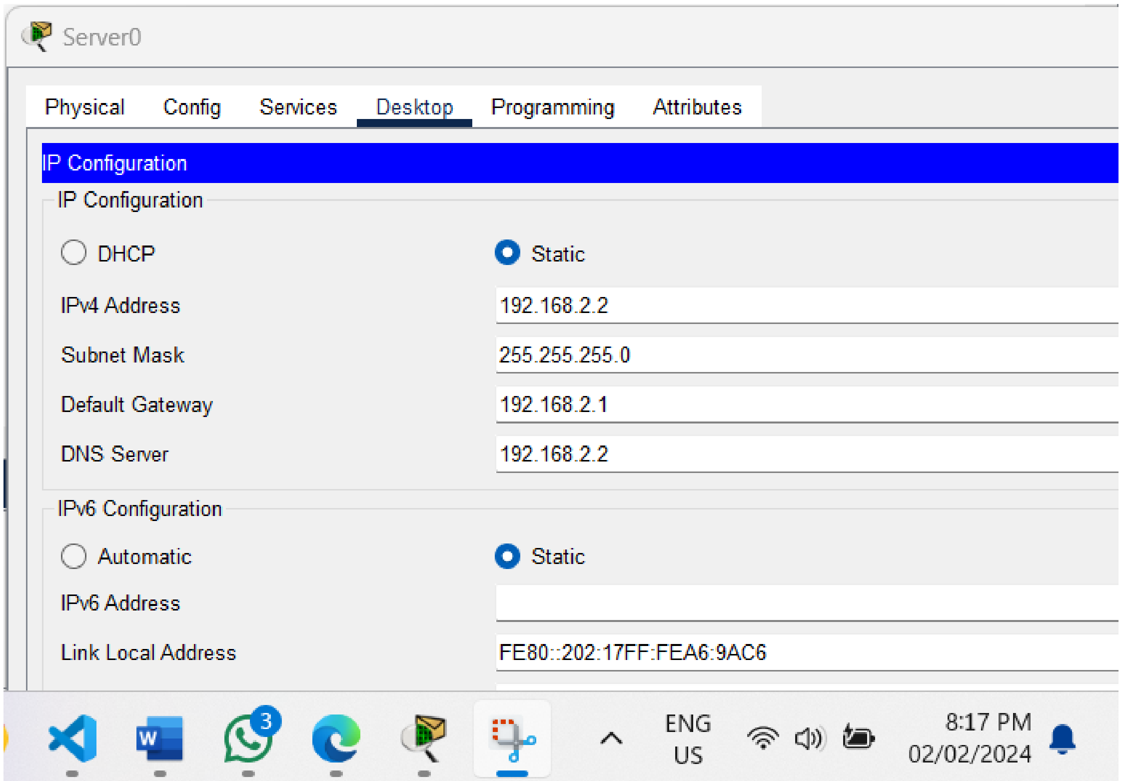
S.No	Device	Model Name	Qty.
01	PC	PC	02
02	Switch	PT-Switch	01
03	Server	Server-PT	01

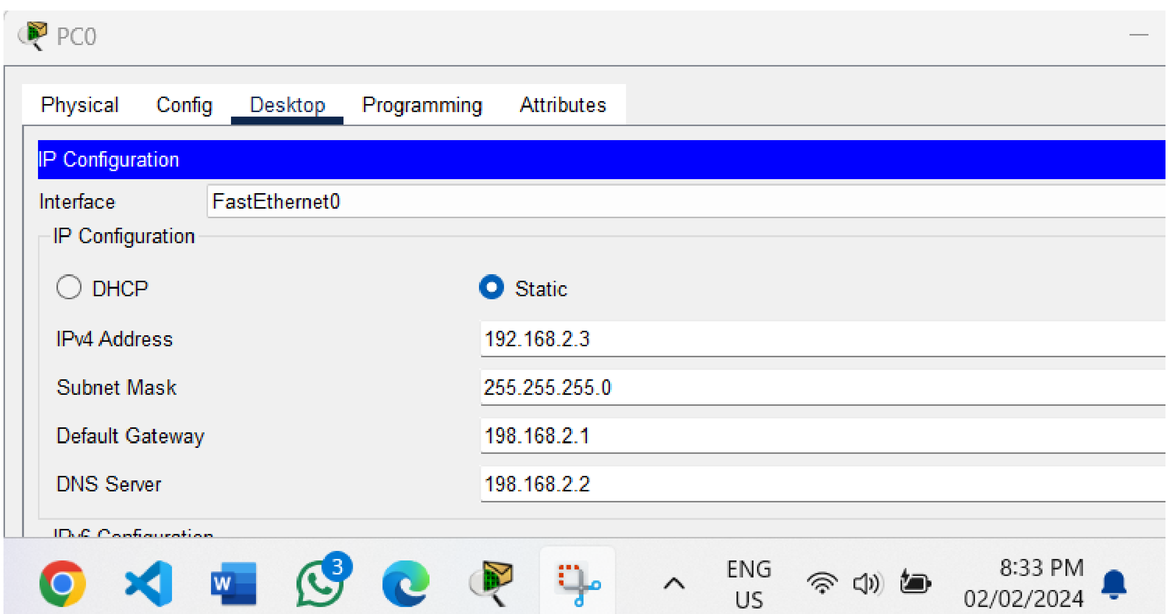
- In the "Devices" tab on the left, select "End Devices" and drag a "Generic" Pc or laptop to the workspace.
- Similarly, drag a "Switch" and "Router" to the workspace.
- Click on the "Connections" icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device's interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.



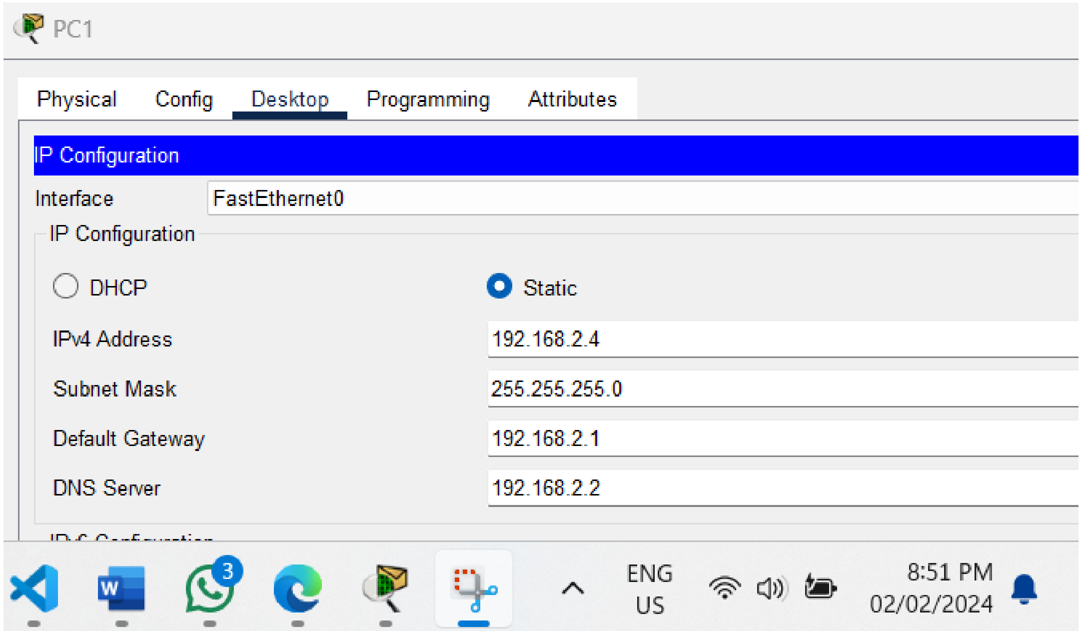
Step 2: Configure static Ip addresses on the PCs and the server.

Name: Sudha Pantha



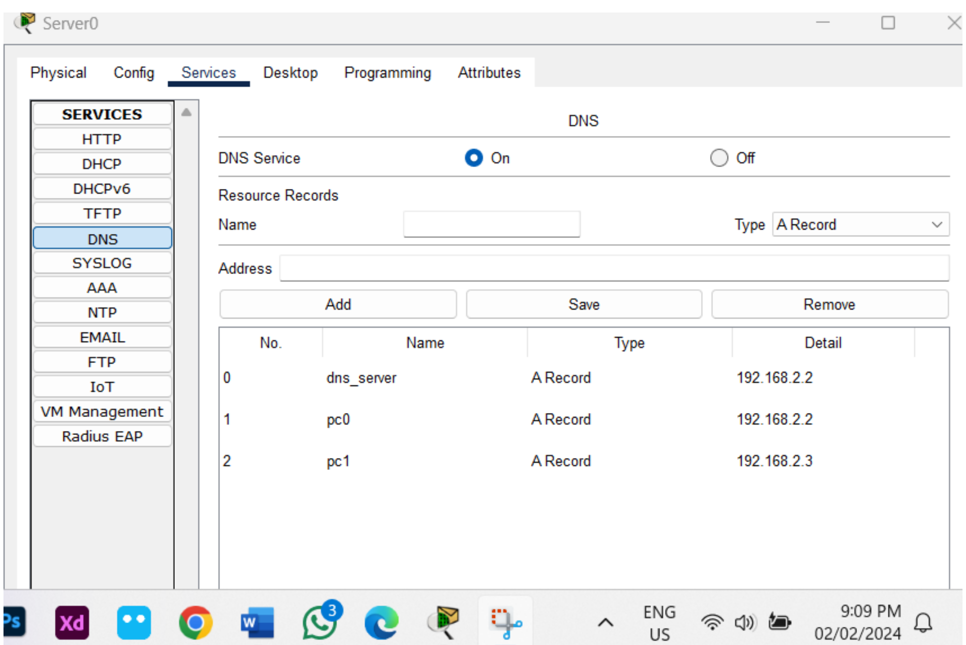


Name: Sudha Pantha



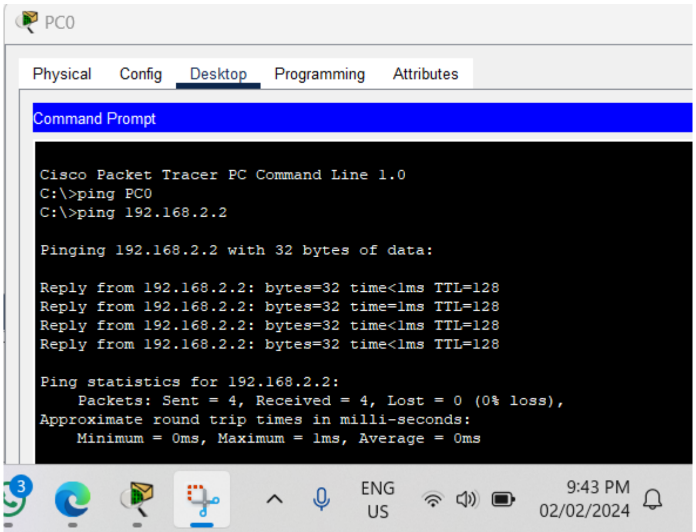
Step 3: Configure DNS service on the generic server.

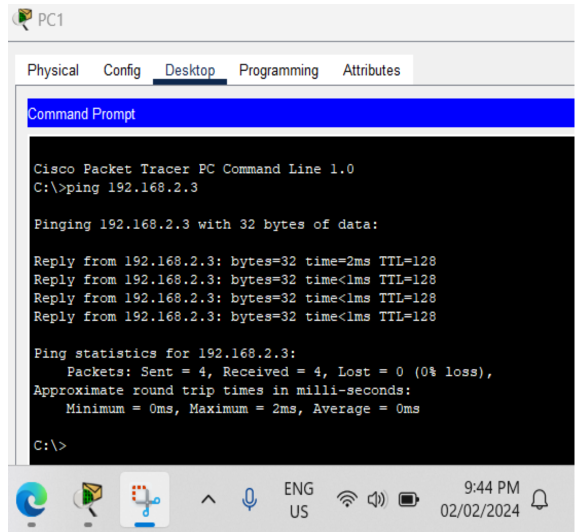
- To do this, click on the server, then Click on **Services** tab.
- Click on **DNS server** from the menu.
- First turn **ON** the DNS service, then define **names** of the hosts and their corresponding **IP addresses.**

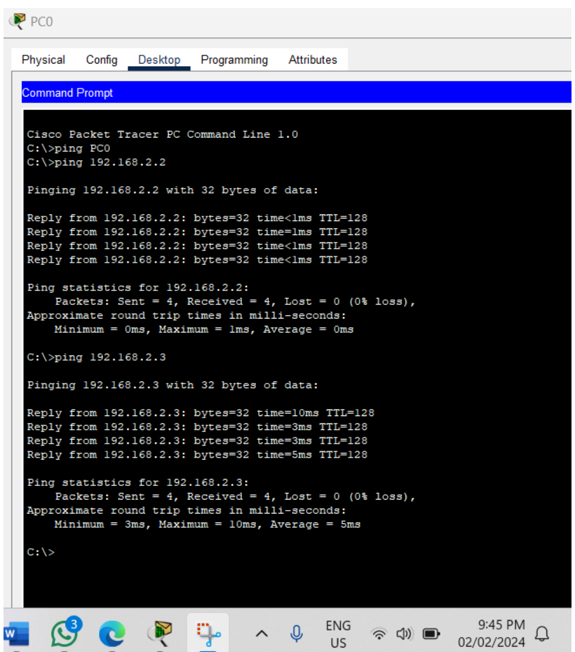


Name: Sudha Pantha

Step 4: Verifying the network by pinging the IP address of any PC.







Conclusion:

Hence, we configured web dns server and verified.

Name: Sudha Pantha

Lab Program Number: …14… ***Date: 2080-10-22***

Title: To set up OSPF in cisco packet Tracer.

Theory:

OSPF stands for Open Shortest Path First a link-state protocol and as the name itself justifies that it is used to find the best and the optimal pathway between the starting point and the destination target router using its own shortest path first algorithm. It sends hello packets every 10 seconds. It varies under (LSRP) Link State Routing Protocol. It automatically generates or decides the optimal path and it is used to send router packets with the auto path decision method. It is one of the Interior gateway protocols that aim to move the packet within a large autonomous system. OSPF works on port no. 89 and uses [AD value 110].

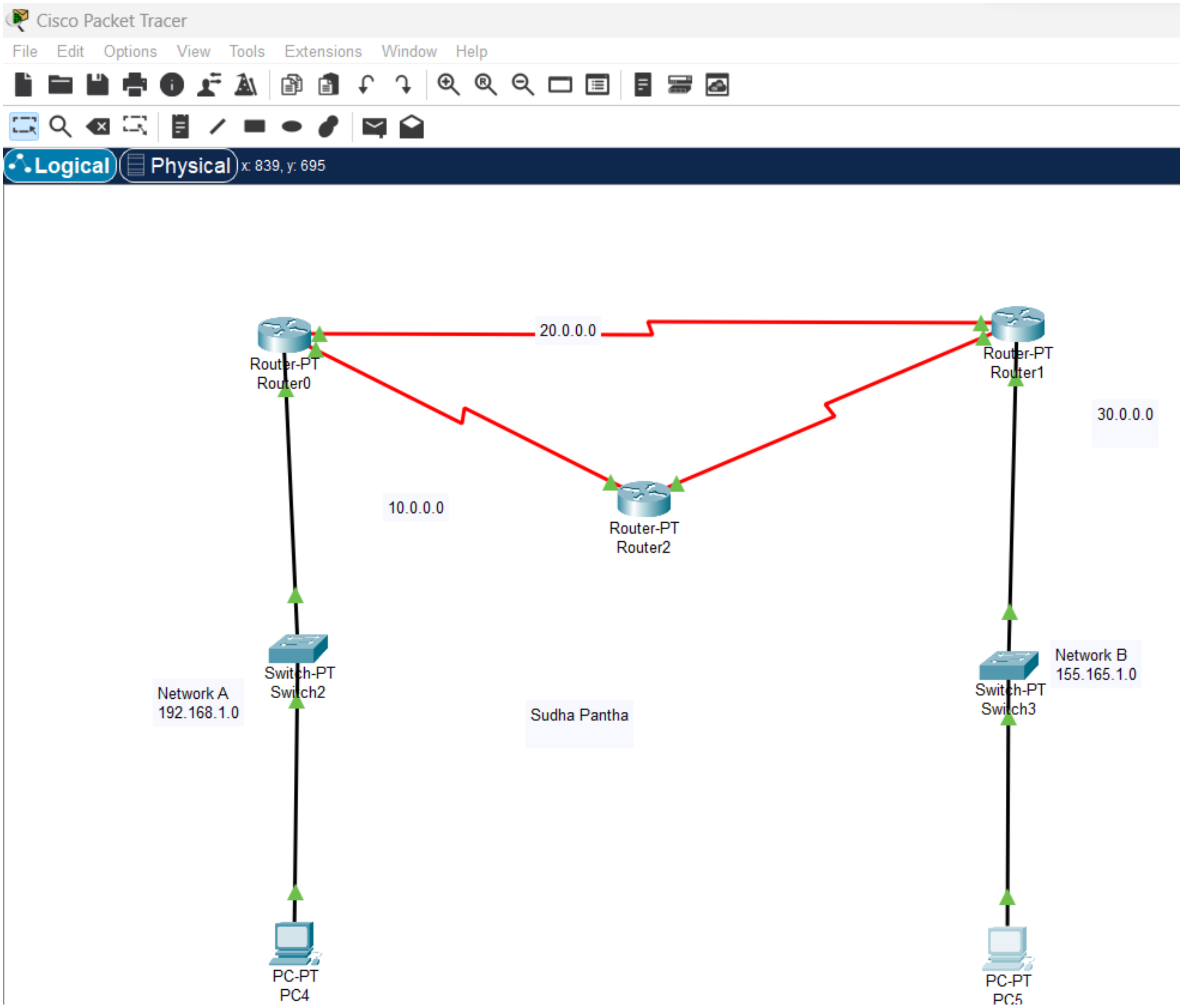
Procedure:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	02
02	Switch	PT-Switch	02
03	Router	Router-PT	03

- In the "Devices" tab on the left, select "End Devices" and drag a "Generic" Pc or laptop to the workspace.
- Similarly, drag a "Switch" and "Router" to the workspace.
- Click on the "Connections" icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device's interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

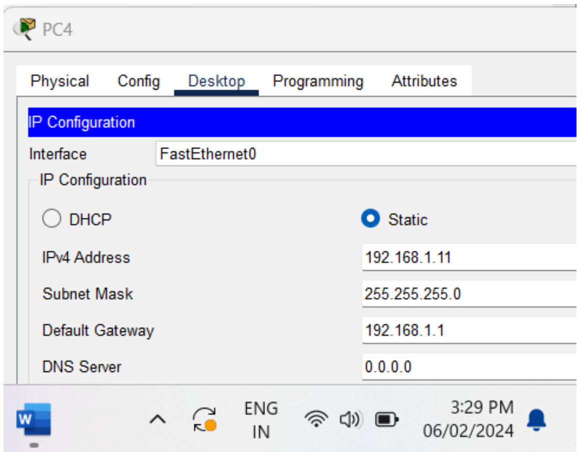
Name: Sudha Pantha

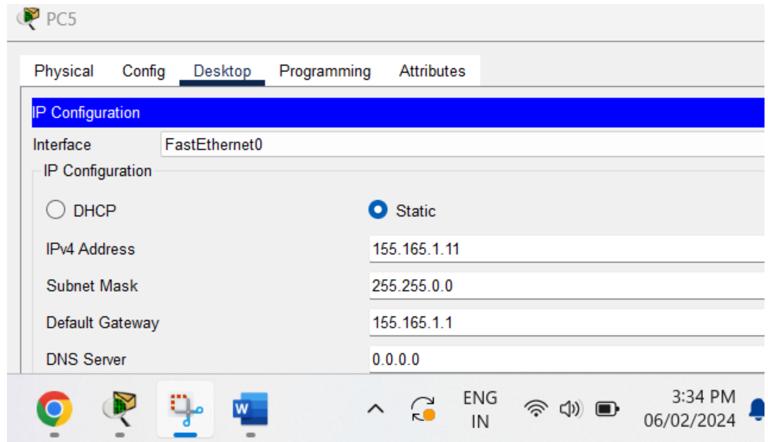


Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

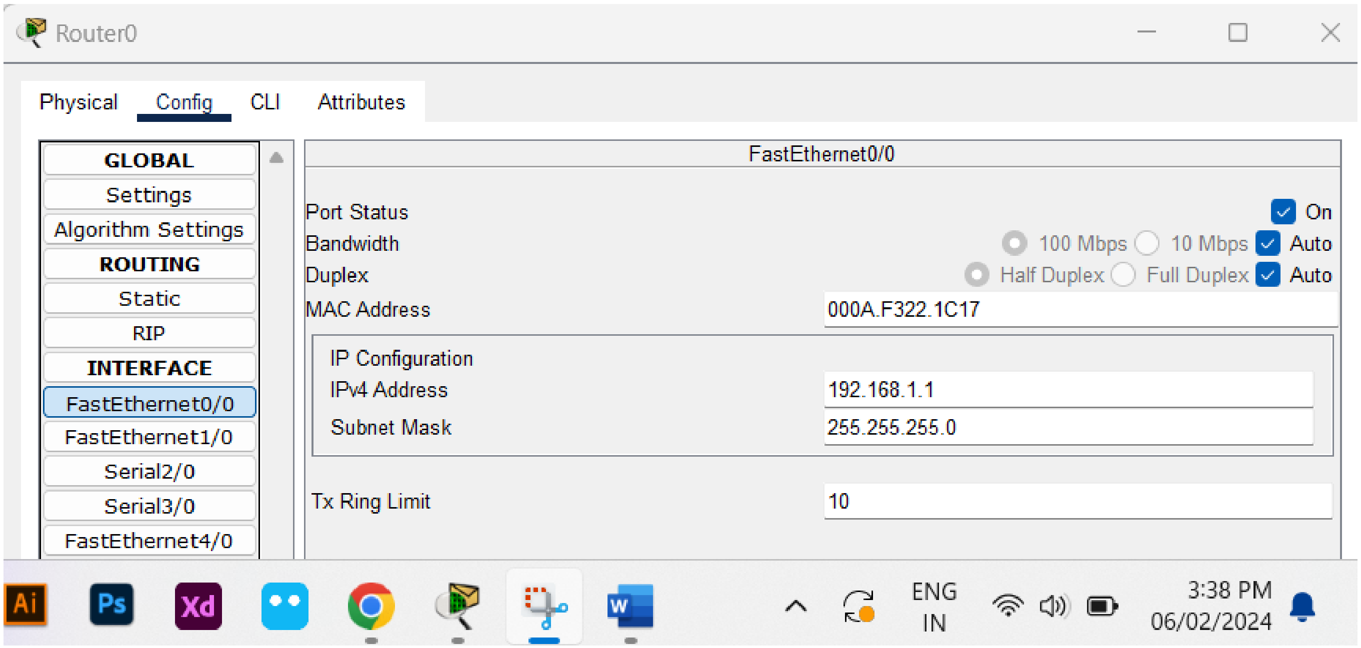
Name: Sudha Pantha

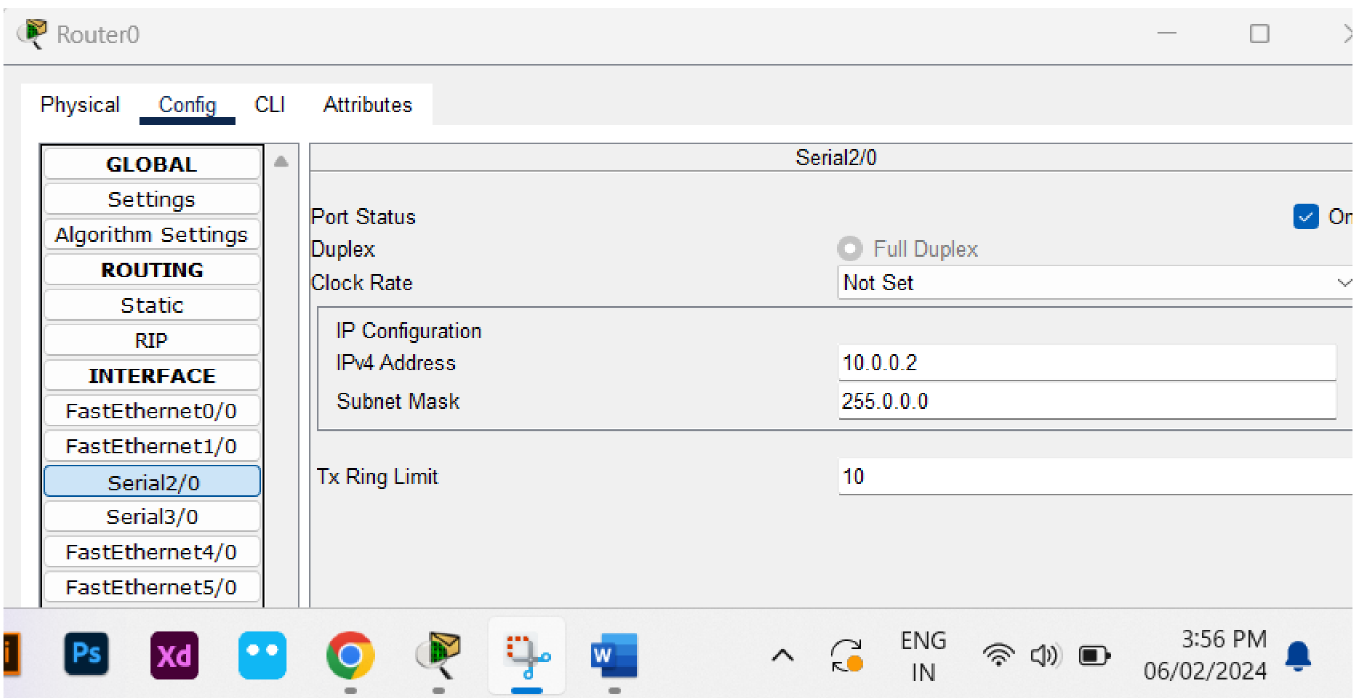




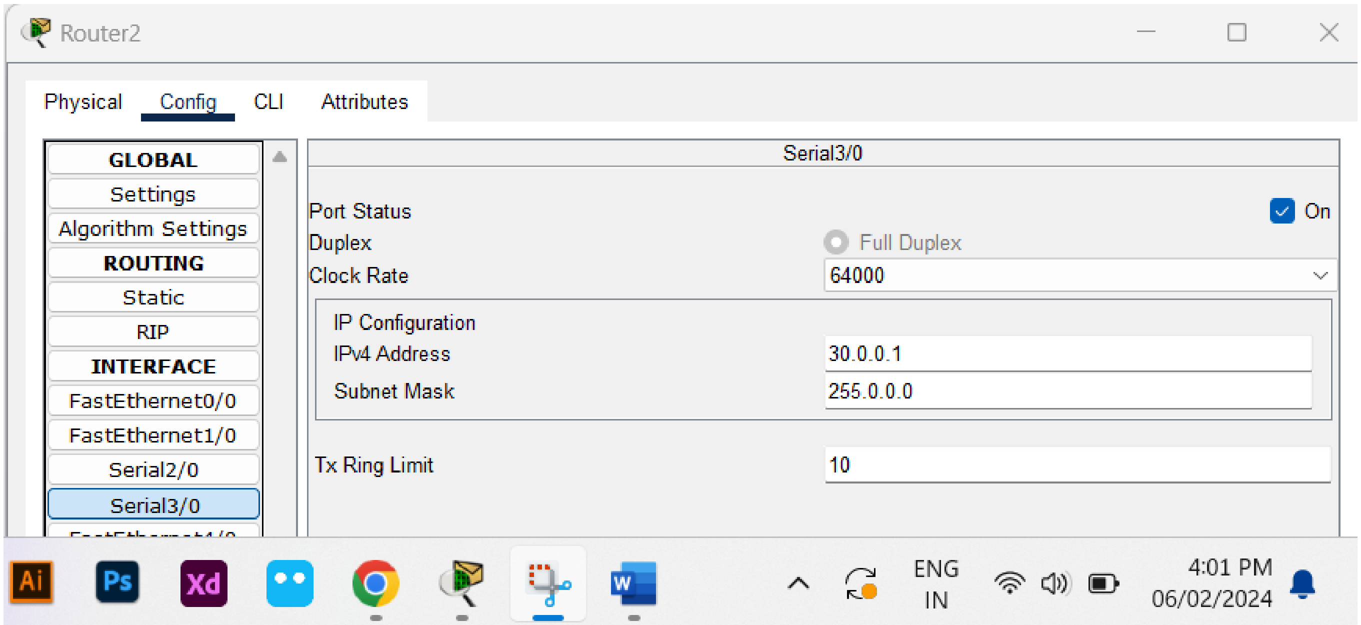
Step 3: Configure router with IP address and Subnet mask.

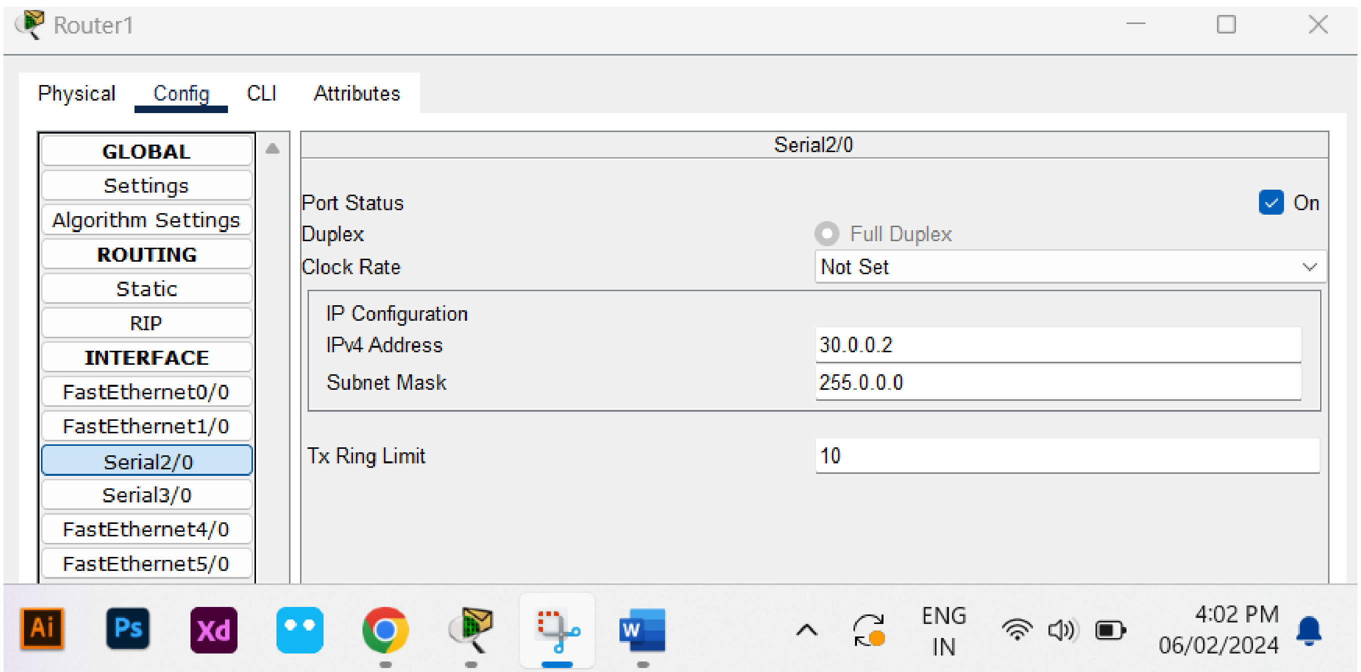
- To assign an IP address in router, click on router.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in FastEthernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask for all router.



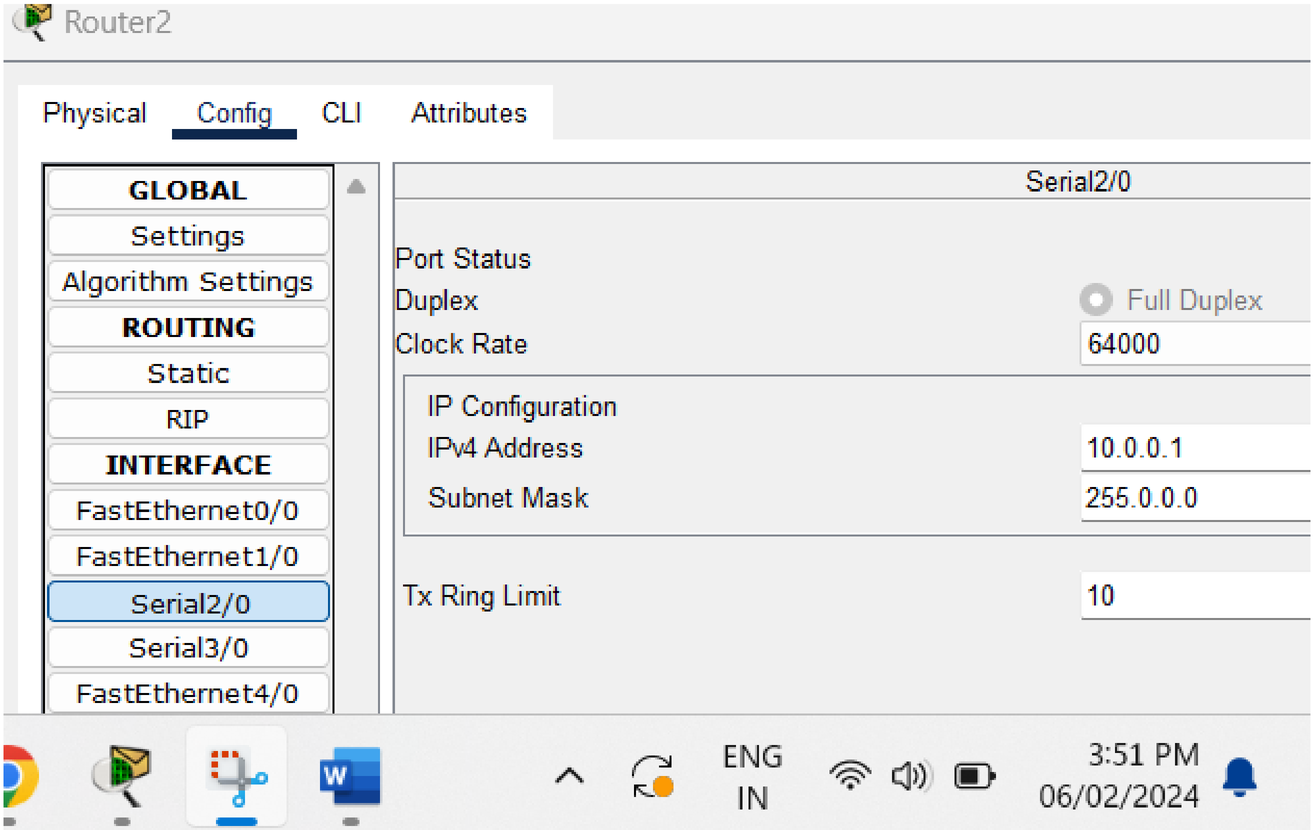


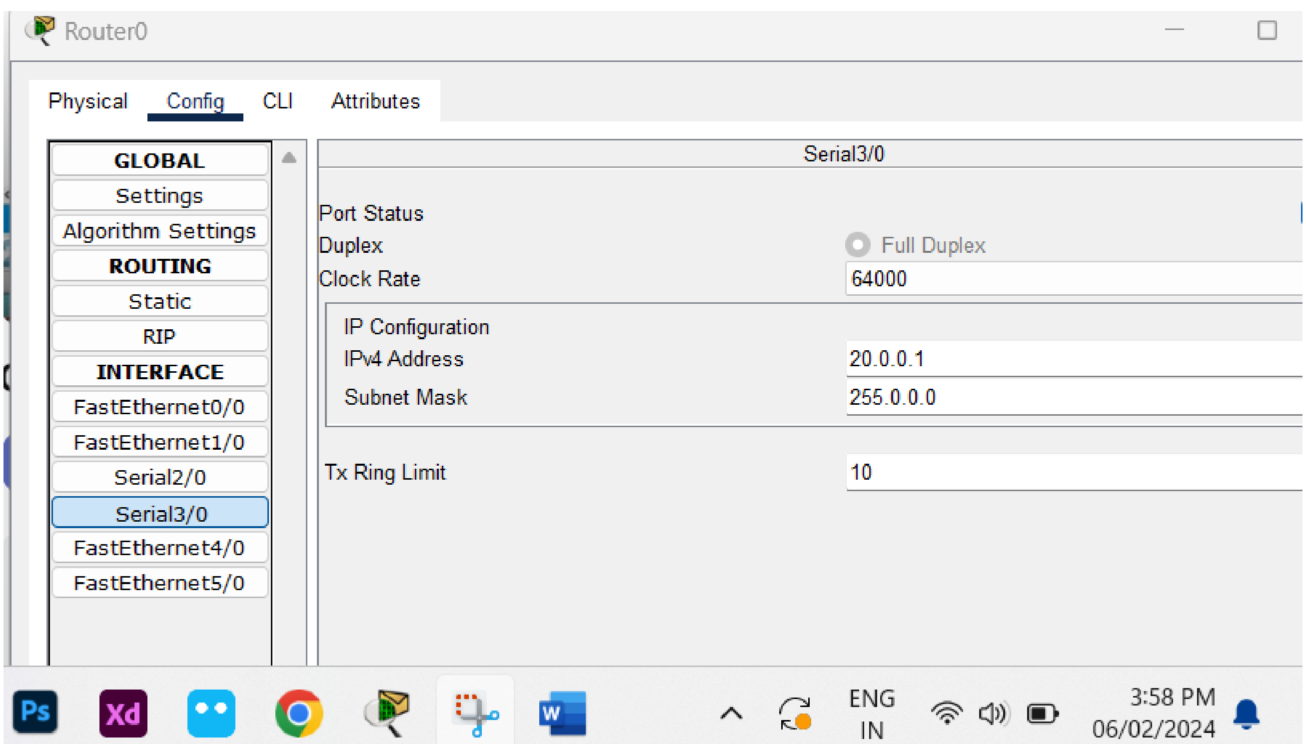
Name: Sudha Pantha



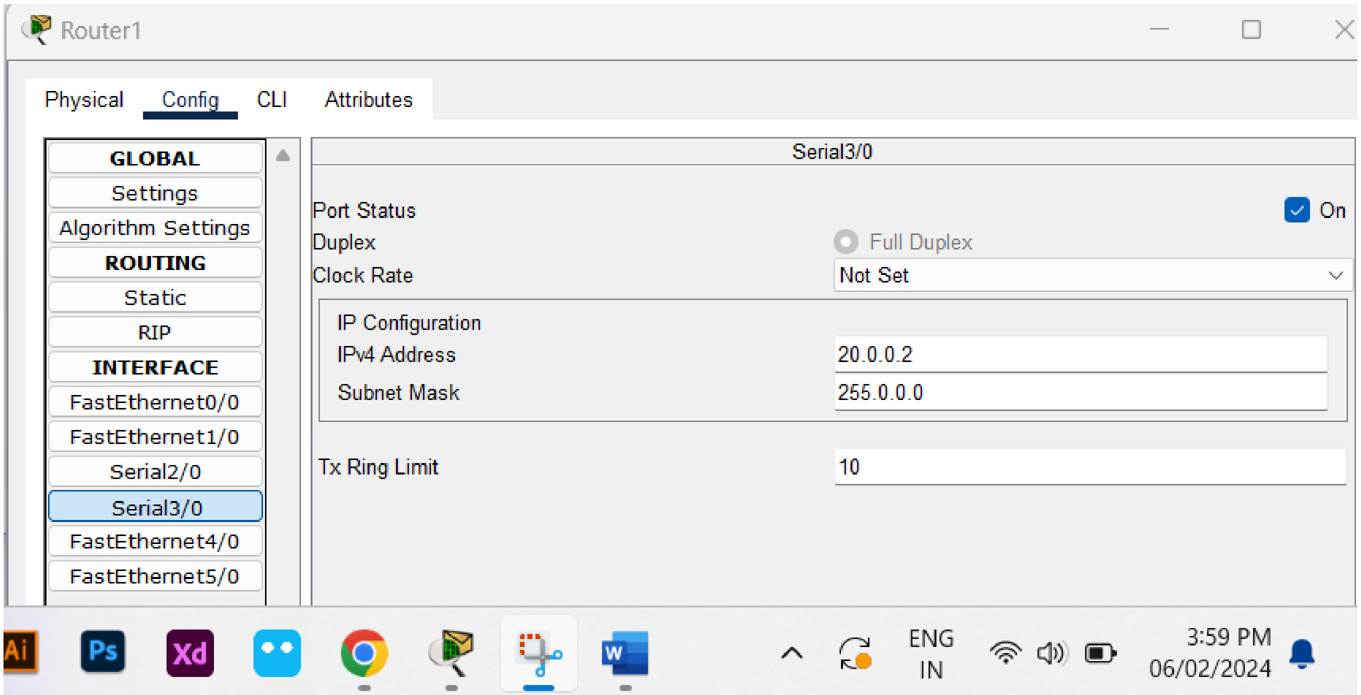


Name: Sudha Pantha





Name: Sudha Pantha

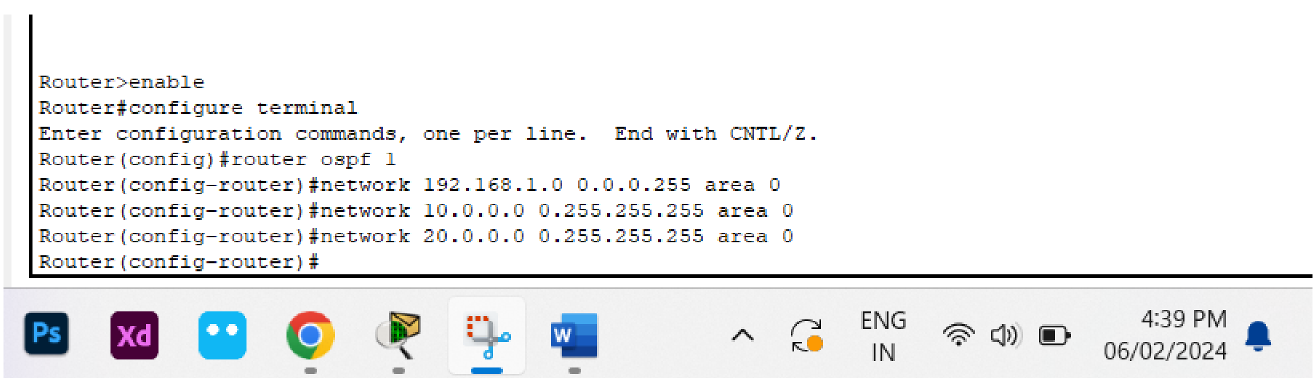


Step 4: After configuring all of the devices we need to assign the routes to the router.

To assign RIP routes to the particular router:

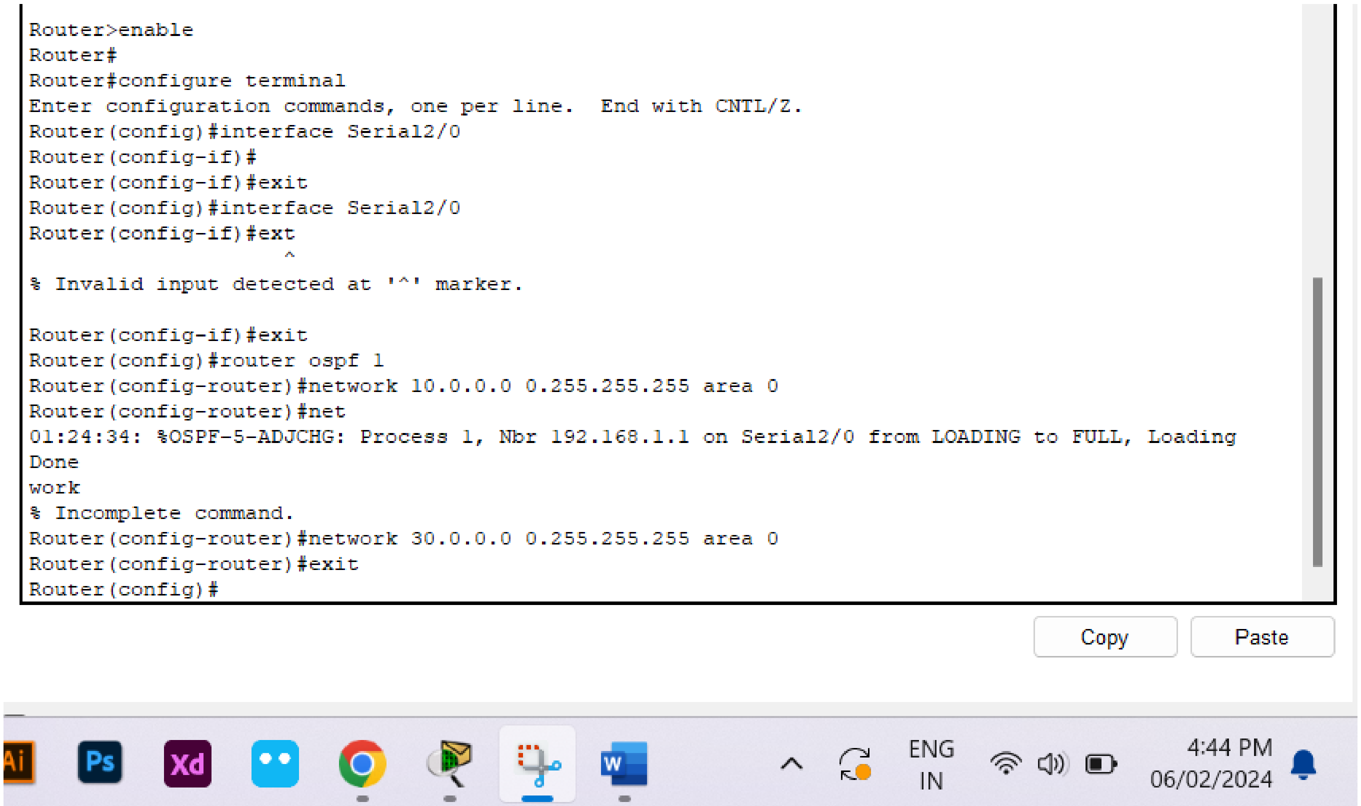
- First, click on router then Go to CLI.
- Then type the commands and IP information given below.
  - ➢ CLI command: router rip
  - ➢ CLI command: network <network id>

In Router 0:



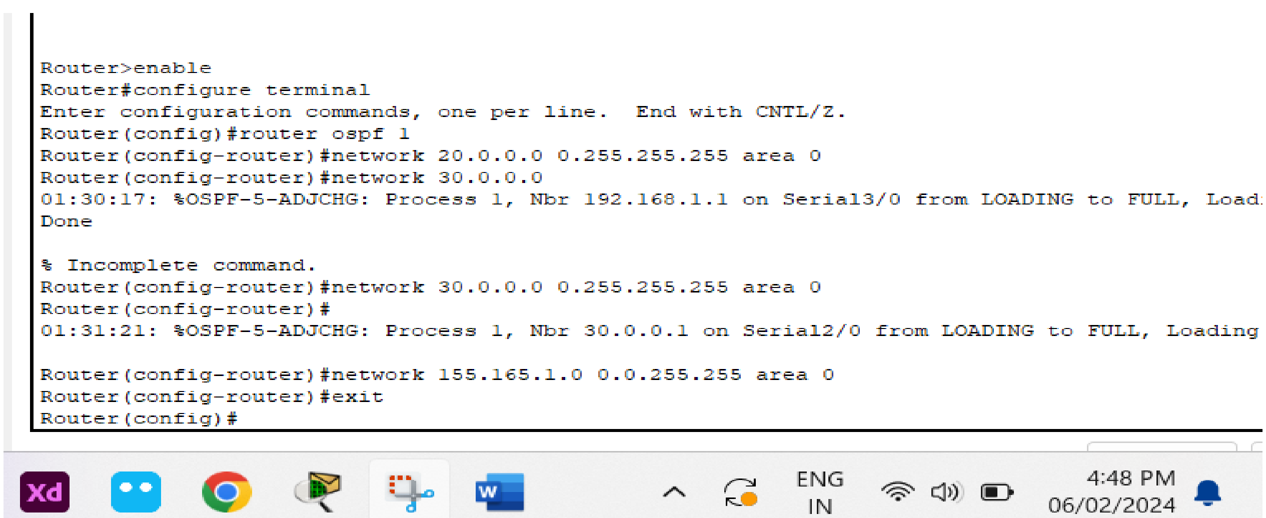
Name: Sudha Pantha

Router 2:



```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ext
                    ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#net
01:24:34: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial2/0 from LOADING to FULL, Loading
Done
work
% Incomplete command.
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```


In Router 1:

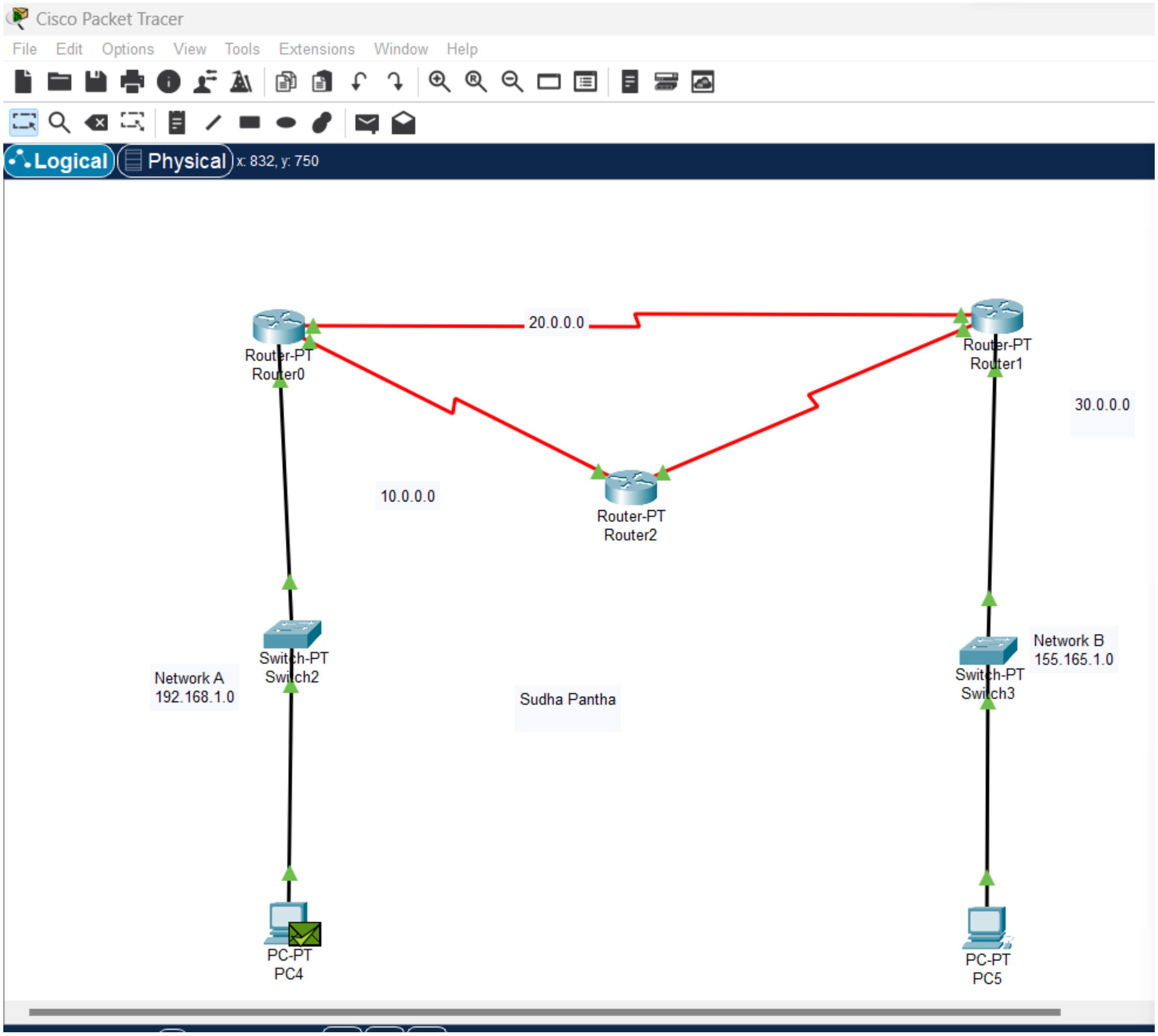


```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0
01:30:17: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial3/0 from LOADING to FULL, Load
Done

% Incomplete command.
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#
01:31:21: %OSPF-5-ADJCHG: Process 1, Nbr 30.0.0.1 on Serial2/0 from LOADING to FULL, Loading

Router(config-router)#network 155.165.1.0 0.0.255.255 area 0
Router(config-router)#exit
Router(config)#
```

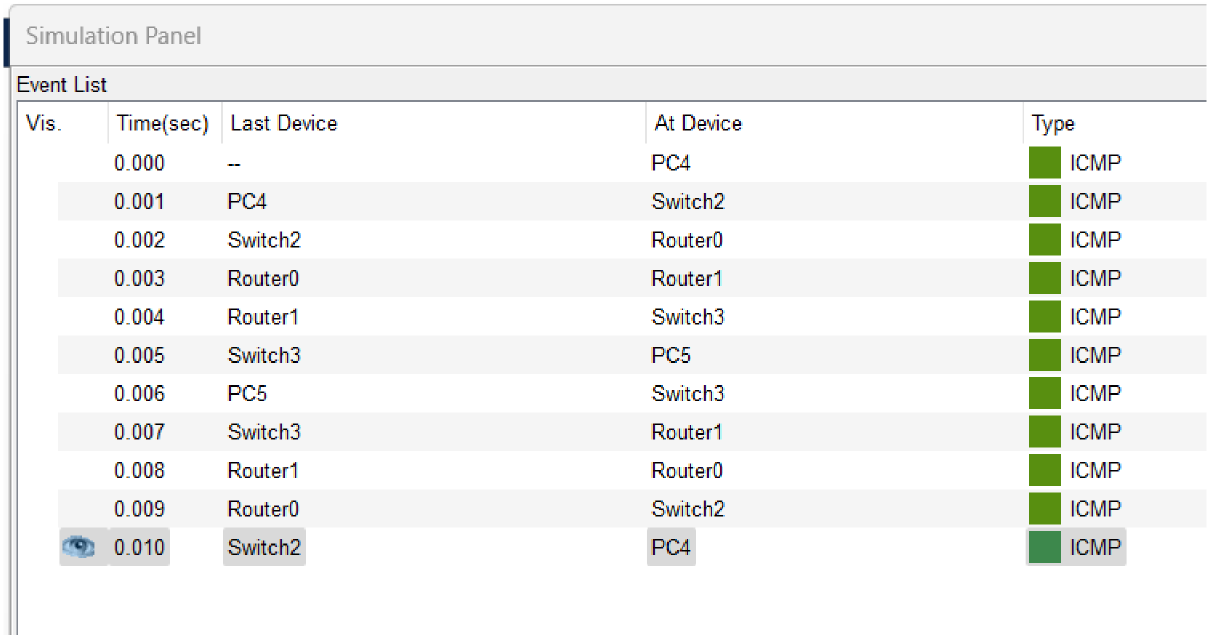

Name: Sudha Pantha

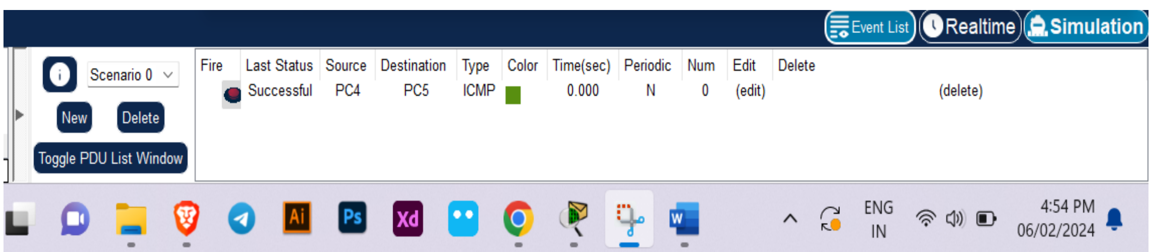
Step 5: Simulation

- A simulation of the experiment is given below we are sending PDU from PC0 to PC1:

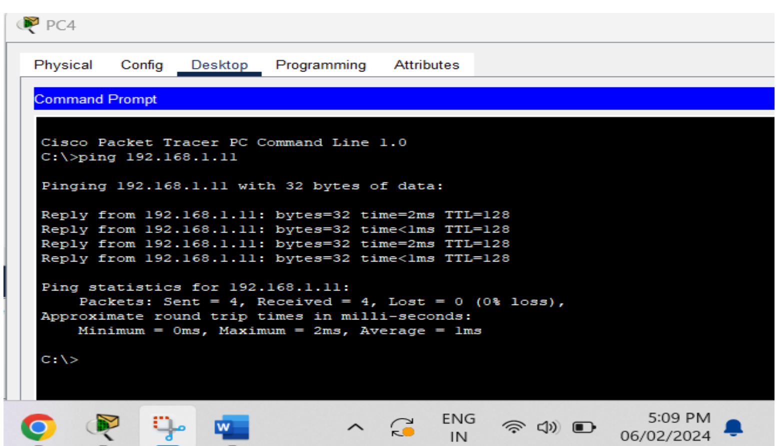


Name: Sudha Pantha

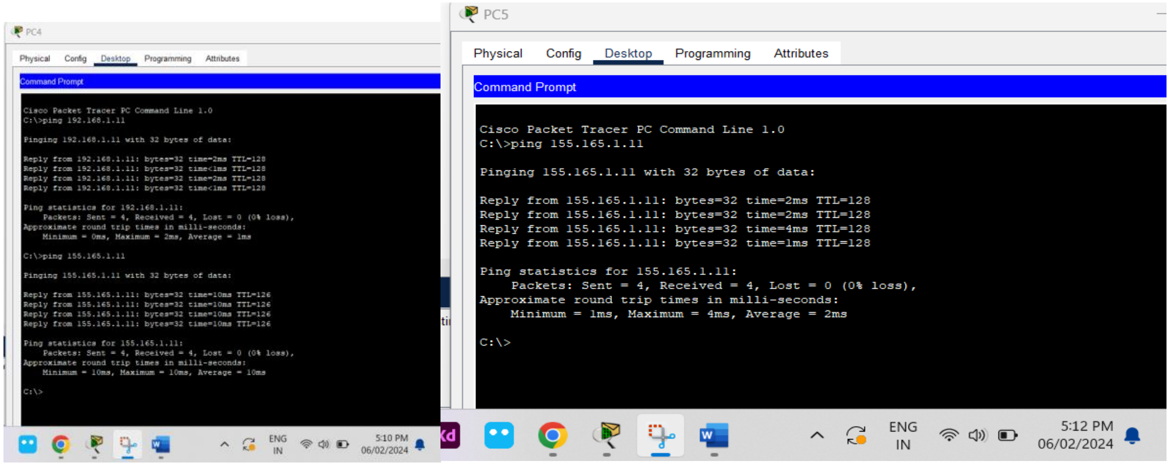




Step 6: Verifying the network by pinging the IP address of any PC.



Name: Sudha Pantha



Conclusion:

Hence, we configured OSPF protocol successfully verified the connection.

Lab Program Number: …15… *Date: 2080-10-22*

Title: To configure and understand about static routing using cisco packet tracer.

Theory:

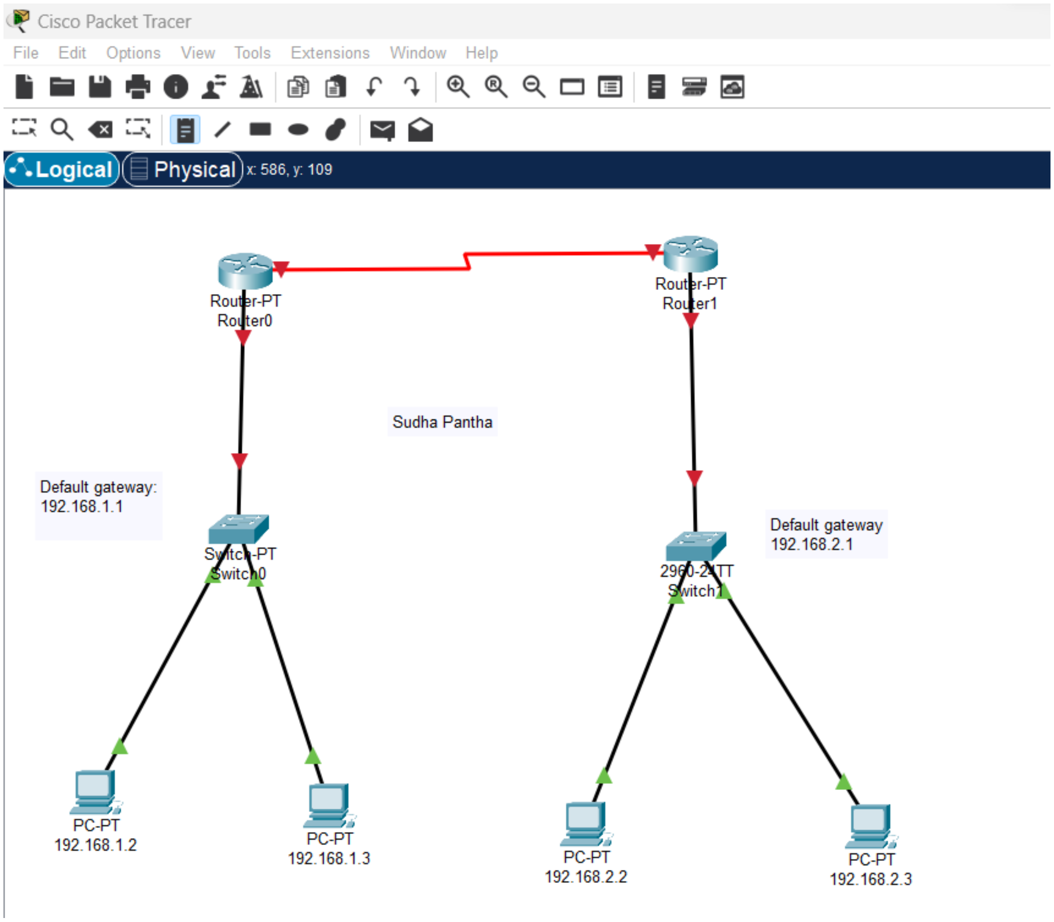
Static routing is a routing protocol that helps to keep your network organized and to optimize routing performance. It enables the router to assign a specific path to each network segment and to keep track of network changes. This helps to improve network stability and continuity. This adds security because a single administrator can only authorize routing to particular networks.

Procedure:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

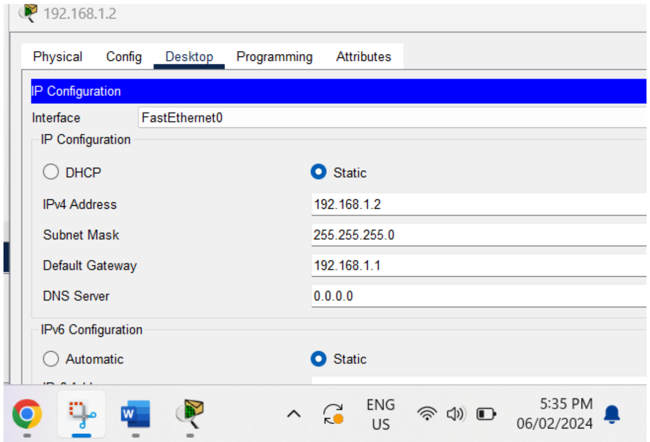
S.No	Device	Model Name	Qty.
01	PC	PC	04
02	Switch	PT-Switch	02
03	Router	Router-PT	02

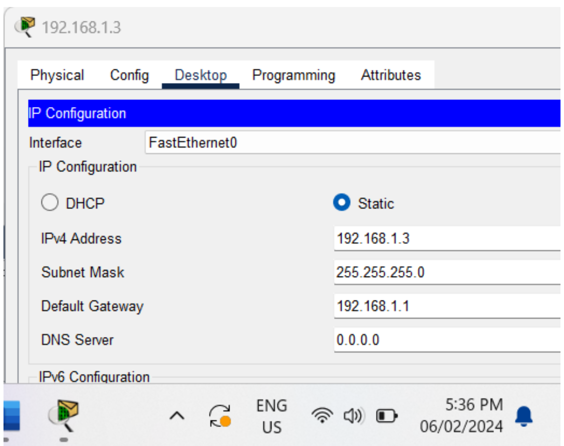
- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.



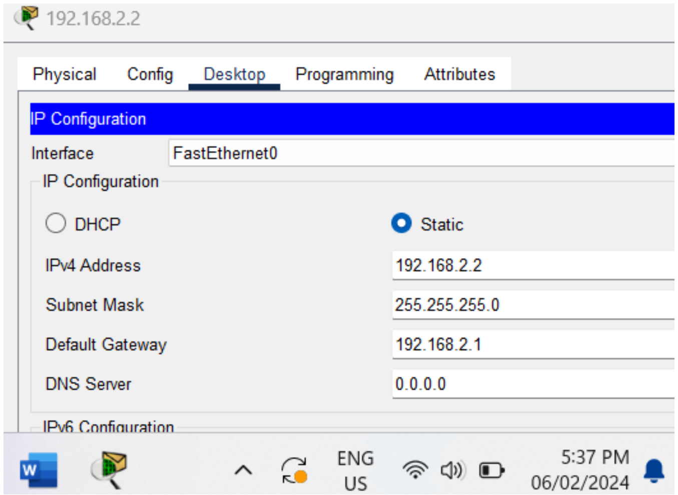
Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

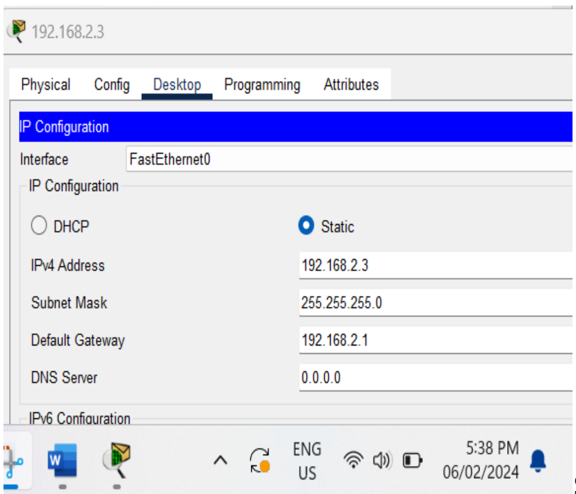
- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.





Name: Sudha Pantha





Step 3: Configure router with IP address and Subnet mask.

- To assign an IP address in router, click on router.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in Fast Ethernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask for all router.

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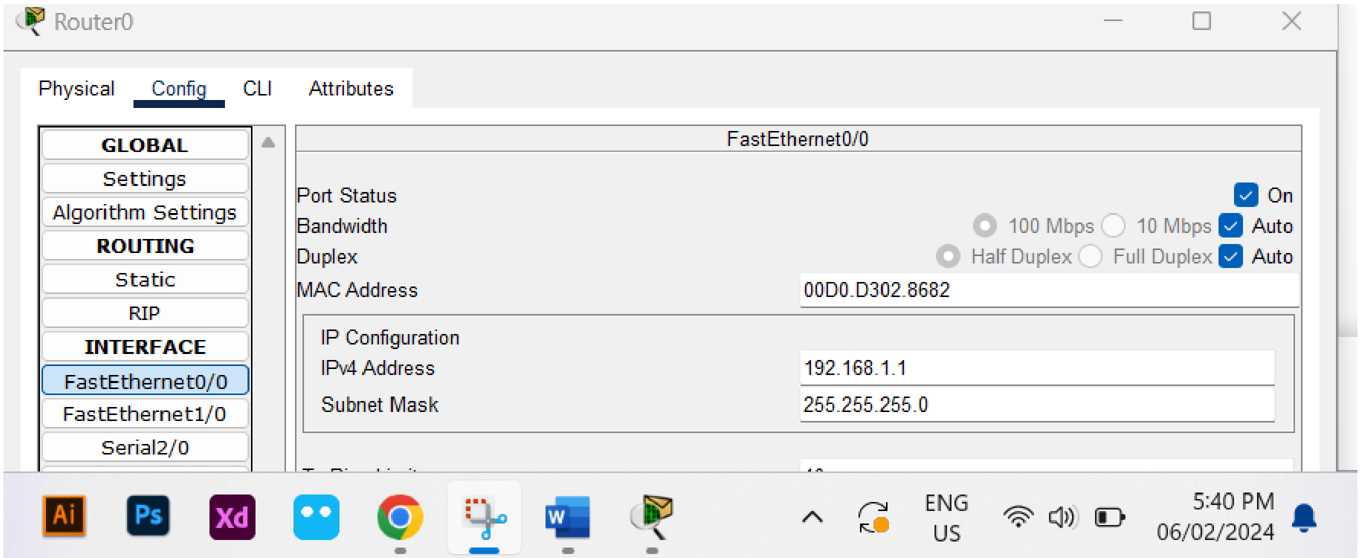

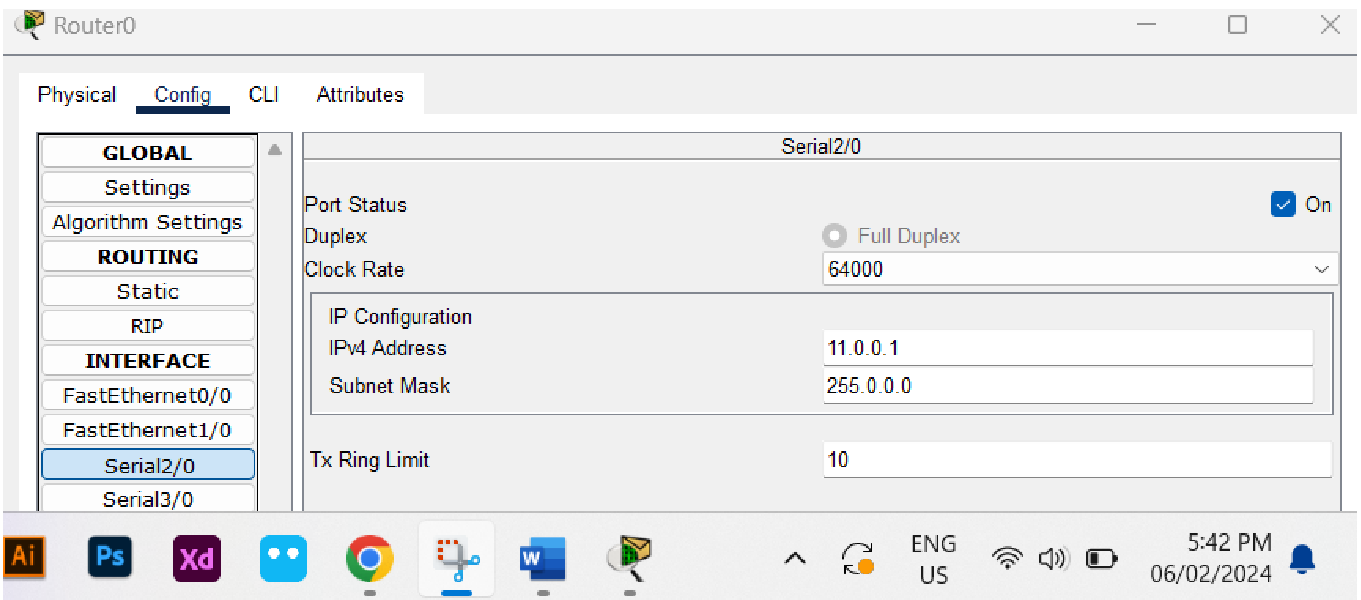

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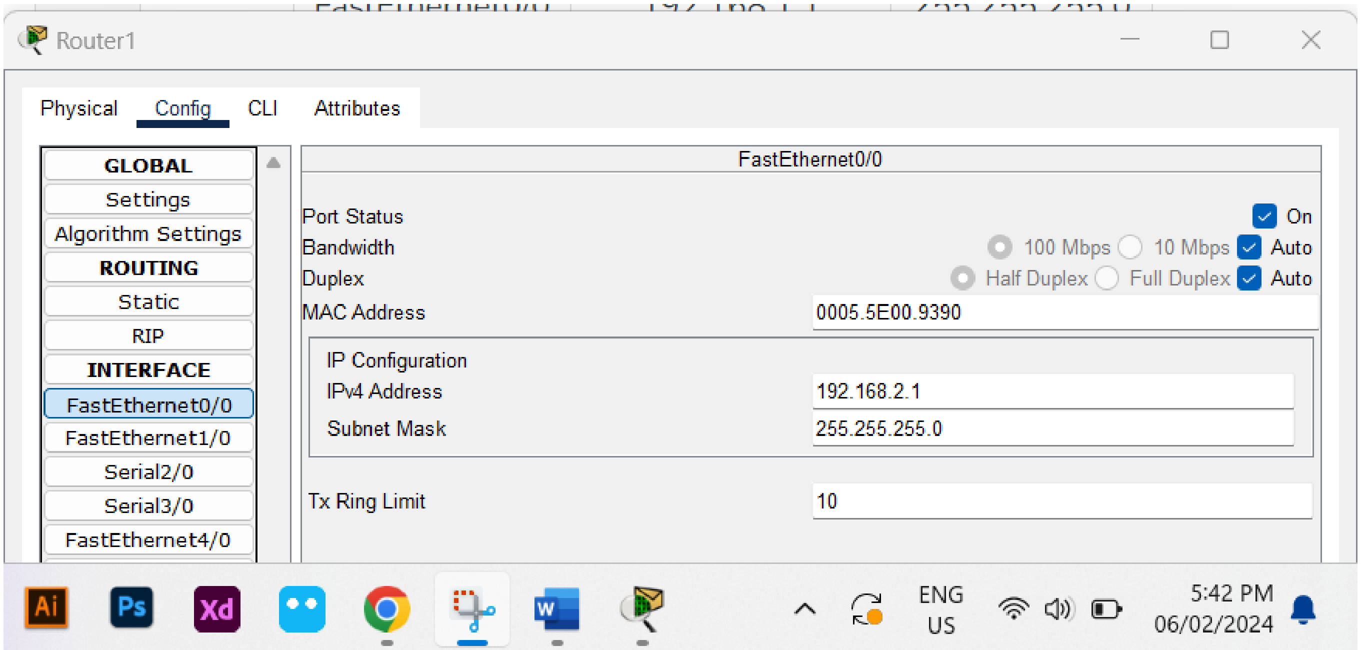

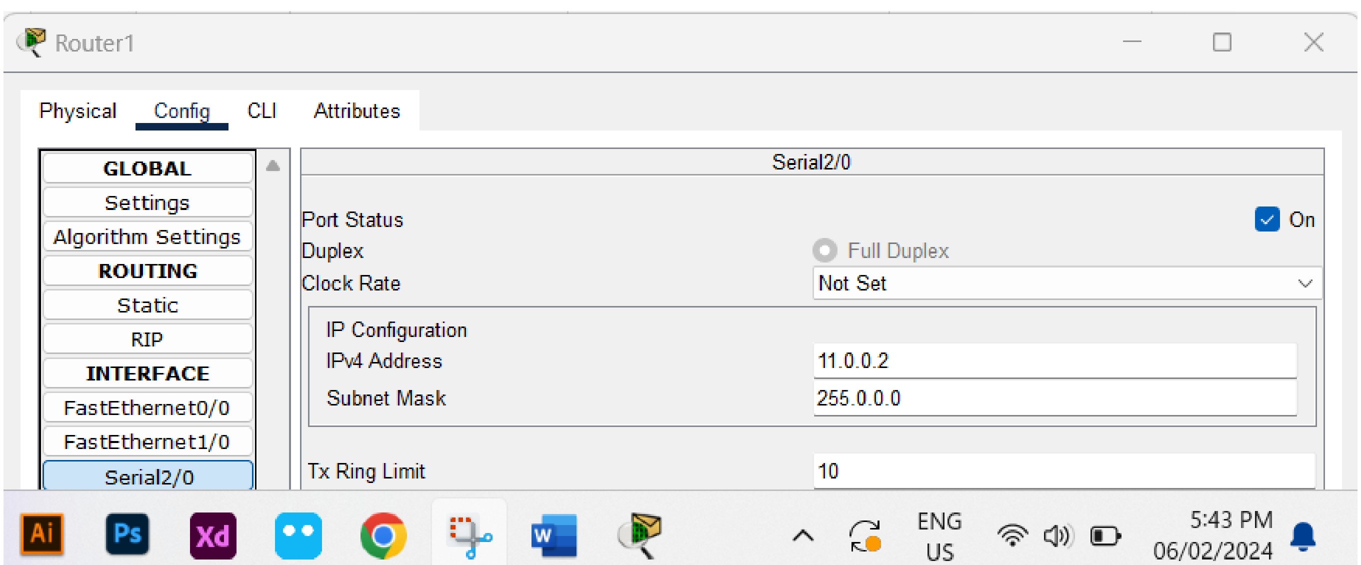

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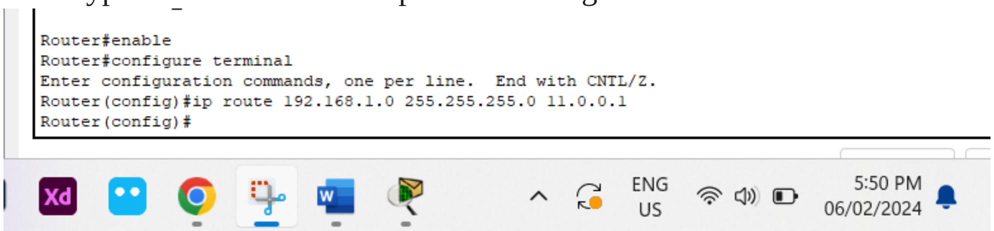
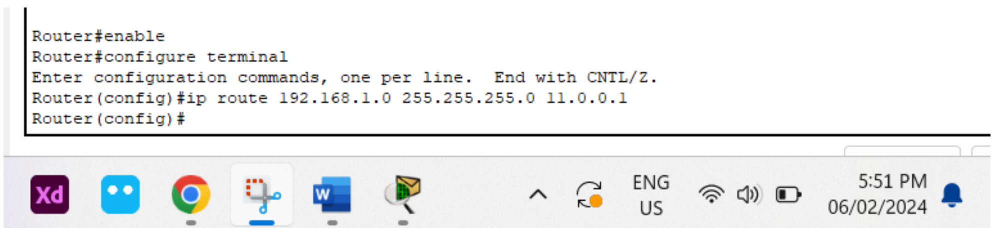
Step 4: After configuring all of the devices we need to assign the routes to the routers.

- To assign static routes to the particular router
- First, click on router0 then Go to CLI.
- Then type the commands and Ip information given below.





Step 5: Verifying the network by pinging the IP address of any PC. We will use the ping command to do so.

- First, click on PC1 then Go to the command prompt
- Then type ping <IP address of targeted node>
- As we can see in the below image we are getting replies which means the connection is working very fine

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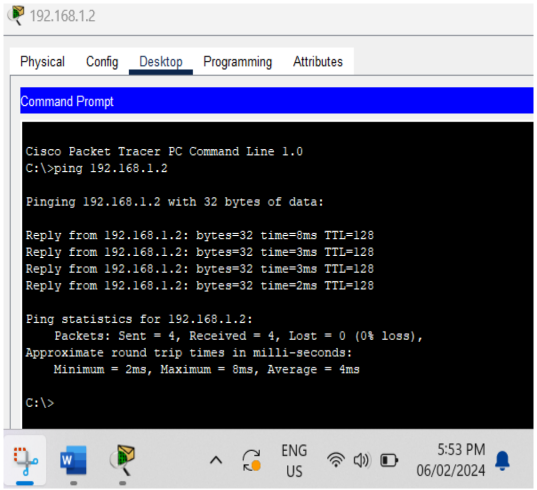

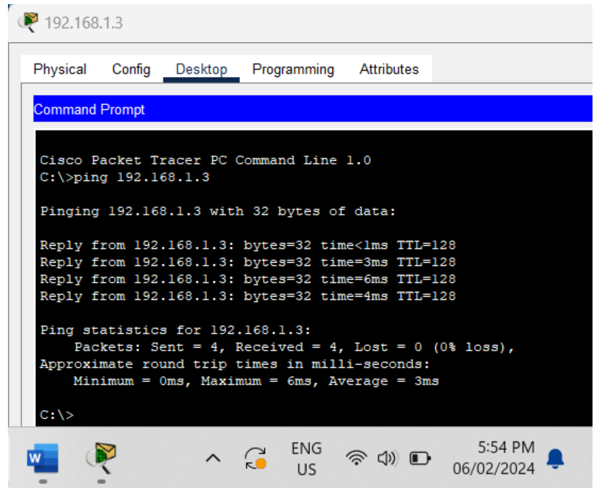

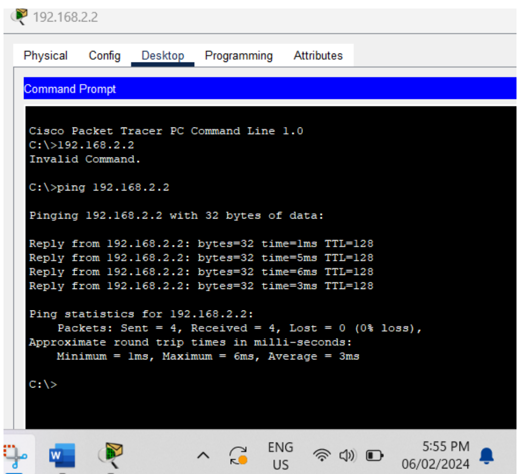

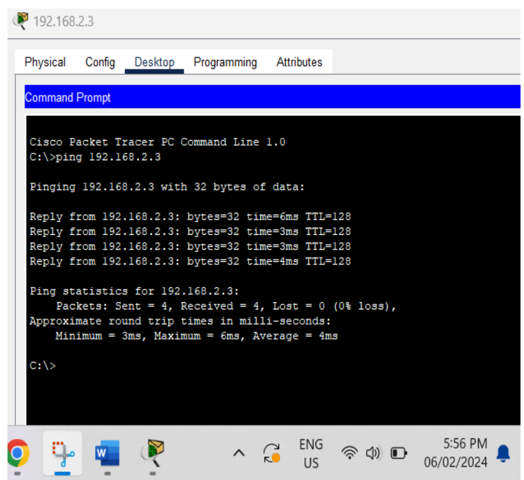

Step 6: We are sending PDU from PC0 to PC1.

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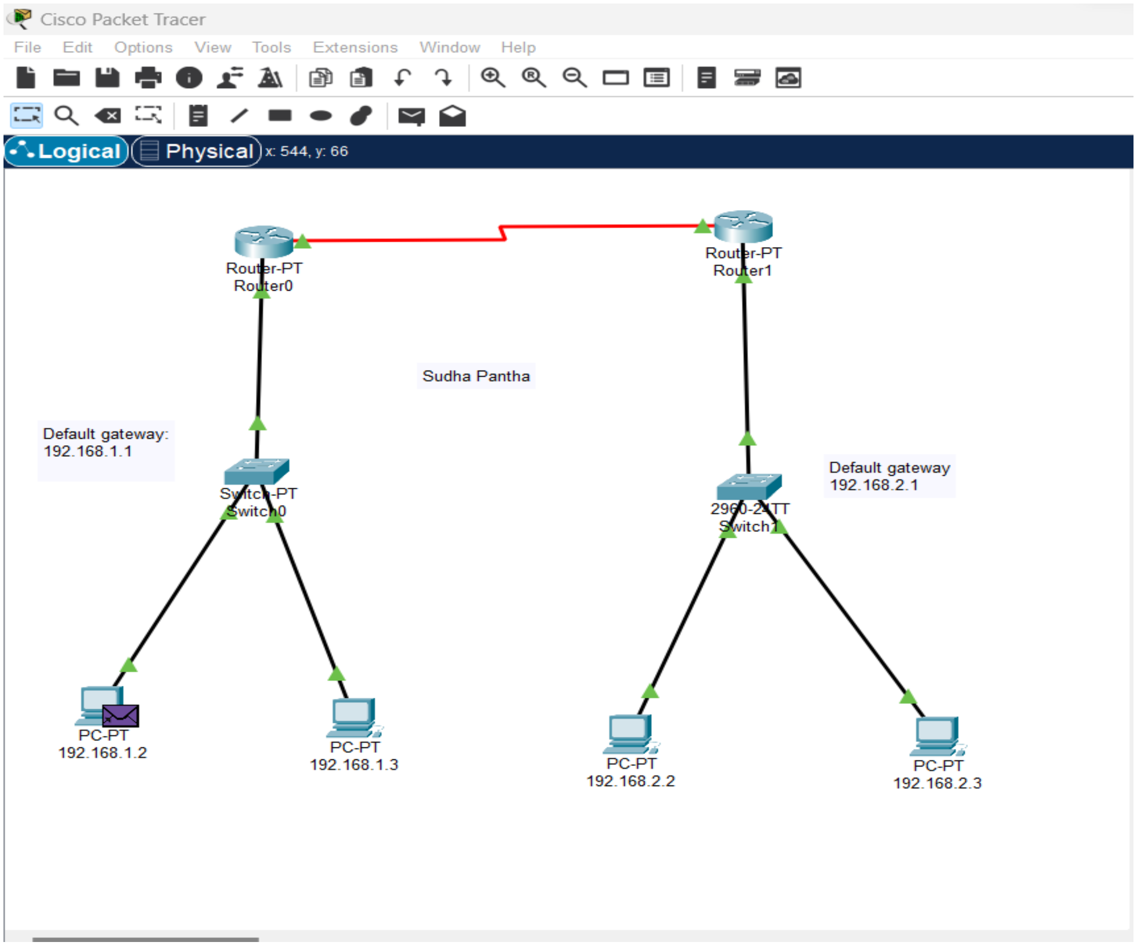

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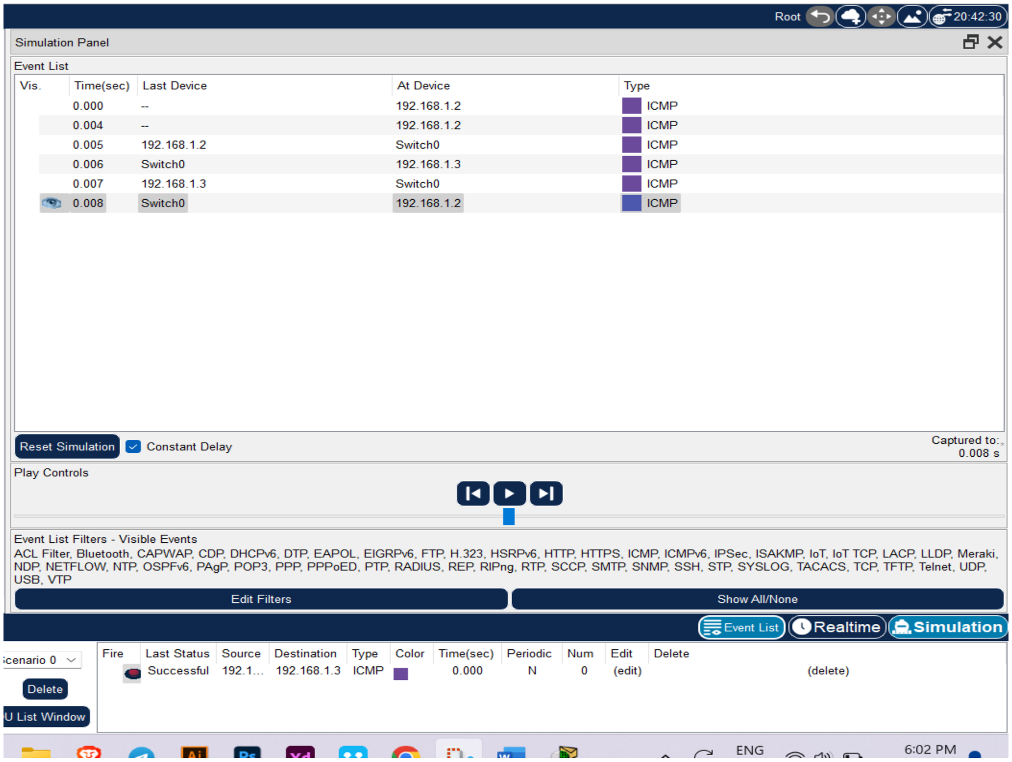

Conclusion:

Hence, we configured static routing protocol successfully verified the connection.

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